भारत दूरसंचार

INDIA TELECOM

सत्यमेव जयते

Department of Telecommunications
Ministry of Communications
Government of India

Compendium on Disaster Management in Telecom Network

(As Updated on 21st January 2026)

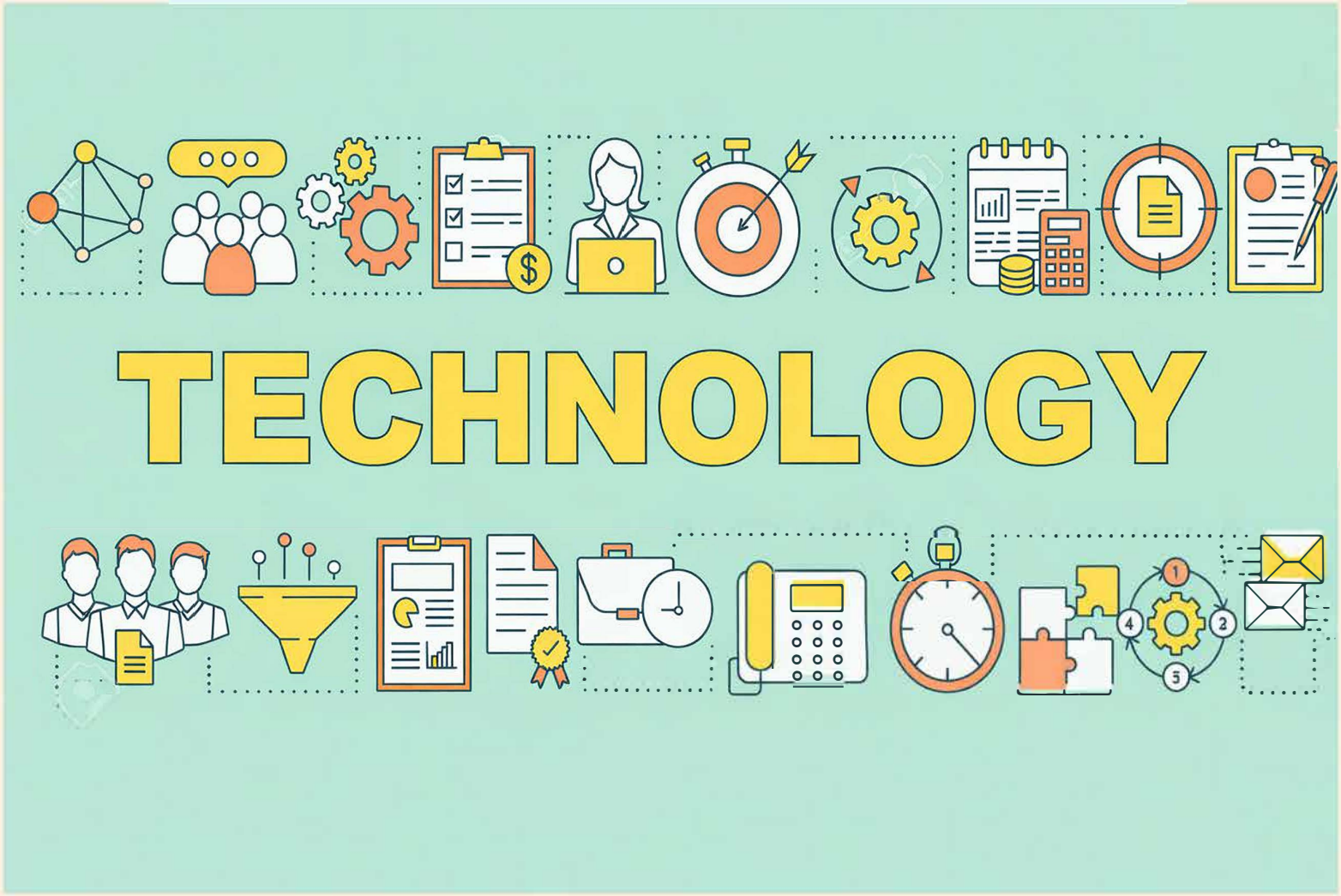
Prepared by:
Technology Vertical, DGT HQ

भारत दूरसंचार

INDIA TELECOM

सत्यमेव जयते

Department of Telecommunications
Ministry of Communications
Government of India



Editorial Board

PATRON

Govind Kewlani

Special DGT, GJ LSA

Chief Editor

Ujwal Gulhane

DDG(Technology), MH LSA

Chief Editor

Prem Chand Sharma

DDG(Technology), DGT HQ

Editor

Sharad Chitale

Director (Technology), MH LSA

Editor

Manish Shukla

Director (Technology), DGT HQ

Associate Editor

Sudhir

ADG (Technology), DGT HQ

Message from Director General Telecom

I extend my sincere congratulations to the officers and staff associated with the compilation and publication of this Compendium of important circulars on Disaster Management for Telecom & Networking products.

The consolidation of various circulars issued from time to time into a single, well-structured Compendium is a significant and timely initiative. It will greatly facilitate ease of reference, ensure uniform understanding, and support effective implementation of policies and instructions across the organisation.

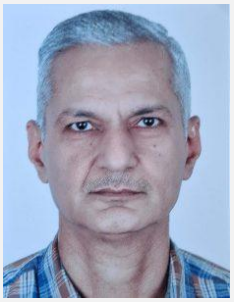
Such an effort reflects institutional foresight and a commitment towards strengthening administrative efficiency, transparency, and informed decision-making. I appreciate the meticulous efforts, coordination, and dedication of all those involved in compiling and presenting the circulars in a systematic and user-friendly manner.

I am confident that this Compendium will serve as a valuable reference document for officers and stakeholders and will contribute meaningfully to the effective discharge of official responsibilities.

I convey my best wishes to the team for continued success in their future endeavours.

Ramesh Kumar Goyal
Director General Telecom

Message from Addl. Director General Telecom



Dear Readers

I am happy to note that Technology Unit of DGT HQ and Maharashtra LSA under the able guidance of the Head of Gujarat LSA has prepared a comprehensive compendium on Disaster Management in Telecom Network.

Disaster management in the telecommunications sector emphasizes advance planning, resilience building, and readiness to ensure continuity of communication services during emergencies and in the aftermath of disasters. Robust telecom networks form the backbone of disaster management efforts, playing a critical role in preparedness, supporting coordinated response operations, and enabling swift recovery through dependable and uninterrupted connectivity. This compendium has been thoughtfully developed to present a comprehensive perspective on disaster management practices within telecom networks.

I commend the team for their meticulous effort in collating this resource, which will serve as a useful reference. It is imperative that this compendium is continually updated and serve as a guiding beacon for disaster management related endeavours in telecom sector.
My best wishes to the editorial team for these excellent initiatives.

Suresh Puri
Addl. Director General Telecom

CONTENTS

Introduction	I
Role of Telecom in Disaster Management	II
Organizational Set-up for Telecommunications Sector	III
Preparedness for Handling Disasters	IV
STDCC meetings	V
Short Code 1070	VI
Sachet-CAP-Integrated Alert System	VII
Satellite Phone	VIII
Annexure: Relevant Circulars, Documents and presentation for Disaster Management in telecom Network	IX

TORNADO

A tornado is a spinning tube of air that touches the ground and a cloud above. They usually travel only a few miles but can destroy everything in their wake.The US has 1200 a year-more than anywhere else in the world.

1. Introduction

A disaster is a serious disruption of the functioning of a community or a society involving widespread human, material, economic or environmental losses and impacts, which exceeds the ability of the affected community or society to cope using its own resources.

The Organization and management of resources and responsibilities for dealing with all humanitarians aspects of emergencies, in particular preparedness, response and recovery in order to lessen the impact of the disasters is the Disaster Management. Disaster management involves a wide range of stakeholders, including government agencies, non-governmental organizations (NGOs), communities, and the private sector.

Disaster Management Act, 2005 : It is a landmark piece of Indian legislation designed to effectively manage disasters and related matters. It establishes a framework for disaster prevention, mitigation, preparedness, and response at the national, state, and local levels. The Act mandates the creation of the **National Disaster Management Authority (NDMA)** and **State Disaster Management Authorities (SDMAs)** to oversee disaster management efforts. It also provides for the formation of the **National Disaster Response Force (NDRF)** for specialized response to disasters.

Key Features of the Act:

- **Institutional Framework** : The Act establishes a multi-layered institutional framework for disaster management, with the NDMA at the national level and SDMAs at the statelevel, according to the Ministry of Home Affairs.
- **National Plan** : The NDMA is responsible for developing and implementing a comprehensive National Plan for Disaster Management.
- **State Plans** : States are required to develop their own Disaster Management Plans in accordance with the National Plan.
- **Preparedness and Mitigation** : The Act emphasizes the importance of disaster prevention, mitigation, and preparedness measures.
- **National Disaster Response Force** : The Act provides for the establishment of the NDRF, a specialized force for responding to disasters.
- **Financial Assistance** : The Act provides for financial assistance to state governments in dealing with disasters.
- **Capacity Building** : The Act promotes capacity building at all levels to enhance disaster management capabilities.

DROUGHT

A drought occurs when there has not been enough rain so that crops and plants can grow. Farm animals die and people don't have enough to eat.

2. Role of Telecom in Disaster Management:

Telecommunications plays a vital role in disaster management, serving as a critical lifeline for communication, coordination, and public safety during and after a disaster. It facilitates emergency response by enabling rapid communication between authorities, rescue teams, and the public. Telecommunication networks also enable early warning systems, disseminating alerts about impending hazards and helping communities prepare.

The role of Department of Telecommunications (DoT) is to act as a Lead/ Primary agency for Emergency Support Function related to provision of telecommunication services by Telecom Service Providers and coordination with other Government agencies.

The DoT is issued Standard Operating Procedures (SOP) for various protocols of disaster management in telecom Network vide Letter No.

<SOP-2015 for responding to Disasters page no (24-46)>

<SOP-2017 for responding to disasters (Revision of SOP-2015) page no (47-75)>

<SOP-2020 for responding to disasters (Revision of SOP-2017) page no (78-103)>

Mission Statement:

"To minimize the impacts of disasters on life, property, environment, and the economy through prompt provisioning and restoration of Telecommunication Services in the disaster affected areas."

Objectives:

- Co-ordination of national actions to ensure the provision of telecommunication support to the centre, state and district administration.
- Co-ordination of the requirement of temporary telecommunication services in the affected areas.
- Co-ordination for restoration of telecommunication services.

EARTHQUAKE

Earthquakes are caused by the movement of the earth's plates. There are 20 which lie under the surface of the earth. 4 out of 5 earthquakes take place in the Pacific Rim.

3.0rganizational Set-up for Telecommunications Sector :

The Department of Telecommunications will render Emergency Support Functions (ESF) for telecommunication services, wherever Central intervention and support are needed by the State Governments. An effective organizational set-up for telecom services during disasters and emergencies at all levels viz. Central, State and District is very important for implementing and monitoring the disaster management plans.

- **National Level :** National Telecom Disaster Co-ordination Committee (NTDCC)
- **State Level** : State Telecom Disaster Co-ordination Committee (STDCC)- At State or telecom circle level, all Disaster related activities will be coordinated
- **District Level :** STDCC shall oversee the sectoral requirement for the rescue, relief and restoration by State/District.

भारत दूरसंचार
INDIA TELECOM

The 'Cell Broadcast Alert System' aims to provide real-time disaster alerts to citizens for scenarios like tsunamis, floods, earthquakes, and more. Emergency phone alerts help keep you informed during critical situations

FLOODS

Floods are one of the worst natural disasters and cost the most money. They occur when storms cause rivers to overflow or by high tides. They cause enormous damage and loss of life.

4. Preparedness for Handling Disasters:

All TSPs shall prepare their Disaster Management Plan and submit the same to DoT HQ Nodal officer i.e. DDG (DM).

- Disaster Response Task Force (DRTF)
- TSPs shall have a Disaster Response Task Force (DRTF) at State level. DRTF teams will be responsible for immediate provisioning of emergency communication and restoration of telecom services in disaster affected areas.

  DRTF shall consist of teams of experts in the following areas:

  - Transmission Team
  - Switching Team
  - Infrastructure Team
  - Mobile Service Team
  - Telecom relief Team (for opening PCOs, helpline response etc.)
  - Rapid Damage Assessment Team (RDAT)
- Resource Centre
- Memoranda of Understanding amongst TSPs
- Training and Drill
- Directory

भारत दूरसंचार
INDIA TELECOM

Later a backronym was created for it in popular usage, and SOS became associated with mnemonic phrases such as **"Save Our Souls"** and **"Save Our Ship".** Moreover, due to its high-profile use in emergencies, the phrase **"SOS"** has entered general usage to informally indicate a crisis or the need for action.

WINDSTORM

A wind that is strong enough to cause at least light damage to trees and buildings and may or may not be accompanied by precipitation.

The preparedness measures taken by TSPs will be reviewed six-monthly.

i. Nodal officer of TSPs of affected telecom circle level shall report to concerned DDG (Technology), DoT (Chairman of STDCC) in that circle, for sharing information and coordination related matters.

ii. LSA of DOT shall be the single nodal point in the disaster region where representatives of TSPs shall also be present to coordinate and oversee communication restoration efforts.

iii. *A* control room will be setup at the state HQ / nearest to affected area, as the case may be, and made operational under control of LSA of affected area.

iv. The DDG (Technology) of concerned LSA shall act as interface between all TSPs and other support agencies including State Government for any coordination related issues.

v. Meeting of STDCC shall be convened to review situations. STDCC shall report to NTDCC.

vi. TSPs shall make helpline numbers operational where the last location of the survivors or missing persons can be intimated to the relatives.

vii. Providing information through helpline numbers about the last location details or sharing of CDRs of the subscribers, in the disaster affected areas, with the state/ central agencies requires special regulatory permission. However, in case of occurrence of disaster, these permissions would be deemed automatically permitted for duration of 2 weeks.

viii. TSPs shall share specialized resources and allow Intra-circle roaming as per MoU for provisioning of services to priority user groups and general public during disaster period.

ix. In order to have wider coverage from a single BTS, TSPs may radiate more radio power from the BTSs located in the disaster affected areas beyond the EMR limits prescribed for 15 days.

x. TSPs shall broadcast messages at regular intervals, in consultation with STDCC/NTDCC, to all the subscribers in the affected areas through SMSs / Cell broadcast free of cost during disaster period based upon instructions of Nodal.

AVALANCHE

An avalanche is a like a mudslide except it is made up of snow and ice. They occur in the mountains after fresh snowfalls and can travel at 80 miles an hour.

authorities as per DM act 2005 i.e. National Executive Committee(NEC)/National Crisis Management Committee(NCMC)/ State Executive Committee(SEC). This shall provide details about:

a. Details of TSPs helpline numbers.

b. Details about rescue and relief activities of state government such as tentative schedule of food / water distribution / nearest shelter/ shelter camp etc. as per need of State agencies.

xi. TSPs shall open sufficient number of PCOs, preferably free of cost, for use of general public in affected area.

xii. TSPs shall also ensure that in the disaster affected areas, no subscriber shall be denied access to voice / SMS communication due to any commercial consideration, whatsoever, including non-payment/insufficient balance/recharge etc. This facility shall continue for at least 15 days.

xiii. TSPs shall submit First Information Report as per <Annexure-III Page No(102)>and thereafter Daily Status Report about their network, help line and PCOs etc. as per <Annexure-IV Page No (103)> to concerned LSA unit on daily basis or as sought by DDG (Technology), Head of STDCC.

xiv. LSA shall submit the status report in respect of overall telecom facilities in affected area to DoT, HQ on daily basis or as sought by DoT, HQ.

xv. The Control room under respective LSA shall remain operational 24X7 till the telecom services are restored to near normalcy or as per instructions from DoT, HQ.

xvi. For the circle wise list of TSPs, website of DoT may be referred.

xvii. Every LSA shall have a Satellite Phone for proper coordination with State Govt. during disasters.

xviii. HAM operators are allowed to operate anywhere in India except those locations restricted by GOI from time to time. Officer in Charge (disaster Management) at State or the District Magistrate of disaster affected district, as the case may be, should prescribe minimum level of officer for signing a message, to be passed on HAM network during disaster.

VOLCANO

Volcanoes are explosions that happen from openings in the earth's surface. Most volcanoes are located in the Pacific Rim. When volcanoes erupt, they spew magma, lava, and ash. There are between 1,300 to 1,500 volcanoes on Earth.

5. STDCC meetings :

To coordinate telecommunications services during and after disaster events, ensuring smooth communication and restoration of services. Meetings often include reviews of disaster management preparedness plans and necessary measures for providing telecommunications services during emergencies.

<STDCC Chairman Directory(158-158)>

FOREST FIRE

Forest fires are fires that burn in forest areas. Lightning can start them but usually careless people do. There are 100,000 fires in the US every year. They can travel at 23 kilometers an hour and will destroy everything as they move. Drought and hot weather are also responsible for many fires.

6. Short Code:

Short code 1070 as Standard Disaster Helpline to all State/UTs vide DoT Letter No. 16-1/2017-AS-III/Short Codes/362 Dated 28.7.2020. This number is of Service category with Unrestricted Accessibility and non metered number.

<Allotment of short code -1070 as disaster help Line to all states/UTS No 16-1/2017- AS-III/Short Codes/362 Dated 28.7.2020 page no (76-77)>

Subject: Allotment of short code '1070' as Standard Disaster Helpline to all State/UTs- reg.

This is in continuation to DoT letter No. 16-03/2011-AS-III//Vol.II/106 dated 27.03.2012 allocating short code '1070' as Disaster Management Helpline to Relief Commissioner of Natural calamities (copy enclosed). In this regard, the undersigned is directed by competent authority to state that short Code '1070' is allocated as Standard Disaster Helpline to all State/UTs with following features:

i. Category-I: Service (i.e. mandatory service to be provided by all the Access providers).
ii. Accessibility: Un-restricted i.e. available with STD code also.
iii. Metered Service: Non-metered

TSUNAMI

A tsunami is a train of gigantic waves that occur after an earthquake or volcanic eruption. They can be as high as 30 meters, travel as fast as a jet plane and go as far inland as 2 kilometers. Most tsunamis happen in the Pacific.

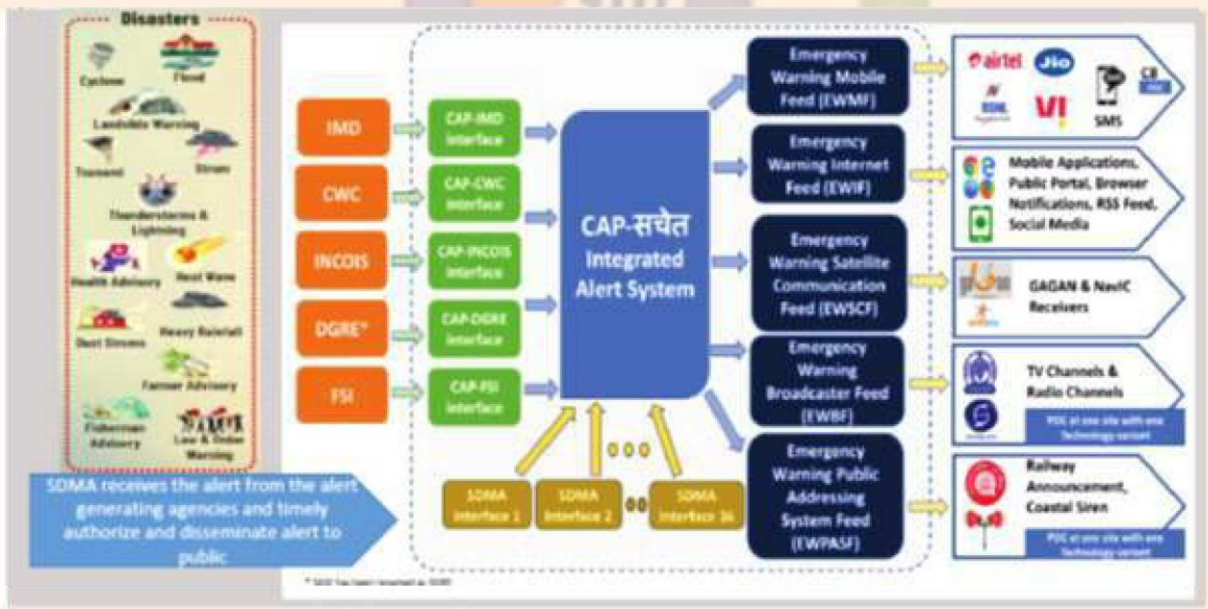
7. Sachet- CAP-Integrated Alert System:

C-DOT's CAP-Integrated Alert System is a state-of-the-art geo-intelligent platform that enables disaster management authorities to send targeted and automated Multi-Hazard Multi-Media alerts to citizens in near real time and in multiple languages. It integrates multiple stakeholders to bridge communication gaps and employs multiple technologies including SMS, Cell Broadcast, Mobile App, TV, Radio, social media, RSS Feed, Browser Notification, and Satellite for enabling effective communication and ensuring last mile reachability in-line with the Hon'ble Prime Minister's Ten Point Agenda on Disaster Risk Reduction.

The platform is based on ITU-x.1303 Common Alerting Protocol (CAP) Standard and aims to realize the United Nations initiative of "Early Warnings for All" to ensure everyone on Earth is protected from hazardous weather, water or climate events through life-saving early warning systems.

MoU signed between NDMA and CDOT on 23rd Aug,2021 for Pan- India implementation of CAP Project.

<Link to Sachet website: https://sachet.ndma.gov.in/>



<Tariff for SMS and cell broadcast Alerts/Messages Disseminated through CAP Platform NO 15-2/2021-DM dated 26.12.2022 page no (106-108)>

<Regarding Tariff for SMS and Cll Broadcast message disseminated through CAP based Integrated Alert System No: C-DOT/DSA/CAP-SACHET /DOT-LSA/2023/1/1 Date 27th January, 2023 page no (109-109)>

<SACHET_CAP_PPT_16.3.2023 page no (126-145)>

SANDSTORM

A sandstorm a strong wind in a desert carrying a large amount of sand. They can reach heights of 10-50ft and are most commonly found in desert regions worldwide

8. Satellite Phone:

Satellite phones play a crucial role in disaster management by providing reliable communication when traditional networks are down or overloaded. They enable emergency responders and relief workers to coordinate efforts, maintain contact with affected areas, and facilitate the flow of information. Satellite phones can also be used to assess damage and plan recovery efforts.

<Procurement of Satellite phone in LSAs.-reg. No- 47-1/2020-DGTHQ/4 dated 12.05.2022 page no (104-105)>

<Procurement of Satellite Phones in LSAs - reg No - 47-1/2022-DGTHQ/4 dated - 18.10.2023 page no (147-147)>

भारत दूरसंचार
INDIA TELECOM

HURRICANE

A hurricane is a strong tropical storm with high winds and heavy rain. Each hurricane has a boy's or girl's name. They cause tremendous damage when they hit land. They occur from June until November in the US.

National Disaster Response Force (NDRF)

(https://www.ndrf.gov.in/en)

A visit was done on dated 22/4/2025 at NDRF HQ at Sudumbare Pune by Sh. Ujwal Gulhane, DDG(T), Sh Sharad Chitle, Director(T) and Smt Sumitra Sarda (AD(T) from MH LSA Pune. Shri Deepak Tiwari 2nd in-Commandant NDRF was present on the occasion. NDRF presented their various activities on Disaster Management.

DUSTSTORM

A dust storm is a wall of dust and debris that is often blown into an area by strong winds from thunderstorms. The wall of dust can be miles long and several thousand feet high.

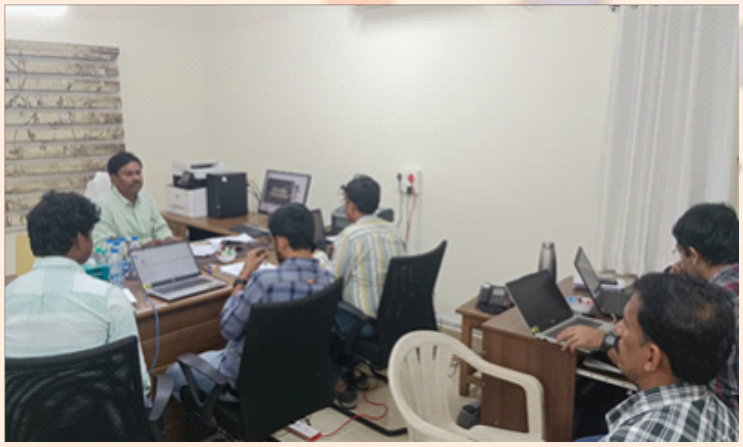
Mock Drill at Goa by NDRF and SDMA Goa

NDRF Maharashtra conducted a Mock Drill at Goa from dated 25/04/2025 wherein all TSP from Goa participated actively. Glimpses of Mock Drill are as above.

HEATWAVE

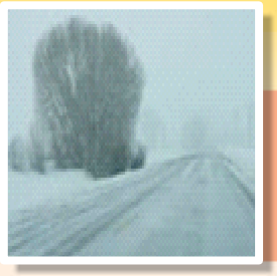
A Heatwave is a period of abnormally high temperatures, more than the normal maximum temperature that occurs during the summer season. It is sometimes described as a period of extreme heat and are primarily caused by high-pressure systems that trap warm air near the surface.

Mock Drill organized in UPE LSA during Maha Kumbh Mela 2025 preparedness



War room established in AP LSA established during Montha Cyclone

STDCC meeting in Odisa LSA to discuss preparation during Dana cyclone with TSPs, State Govt. and NDMA



SNOWSTORM

A blizzard is a severe snowstorm with very high winds, extremely low temperatures, and poor or no visibility. They are especially dangerous for travelers who may get stuck in their cars and could suffer from hypothermia.

Mock drill conducted for disaster preparedness in Sonepur (Saran District), Bihar

STDCC Meeting by Haryana LSA for disaster preparedness before Monsoon Season

Restoration of network underway during Heavy rains & landslides in District Kullu HP

HAILSTORM

Hailstorm is a type of solid rain made up of balls or lumps of ice. They typically last for no more than 15 minutes but can cause injuries to people and damage buildings, vehicles and crops.

Annexure:

Relevant Circulars, Documents and presentation for Disaster Management in telecom Network

Sr. No.	Circular Description	Date	Page No.
1	SOP-2015 for responding to Disasters	05th June 2015	24-46
2	SOP-2017 for responding to Disaster (Revision of SOP-2015)	17th Mar 2017	47-75
3	Allotment of short code-1070 as disaster help line to all states/UTS No.16-1/2017-AS-III/Short codes/362 Dated 28.7.2020	28th July 2020	76-77
4	SOP-2020 for responding to Disasters (Revision of SOP-2017)	14th Dec 2020	78-103
5	Procurement of Satellite phone in LSAs.-reg. No.-47-1/2020-DGTHQ/4 dated 12.05.2022.	12th May 2022	104-105
6	Tarrif for SMS and cell broadcast Alerts/Messages Disseminated through CAP Platform No15-2/2021-DM dated 26.12.2022.	26th Dec 2022	106-108
7	Regarding Tarrif of SMS and CII broadcast message disseminated through CAP based Integrated Alert System No.:C-DOT/DSA/CAP-SACHET/DOT-LSA/2023/1/1.	27th Jan 2023	109-109
8	Standard Operating System(SOP) for maintaining Quality of Telecom Services during Significant Social/Cultural/Religious/Commercial/International events etc. Dated 10.07.2024	10th July 2024	110-122
9	Clarification regarding Para C(3) " Emergency Preparedness" of Standard Operating Procedure(SOP) dated10.07.2024.	22nd Jan 2025	123-124
10	NoC OM dated 08.12.2025.	08th Dec 2025	125-125
11	SACHET_CAP_PPT_16.03.2023	16th Mar 2023	126-145
12	Procurement of Satellite Phones in LSA-reg No-47-1/2022-DGTHQ/4 Dated 18.10.2023	18th Oct 2023	146-146

SINKHOLE

A sinkhole is a depression or hole in the ground caused by some form of collapse of the surface layer. It is caused by removal of material below the ground and either collapse or gradual subsidence of the surface into the resulting void.

Sr. No.	Circular Description	Date	Page No.
13	HAM Radio in DM_PPT_29.05.2025	29th May 2025	147-157
14	STDCC Chairman Directory	31st May 2025	158-158
15	List of Satellite Phones	1st May 2025	159-159

भारत दूरसंचार
INDIA TELECOM

Mudslide/Landslide

Mudslides are fast moving landslides that occur after heavy rains. They can destroy and kill everything in their path.

Disclaimer:

This compendium has been compiled based on available circulars, guidelines, and field-level inputs as of the date of publication. While every effort has been made to ensure its accuracy, DoT Maharashtra LSA does not assume responsibility for policy changes or any administrative modifications. The information provided on this compendium is for general informational purposes only. This office assumes no responsibility for errors or omissions in the content or for any actions taken based on the information provided. The information contained in this document and any attachments are confidential and strictly for internal circulation only. Further copying, reproducing, or republishing any part of the content, including text, images, and other materials, is strictly prohibited.

STANDARD OPERATING PROCEDURE

(SOP-2015)

for

TELECOMMUNICATION SERVICES

for Responding to

DISASTERS

GOVERNMENT OF INDIA

MINISTRY OF COMMUNICATION & IT

DEPARTMENT OF TELECOMMUNICATIONS

TABLE OF CONTENTS

Sl. No.	Content	Page No.
1	INTRODUCTION	1-3
2	INSTITUTIONAL STRUCTURE	4-5
3	MITIGATION and PREPAREDNESS	6-10
4	CONTROL ROOM	11
5	TRIGGER MECHANISM	12-13
6	RESPONSE and RECOVERY	14-16
7	ABBREVIATIONS	17-18
8	Annexure – I : Directory of Nodal Officers	19
6	Annexure-II : Format for First Information Report	20
7	Annexure – III : Daily Situation Report	21

INTRODUCTION

The disasters may cause enormous loss of life and property. Even though the global average of number of disasters has been rising rapidly, improved warnings & mitigation programme and quick response have reduced significantly, the loss of human life. In the aftermath of a disaster, the telecommunication services are required on top priority and play a pivotal role in undertaking rescue, relief and rehabilitation measures. The role of Department of Telecommunications (DOT) is to act as a Lead/Primary agency for Emergency Support Function related to provision of telecommunication services by Telecom Service Providers and coordination with other Government agencies.

1.2. Disaster Management is broadly categorized into following 4 phases :

- Mitigation - includes any activity that prevents a disaster, reduces the chance of a disaster happening, or reduces the damaging effects of unavoidable disasters.
- Preparedness - includes plans or preparations made to save life or property, and to help the response and rescue service operations.
- Response - includes actions taken to save lives and prevent property damage, and to preserve the environment during emergencies or disasters. The response phase is the implementation of action plans.
- Recovery – includes actions that assist a community to return to a sense of normalcy after a disaster.

1.3. The New Telecom Policy - 2012 (NTP-2012) also emphasizes importance of Disaster Management and various provisions in this regard are as mentioned below :

- Creation of the robust and resilient telecom networks for adequately addressing the need for proactive support for mitigating disasters - natural and manmade (Para 15 of Preamble).
- To prescribe sectoral Standard Operating Procedures for aiding effective and early mitigation during disasters and emergencies (Para 5.12 of Strategies).
- To create appropriate regulatory framework for provision of reliable means of public communication by Telecom Service Providers during disasters (Para 5.13 of Strategies).
- To encourage use of ICTs in prediction, monitoring and early warning of disasters and dissemination of information (Para 5.14 of Strategies).
- To facilitate an institutional framework to establish nationwide Unified Emergency Response Mechanism by providing nationwide single access number for emergency services (Para 5.15. of Strategies).

1.4. These provisions made in preamble and strategies of NTP-2012 have far reaching implications for the nation, being prone to natural disasters like floods, earth-quakes, coastal cyclones etc. Therefore, putting in place processes and structures by developing a holistic, proactive, multi disaster oriented and technology driven strategy through a culture of prevention, mitigation and preparedness in the country will definitely help in better response and recovery.

Therefore, this document deals with Standard Operating Procedures (SOP) to be followed by various stake holders providing Telecommunication services in the country, for effective mitigation and recovery during disaster and emergency.

1.5. Mission

"To minimize the impacts of disasters on life, property, environment, and the economy through prompt provisioning and restoration of Telecommunication Services in the disaster affected areas."

1.6. Objectives

- Co-ordination of national actions to ensure the provision of telecommunication support to the centre, state and district administration.
- Co-ordination of the requirement of temporary telecommunication services in the affected areas.
- Co-ordination for restoration of telecommunication services.

1.7. "Disaster Management Guidelines for Telecommunications" were issued by the Telecom Engineering Centre in Feb' 2008. These Guidelines, as amended from time to time, may also be referred for the technical details by various stake holders.

1.8. On occurrence of disasters, all stake holders make all out efforts to their own judgment for the restoration of damages caused to telecom network. However, in such times a coherent and coordinated approach among all stakeholders is the key to reap the full benefit of the efforts made for restoration of telecom networks in such emergent situations. It is, therefore, necessary to standardize the processes and actions that need to be taken.

1.9. Accordingly, in order to optimize the efforts of all stakeholders, "Standard Operating Procedure" (SOP) is required to be adopted covering all aspects of disaster management from mitigation to recovery to minimize loss due to disaster and to ensure efficient response.

1.10. For effective provisioning of emergency telecommunication services and restoration of normal telecommunication services, DoT will need support of following Ministries/ Departments/ Organizations:

(i) Telecom Service Providers / Telecom Infrastructure Providers
(ii) Department of Electronics and Information Technology
(iii) Directorate of Communication and Police Wireless
(iv) Ministry of Railways
(v) National Disaster Response Force
(vi) Ministry of Civil Aviation
(vii) Ministry of Defence
(viii) Ministry of Power
(ix) Ministry of Road Transport & Highways
(x) Ministry of Petroleum & Natural Gas
(xi) Power Grid Corporation of India Ltd.
(xii) Department of Space
(xiii) Ministry of Information & Broadcasting
(xiv) State and District Administration / Agencies
(xv) Indian Meteorological Department

INSTITUTIONAL STRUCTURE

2.1 In the aftermath of a disaster, the primary responsibility for organizing response and relief as well as recovery in disaster affected areas lies with the state government. However, sectoral institutional structures are needed to strengthen and supplement the efforts of rescue, relief and restoration.

2.2 The Disaster Management Act, 2005, lays down a three tier institutional structure for disaster management at the national, state and district levels in the form of NDMA, SDMA and DDMA. National Policy on Disaster Management (NPDM) has further specified the roles and responsibilities of various organizations for disaster response. Accordingly, to fulfill roles and responsibility of Department of Telecommunications institutional structure has been set up in the following paragraphs.

2.3 Organizational Set-up for Telecommunications Sector

The Department of Telecommunications will render Emergency Support Functions (ESF) for telecommunication services, wherever Central intervention and support are needed by the State Governments. An effective organizational set-up for telecom services during disasters and emergencies at all levels viz. Central, State and District is very important for implementing and monitoring the disaster management plans.

2.3.1 NATIONAL LEVEL

National Telecom Disaster Co-ordination Committee (NTDCC) :

At National or central level, all Disaster related activities will be coordinated and monitored by **National Telecom Disaster Co-ordination Committee (NTDCC)**. The committee will be chaired by the Member (T), DOT and will have the following constitution:

- Member (Technology) - Chairman
- Sr.DDG (Term-Security), DOT HQ
- DDG (AS), DOT HQ
- DDG (CS), DOT HQ
- DDG (DS), DOT HQ – Member Convener
- Members from Telecom Service Providers (Public & Private)
- Representative from MHA/NDMA

It will issue guidelines from time to time as required for effective preparedness and response to disasters.

2.3.2 STATE LEVEL

State Telecom Disaster Co-ordination Committee (STDCC) :

At State or telecom circle level, all Disaster related activities will be coordinated and monitored by **State Telecom Disaster Co-ordination Committee (STDCC).** The Committee will be chaired by the respective DDG (TERM) and will have members from :

(i) All the Telecom Service Providers.
(ii) Representative from concerned disaster management unit of State Government .

The Telecom Service Providers shall comply with the instructions of STDCC. The head of STDCC will also co-ordinate with State Government, its concerned organizations and other agencies for support functions including transportation and allocation of petrol/diesel on priority to the TSPs.

Further, the STDCC, in consultation with District Collector, will earmark petrol pumps in the districts which will supply petrol/diesel on priority to TSPs.

2.3.3 DISTRICT LEVEL

In the telecom sector, the organizational set up of most of the Telecom Service Providers is at State / Telecom circle level. It is therefore expected that in the eventuality of a disaster, STDCC shall oversee the sectoral requirement for the rescue, relief and restoration by State / District. STDCC shall also promptly establish a camp office at the respective District HQ., if needed, for effective co-ordination and support with District administration in disaster management activities. Such Camp office of STDCC shall be headed by the concerned DDG TERM or his nominated officer, as the case may be.

All the above Co-ordination committees shall meet at least once in 6 months to review the plans and activities related to disaster management.

MITIGATION and PREPAREDNESS

3.1 Telecommunication Systems/Networks

Robustness and preventive measures of the telecommunications systems are keys for their effective utilization during disaster. Telecom Service providers (TSPs) are required to implement the following measures in order to increase the robustness and prevent failures of their networks during disasters and emergencies.

3.1.1 **Physical Infrastructure Safety**

TSPs should follow norms and standards in building their physical infrastructure. In addition, the following practices help in the safety of the primary network elements and infrastructure:

- Telecommunication equipments should be installed at suitable locations in disaster prone areas to be able to withstand impacts of any disaster. e.g. in flood prone areas location of exchanges/ critical equipment to be preferably at higher altitude area to avoid inundation of water. The plinth should be kept high in coastal and flood prone areas.

- Wherever feasible, critical equipment should not be concentrated in one building.

- All buildings, towers and equipment sites should be equipped with adequate fire protection measures like detection and extinguishing systems etc.

- All buildings, towers and equipment site structure should comply with building bylaws prescribed for earthquake resistant building depending upon seismic zones.

- As far as possible, communication cables should be buried underground in ducts to reduce their vulnerability (it is also advisable to have all disaster management centers connected through underground cables).

3.1.2 **Redundancy**

It is well established that redundancy in traffic management is essential to the integrity and robustness of the telecom networks. Sufficient redundancy prevents total network failure due to a single point of failure.

TSPs should ensure that transmission links between main Network Elements and switching equipment are redundant through two distinct geographical paths. Some of the key aspects of a robust and resilient telecom network are:

- Alternative telecommunications links (such as SDH ring on optical fibre) between primary switches
- Connection of main switches and Network Elements through mesh and ring transmission networks
- Redundant microwave, aerial or underground links and other network elements such as switches etc. should be secured in alternative locations.
- In hilly and remote area satellite connectivity should be preferred.

3.1.3 **Vulnerable and critical network components**

- According to hazard profile of the area, TSPs will identify vulnerability of their respective telecom infrastructure and accordingly prepare plan for emergency situations. All the vulnerable critical network components should have sufficient redundancy including transmission links and power backups in terms of battery storage capacity and diesel / fuel availability.

- Low power consumption equipment should be preferred at all vulnerable / critical locations.

3. 1.4 **Backup of network elements**

Provision of sufficient backups of network elements, gensets/batteries and fuel can prevent total failures from minor equipment damage. TSPs should ensure sufficient fuel, power and essential equipment backups. Some of the key actions recommended are:

- Provide an uninterrupted power supply (UPS) along with sufficient External Battery support to ensure that the power supply is not interrupted to the key equipments in the event of a main power supply failure.

- Ensure supply of fuel for back-up generators.

- Ensure availability of spares on site during emergency.

- Ensure enough spares in air conditioning equipment to serve the peak hours load.

- Store backup spares and fuel in an accessible and secured area.

- Use alternate means of power like solar panel etc., wherever possible.

3.1.5 **Overload Prevention Measures**

Emergency situation often triggers overload of the network due to high traffic, anxiety calls and repeated call attempts. TSPs should ensure provision of an effective solution to prevent the crash of the network in such cases and develop effective congestion management processes which should be reviewed and tested periodically.

Public is also required to be made aware to use alternate mode of communications such as SMS or internet media whenever congestion in the voice calling in mobile network is experienced.

3.2 Preparedness for Handling Disasters

All TSPs shall prepare their Disaster Management Plan and submit the same to DoT HQ Nodal officer i.e. DDG (DS).

3.2.1 **Nodal Officers:**

TSPs shall identify Main and alternate Nodal officers at central level and at every telecom circle level and publicize their full contact details for coordination related to disaster management prominently on their website. Same shall also be informed to DoT HQ Nodal officer and respective TERM cell head.

3.2.2 **Disaster Response Task Force (DRTF)**

TSPs shall have a Disaster Response Task Force (DRTF) at State level. DRTF teams will be responsible for immediate provisioning of emergency communication and restoration of telecom services in disaster affected areas.

DRTF shall consist of teams of experts in the following areas:

- Transmission Team
- Switching Team
- Infrastructure Team
- Mobile Service Team
- Telecom relief Team (for opening PCOs, helpline response etc.)

All teams, shall have self contained minimum infrastructure (e.g. Satellite terminal, system related spares & stores, tools & testers, portable gen-set, vehicles, arrangements for logistics support like food, water, tents, beds blankets etc). Additional resource requirements shall be augmented by TSPs as and when required.

3.2.3 Rapid Damage Assessment Team (RDAT)

In addition to DRTF, in case the disaster affected area is large or remote area, wherever required, TSPs shall also have the Rapid Damage Assessment Team (RDAT). RDAT shall work to determine the precise nature and extent of damage so that the planning for restoration of telecommunication services can be done in the efficient and effective manner. The initial focus of the RDAT will be to identify:

- Operational telecommunications assets available for use within the affected area;
- Damaged communication facilities.
- Telecommunications assets not within the affected area that may be brought physically or employed electronically to support the affected area.

Preliminary assessment shall be carried out immediately within 24 hours for planning the response.

RDAT shall consist of members from various telecom fields such as mobile communication, switching, transmission, civil, electrical experts etc. This team shall also be well equipped with tools for quick assessment. RDAT must be a self contained team having arrangements for logistic support like water, food [for at least 2 days], blankets, bed, satellite phone, mobile phone etc.

3.2.4 Resource center

- Locations of the designated resource centers are to be identified at central and regional / Sate level (to be decided) for keeping the inventory taking into account the location of pre identified disaster prone areas. The inventory should include following:

    - Satellite terminal (e.g. DSPT etc.)
    - Satellite based Emergency Communication Terminals
    - Portable/ air-transportable BTSs and satellite equipment for connecting to BSC/MSC
    - OFC cable with restoration kit
    - Tool kit
    - Portable gen-sets/batteries etc.

  At the resource center locations, concerned responsible officers are required to be identified along with participation of nodal officer from the TSPs.

- In order to restore the mobile services disrupted due to damaged BTSs, TSPs should keep, as inventory, certain minimum number of portable BTSs and satellite equipment for connecting to BSC/MSC in case of disaster. Suggested number could be around 10 per TSP so that collectively sufficient numbers of portable BTSs with VSAT connectivity are available.

3.2.5 Memoranda of Understanding amongst TSPs

For restoration of Telecom services in emergencies and disaster conditions, TSPs may enter into Memoranda of Understanding (MoU) among themselves for sharing specialized resources and Intra-circle roaming for provisioning of services.

Priority user groups are to be identified, who might be involved in rescue, relief and restoration activities. These might include: different government agencies, police, Fire, Medical, civil defense, Red Cross, Army, financial institutions, NGOs, all officers and staffs engaged in restoration of telecommunication services, etc. Priority is to be given to these groups for provisioning of additional communication facilities and restoration of telecom services, if required, for quick and efficient rescue and relief operations.

3.2.5 Training and Drill

The quick & efficient response to disaster depends on availability of trained staff and inventory in immediate deployable condition. Periodic training ensures a continuous awareness of the additional demands which each individual might be confronted with in case of disaster. Also, it is necessary that the equipment meant for restoration should be kept in working condition. TSPs are required to conduct periodic mock-drills within their network and in coordination with other support agencies.

3.2.6 Directory

A directory of Nodal officers responsible for telecommunication services during disaster at various levels shall be prepared for National and State levels giving their names, addresses, telephone numbers, mobile numbers, email address, Fax numbers. Such directory will be widely circulated and updated annually. The current directory is at **Annexure –III.**

3.2.7 The preparedness measures taken by TSPs will be reviewed six-monthly.

CONTROL ROOM (CR)

4.1. Emergency Control rooms will be set up at National and State levels with requisite facilities. The Control Rooms already in existence at these levels will be suitably upgraded.

4.2 Objectives of the Control Room

4.2.2 The Control Rooms at National and State levels will be the nerve centres for coordination and management of disasters. The objectives of the control rooms shall be to provide centralized direction and control of any or all of the following functions:

- Receive and process disaster alerts and warnings from nodal agencies and other sources and communicate the same to all designated authorities.
- Monitor emergency operations
- Facilitate Coordination among Ministries/Departments/Agencies.
- Requisitioning additional resources during the disaster phase
- Issuing disaster/incident specific information and instructions to all concerned;
- Consolidation, analysis, and dissemination of damage, loss and assessment data;
- Forwarding of consolidated reports to all designated authorities.

4.3 Location of Control Room

The control room will be set up at a suitable location and the building should be disaster resistant so as to withstand the impact of disasters and remain functional during the emergency phase.

TRIGGER MECHANISM

5.1 This Trigger Mechanism prescribes the manner in which the disaster response system shall be automatically activated after receiving early warning signals of a disaster happening or likely to happen or on receipt of information of an incident. Activities envisaged in this SOP under the response phase shall be initiated simultaneously without loss of time to minimize the loss and damage and mitigate the impact of disaster.

5.2 The objective of having a trigger mechanism for natural disasters is to have a suo-motto activation mechanism for spontaneous response to set in motion command, control and management of the situation.

There shall be two types of situation with different trigger mechanisms for natural disasters:

(i) Situation I – Where Early Warning signals are available
(ii) Situation II- Where Disaster occurs without early warning

5.2.1 Where Early Warning signals are available

i. At the National Level Nodal Agencies have been designated by MHA for generating/forecasting of events of natural disasters. Onset of disaster shall be indicated through forecasting by the Nodal Agencies.

ii. NTDCC, STDCC shall be fully activated and activities envisaged in this SOP under the response phase shall be initiated simultaneously without loss of time to minimize the loss and damage due to impact of likely disaster.

iii. National and State Control Centers shall be fully activated.

iv. TSPs shall inform their customers via SMS / Cell Broadcast or recorded voice messages as per instructions of Government in the affected areas.

v. TSPs shall keep all the required inventory and personnel in readiness.

vi. TSPs shall take all pre-emptive measures (based on the nature of warning) as mentioned in this SOP or in any specific instructions/ orders issued by Government and report the same to SDTCC. SDTCC shall compile the details of such measures and report to NDTCC.

5.2.2 Where Disaster occurs without early warning

In disaster situations where no early warning signals are available, the primary objective of the trigger mechanism shall be to mount immediate rescue and relief operations and set the process in as quickly as possible as per this SOP.

The following procedure shall be followed in such situations:

1. State Control Room shall be fully activated for managing the incident.
2. National Control Room shall be informed. First Information Report shall be submitted to National Control Room.
3. DRTF and RDAT teams shall be deployed by TSPs.

RESPONSE and RECOVERY

6.1 NATIONAL LEVEL

Following shall be the sequence of actions at the national level:

i. Convening of NTDCC meeting to discuss and take stock of the situation arising due to disaster.
ii. NTDCC shall also discuss the response of various agencies.
iii. It may also depute a team of Officials to visit the affected areas for on the spot assessment of the situation and coordinate with STDCC, if required.
iv. It shall monitor and review the situation on a regular basis.
v. Central level Nodal officers of TSPs shall report to Convener, NTDCC, DoT, HQ, Delhi.

List of Satellite Phones

6.2 STATE LEVEL

Following shall be the sequence of actions at the State level:

i. Where Disaster strikes with/ without early warning signals, TSPs shall immediately assess damage to their network and deploy Rapid Damage Assessment Team & Disaster Response Task Force Teams (DRTF) with required inventory to provide emergency communication to priority callers like police, Fire, Medical, civil defense, Red Cross, Army, financial institutions, NGOs, all officers and staffs engaged in restoration of telecommunication services, etc.

ii. If required portable / vehicle mounted / air-transportable BTSs / BSCs with backhaul on satellite media may be installed by TSPs.

iii. Nodal officer of TSPs of affected telecom circle level shall report to concerned DDG (TERM), DoT (Chairman of STDCC) in that circle, for sharing information and coordination related matters.

iv. TERM units of DOT shall be the single nodal point in the disaster region where representatives of TSPs shall also be present to coordinate and oversee communication restoration efforts.

v. A control room will be setup at the state HQ / nearest to affected area, as the case may be, and made operational under control of TERM cell of affected area.

vi. The DDG of concerned TERM cell shall act as interface between all TSPs and other support agencies including State Government for any coordination related issues.

vii. Meeting of STDCC shall be convened to review situations. STDCC shall report to NTDCC.

viii. TSPs shall make helpline numbers operational where the last location of the survivors or missing persons can be intimated to the relatives.

ix. Providing information through helpline numbers about the last location details or sharing of CDRs of the subscribers, in the disaster affected areas, with the state/ central agencies requires special regulatory permission. However, in case of occurrence of disaster, these permissions would be deemed automatically permitted for duration of 2 weeks.

x. TSPs shall share specialized resources and allow Intra-circle roaming as per MoU for provisioning of services to priority user groups and general public during disaster period.

List of Satellite Phones

xi. In order to have wider coverage from a single BTS, TSPs may radiate more radio power from the BTSs located in the disaster affected areas beyond the EMR limits prescribed for 15 days.

xii. TSPs shall broadcast messages at regular intervals, in consultation with STDCC/NTDCC, to all the subscribers in the affected areas through SMSs / Cell broadcast giving details about:

(i) Details of TSPs helpline numbers.
(ii) Details about rescue and relief activities of state government such as tentative schedule of food / water distribution / nearest shelter/ shelter camp etc. as per need of State agencies.

xiii. TSPs shall open sufficient number of PCOs, preferably free of cost, for use of general public in affected area.

xiv. TSPs shall also ensure that in the disaster affected areas, no subscriber shall be denied access to voice / SMS communication due to any commercial consideration, whatsoever, including non-payment / insufficient balance/recharge etc. This facility shall continue for atleast 15 days.

xv. TSPs shall submit First Information Report as per **Annexure-II** and thereafter Daily Status Report about their network, help line and PCOs etc. as per **Annexure-III** to concerned TERM cell on daily basis or as sought by DDG(TERM), Head of STDCC.

xvi. TERM Cell shall submit the status report in respect of overall telecom facilities in affected area to DoT, HQ on daily basis or as sought by DoT, HQ.

xvii. The Control room under respective TERM cell shall remain operational 24X7 till the telecom services are restored to near normalcy or as per instructions from DoT, HQ.

****

ABBREVIATIONS

AS	- Access Services
BSNL	- Bharat Sanchar Nigam Limited
BW	- Building Works
CS	- Carrier Services
DOT	- Department of Telecommunications
DRTF	- Disaster Response Task Force
DS	- Data Services

List of Satellite Phones

ESF	- Emergency Support Function
GOI	- Government of India
GSM	- Global System for Mobile communications
ICT	- Information and Communication Technology
MHA	- Ministry of Home Affairs
MTNL	- Mahanagar Telephone Nigam Limited
NDMA	- National Disaster Management Authority
NEOC	- National Emergency Operation Centre
NPDM	- National Policy on Disaster Management
NTDCC	- National Telecom Disaster Co-ordination Committee
NTP	- National Telecom Policy
PMRTS	- Public Mobile Radio trunk Service
PSU	- Public Sector Undertaking
RDAT	- Rapid Damage Assessment Team
SDH	- Synchronous Digital Hierarchy
SDMA	- State Disaster Management Authority

SMS	- Short Message Service
SOP	- Standard Operating Procedure
STDCC	- State Telecom Disaster Co-ordination Committee
TERM	- Telecom Enforcement, Resource and Monitoring
TSP	- Telecom Service Provider
UPS	- Uninterrupted Power Supply
WLL	- Wireless in Local Loop

146-146

Annexure-I

Directory of Nodal officers responsible for telecommunication services during disaster

Sl.No.	Name & Designation	Organization / Company	Contact details

Annexure-II

Format for First Information Report on occurrence of Disaster

(To be sent to DOT , Government of India within maximum of 24 hours of occurrence of disaster)

From: State ---------- Date of Report ---------- Name of TSP ----------

To

The DDG(TERM), …….Concerned State (fax : ____ ;email :_____)
Director(DM), DOT (fax : _____ ; email :_____)
Convener, NTDCC, DOT (fax: _____; email:_____)

a. Nature of Disaster:_____
b. Date and time of occurrence:_____
c. Affected area (number and names of affected districts):_____
d. Initial Risk Assessment:

Sl. No.	Item	Present consequences	Future consequences	Worst case Scenario
1	Number of Priority users affected			
2	Number of subscribers affected			
3	Number of Exchanges affected			
4	Number of Towers(BTSs) affected			
5	Number of OFC/MW/Satellite links affected			
6	Any major switches affected			
7	Services affected			

e. Actions taken for restoration, alternate communication backups and opening of help lines in brief

f. Immediate resource assistance required from State / Central agencies:

g. Any other critical information:_____

h. Next information update time/date:_____

Name : Contact :

Signature Time Report Logged :

TSPs Name:

Annexure-III

Daily Status Report

State/Circle : ---------- Date : ---------- Name of TSP :----------
To,
---------- (fax ---------- Email _____)

- Mobile Services:

Sl. No.	Total No. of BTSs	Total No. of BTSs affected	No. of BTSs Restored	No. of BTSs yet to be restored	Remarks

- Landline Services :

Sl. No.	Total no. of Exchanges	Total No. of Exchanges affected	No. of Exchanges restored	No. of Exchanges yet to be restored	Remarks

- Transmission Links :

Sl. No.	Media	Total no. of links affected	No. of links restored	No. of links yet to be restored	Remarks
1	OFC				
2	Microwave				
3	Satellite				

- Help Lines & Location Services :

No. of queries on help line till date	No. of queries answered	No. of queries related to location	No of queries answered related to location	No. of free public booths opened if any

- Any Assistance required from local administration :

Signature
Name :----------
Contact----------
Time report logged:----------
Name of TSP----------

STANDARD OPERATING PROCEDURE

(SOP-2017)

for

TELECOMMUNICATION SERVICES

for Responding to

DISASTERS

GOVERNMENT OF INDIA

MINISTRY OF COMMUNICATIONS

DEPARTMENT OF TELECOMMUNICATIONS

TABLE OF CONTENTS

Sl. No.	Content	Page No.
1	INTRODUCTION	3-5
2	INSTITUTIONAL STRUCTURE	6-7
3	MITIGATION and PREPAREDNESS	8-12
4	CONTROL ROOM	13
5	TRIGGER MECHANISM	14-15
6	RESPONSE and RECOVERY	16-18
7	ABBREVIATIONS	19-20
8	Annexure – I : Directory of Nodal Officers	21-24
9	Annexure –II : Directory of Nodal Officers of various TSPs	25-27
10	Annexure-III : Format for First Information Report	28
11	Annexure – IV : Daily Situation Report	29

CHAPTER 1

INTRODUCTION

The disasters may cause enormous loss of life and property. Even though the global average of number of disasters has been rising rapidly, improved warnings & mitigation programme and quick response have reduced significantly, the loss of human life. In the aftermath of a disaster, the telecommunication services are required on top priority and play a pivotal role in undertaking rescue, relief and rehabilitation measures. The role of Department of Telecommunications (DoT) is to act as a Lead/Primary agency for Emergency Support Function related to provision of telecommunication services by Telecom Service Providers and coordination with other Government agencies.

1.2. Disaster Management is broadly categorized into following 4 phases:

- Mitigation - includes any activity that prevents a disaster, reduces the chance of a disaster happening, or reduces the damaging effects of unavoidable disasters.
- Preparedness - includes plans or preparations made to save life or property, and to help the response and rescue service operations.
- Response - includes actions taken to save lives and prevent property damage, and to preserve the environment during emergencies or disasters. The response phase is the implementation of action plans.
- Recovery – includes actions that assist a community to return to a sense of normalcy after a disaster.

1.3. The New Telecom Policy - 2012 (NTP-2012) also emphasizes importance of Disaster Management and various provisions in this regard are as mentioned below:

- Creation of the robust and resilient telecom networks for adequately addressing the need for proactive support for mitigating disasters - natural and manmade (Para 15 of Preamble).
- To prescribe sectoral Standard Operating Procedures for aiding effective and early mitigation during disasters and emergencies (Para 5.12 of Strategies).
- To create appropriate regulatory framework for provision of reliable means of public communication by Telecom Service Providers during disasters (Para 5.13 of Strategies).
- To encourage use of ICTs in prediction, monitoring and early warning of disasters and dissemination of information (Para 5.14 of Strategies).

- To facilitate an institutional framework to establish nationwide Unified Emergency Response Mechanism by providing nationwide single access number for emergency services (Para 5.15. of Strategies).

1.4. These provisions made in preamble and strategies of NTP-2012 have far reaching implications for the nation, being prone to natural disasters like floods, earth-quakes, tsunami, coastal cyclones, urban flooding, landslides etc. Therefore, putting in place processes and structures by developing a holistic, proactive, multi disaster oriented and technology driven strategy through a culture of prevention, mitigation and preparedness in the country will definitely help in better response and recovery.

Therefore, this document deals with Standard Operating Procedures (SOP) to be followed by various stake holders providing Telecommunication services in the country, for effective mitigation and recovery during disaster and emergency.

1.5. Mission

"To minimize the impacts of disasters on life, property, environment, and the economy through prompt provisioning and restoration of Telecommunication Services in the disaster affected areas."

1.6. Objectives

- Co-ordination of national actions to ensure the provision of telecommunication support to the centre, state and district administration.
- Co-ordination of the requirement of temporary telecommunication services in the affected areas.
- Co-ordination for restoration of telecommunication services.

1.7. "Disaster Management Guidelines for Telecommunications" were issued by the Telecom Engineering Centre in Feb' 2008. These Guidelines, as amended from time to time, may also be referred for the technical details by various stake holders.

1.8. On occurrence of disasters, all stake holders make all out efforts to their own judgment for the restoration of damages caused to telecom network. However, in such times a coherent and coordinated approach among all stakeholders is the key to reap the full benefit of the efforts made for restoration of telecom networks in such emergent situations. It is, therefore, necessary to standardize the processes and actions that need to be taken.

1.9. Accordingly, in order to optimize the efforts of all stakeholders, “Standard Operating Procedure” (SOP) is required to be adopted covering all aspects of disaster management from mitigation to recovery to minimize loss due to disaster and to ensure efficient response.

1.10. For effective provisioning of emergency telecommunication services and restoration of normal telecommunication services, DoT will need support of following Ministries/ Departments/ Organizations:

(i) Telecom Service Providers / Telecom Infrastructure Providers
(ii) Department of Electronics and Information Technology
(iii) Directorate of Communication and Police Wireless
(iv) Ministry of Railways
(v) National Disaster Response Force
(vi) Ministry of Civil Aviation
(vii) Ministry of Defence
(viii) Ministry of Power
(ix) Ministry of Road Transport & Highways
(x) Ministry of Petroleum & Natural Gas
(xi) Power Grid Corporation of India Ltd.
(xii) Department of Space
(xiii) Ministry of Information & Broadcasting
(xiv) State and District Administration / Agencies
(xv) Indian Meteorological Department
(xvi) Ministry of Water Resources
(xvii) Ministry of Earth Sciences (INCOIS)

CHAPTER 2

INSTITUTIONAL STRUCTURE

2.1 In the aftermath of a disaster, the primary responsibility for organizing response and relief as well as recovery in disaster affected areas lies with the state government. However, sectoral institutional structures are needed to strengthen and supplement the efforts of rescue, relief and restoration.

2.2 The Disaster Management Act, 2005, lays down a three tier institutional structure for disaster management at the national, state and district levels in the form of NDMA, SDMA and DDMA. National Policy on Disaster Management (NPDM) has further specified the roles and responsibilities of various organizations for disaster response. Accordingly, to fulfil roles and responsibility of Department of Telecommunications, institutional structure has been set up in the following paragraphs.

2.3 Organizational Set-up for Telecommunications Sector

The Department of Telecommunications will render Emergency Support Functions (ESF) for telecommunication services, wherever Central intervention and support are needed by the State Governments. An effective organizational set-up for telecom services during disasters and emergencies at all levels viz. Central, State and District is very important for implementing and monitoring the disaster management plans.

2.3.1 NATIONAL LEVEL

National Telecom Disaster Co-ordination Committee (NTDCC):

At National or central level, all Disaster related activities will be coordinated and monitored by **National Telecom Disaster Co-ordination Committee (NTDCC)**. The committee will be chaired by the Member (T), DOT and will have the following constitution:

- Member (Technology) - Chairman
- Sr. DDG (TERM), DOT HQ
- Sr. DDG (AS), DOT HQ
- DDG (CS), DOT HQ
- DDG (DS), DOT HQ – Member Convener
- Members from Telecom Service Providers (Public & Private)
- Representative from MHA/NDMA

It will issue guidelines from time to time as required for effective preparedness and response to disasters.

2.3.2 STATE LEVEL

State Telecom Disaster Co-ordination Committee (STDCC):

At State or telecom circle level, all Disaster related activities will be coordinated and monitored by **State Telecom Disaster Co-ordination Committee (STDCC).** The Committee will be chaired by the respective DDG (TERM) and will have members from:

(i) All the Telecom Service Providers.
(ii) Representative from concerned disaster management unit of State Government.
(iii) Representative of concerned NDRF Battalion.

The Telecom Service Providers shall comply with the instructions of STDCC. The head of STDCC will also co-ordinate with State Government and State District Management Authority (SDMA), its concerned organizations and other agencies for support functions including transportation and allocation of petrol/diesel on priority to the TSPs.

Further, the STDCC, in consultation with District Collector, will earmark petrol pumps in the districts which will supply petrol/diesel on priority to TSPs.

2.3.3 DISTRICT LEVEL

In the telecom sector, the organizational set up of most of the Telecom Service Providers is at State / Telecom circle level. It is therefore expected that in the eventuality of a disaster, STDCC shall oversee the sectoral requirement for the rescue, relief and restoration by State / District. STDCC shall also promptly establish a camp office at the respective District HQ., if needed, for effective co-ordination and support with District administration and District Disaster Management Authority (DDMA) in disaster management activities. Such Camp office of STDCC shall be headed by the concerned DDG TERM or his nominated officer, as the case may be.

All the above Co-ordination committees shall meet at least once in 6 months to review the plans and activities related to disaster management.

CHAPTER 3

MITIGATION AND PREPAREDNESS

3.1 Telecommunication Systems/Networks

Robustness and preventive measures of the telecommunications systems are keys for their effective utilization during disaster. Telecom Service providers (TSPs) are required to implement the following measures in order to increase the robustness and prevent failures of their networks during disasters and emergencies.

3.1.1 Physical Infrastructure Safety

TSPs should follow norms and standards in building their physical infrastructure. In addition, the following practices help in the safety of the primary network elements and infrastructure:

- Telecommunication equipment should be installed at suitable locations in disaster prone areas to be able to withstand impacts of any disaster. e. g. in flood prone areas location of exchanges/ critical equipment to be preferably at higher altitude area to avoid inundation of water. The plinth should be kept high in coastal and flood prone areas.

- Wherever feasible, critical equipment should not be concentrated in one building.

- All buildings, towers and equipment sites should be equipped with adequate fire protection measures like detection and extinguishing systems etc.

- All buildings, towers and equipment site structure should comply with building bylaws prescribed for earthquake resistant building depending upon seismic zones.

- As far as possible, communication cables should be buried underground in ducts to reduce their vulnerability (it is also advisable to have all disaster management centers connected through underground cables).

3.1.2 Redundancy

It is well established that redundancy in traffic management is essential to the integrity and robustness of the telecom networks. Sufficient redundancy prevents total network failure due to a single point of failure.

TSPs should ensure that transmission links between main Network Elements and switching equipment are redundant through two distinct geographical paths. Some of the key aspects of a robust and resilient telecom network are:

- Alternative telecommunications links (such as SDH ring on optical fibre) between primary switches
- Connection of main switches and Network Elements through mesh and ring transmission networks
- Redundant microwave, aerial or underground links and other network elements such as switches etc. should be secured in alternative locations.
- In hilly and remote area satellite connectivity should be preferred.

3.1.3 Vulnerable and critical network components

- According to hazard profile of the area, TSPs will identify vulnerability of their respective telecom infrastructure and accordingly prepare plan for emergency situations. All the vulnerable critical network components should have sufficient redundancy including transmission links and power backups in terms of battery storage capacity and diesel / fuel availability.

- Low power consumption equipment should be preferred at all vulnerable / critical locations.

3.1.4 Backup of network elements

Provision of sufficient backups of network elements, gen-sets/batteries and fuel can prevent total failures from minor equipment damage. TSPs should ensure sufficient fuel, power and essential equipment backups. Some of the key actions recommended are:

- Provide an uninterrupted power supply (UPS) along with sufficient External Battery support to ensure that the power supply is not interrupted to the key equipment in the event of a main power supply failure.

- Ensure supply of fuel for back-up generators.

- Ensure availability of spares on site during emergency.

- Ensure enough spares in air conditioning equipment to serve the peak hour's load.

- Store backup spares and fuel in an accessible and secured area.

- Use alternate means of power like solar panel etc., wherever possible.

3.1.5 Overload Prevention Measures

Emergency situation often triggers overload of the network due to high traffic, anxiety calls and repeated call attempts. TSPs should ensure provision of an effective solution to prevent the crash of the network in such cases and develop effective congestion management processes which should be reviewed and tested periodically.

Public is also required to be made aware to use alternate mode of communications such as SMS or internet media whenever congestion in the voice calling in mobile network is experienced.

3.2 Preparedness for Handling Disasters

All TSPs shall prepare their Disaster Management Plan and submit the same to DoT HQ Nodal officer i.e. DDG (DS).

3.2.1 Nodal Officers:

TSPs shall identify Main and alternate Nodal officers at central level and at every telecom circle level and publicize their full contact details for coordination related to disaster management prominently on their website. Same shall also be informed to DoT HQ Nodal officer and respective TERM cell head.

3.2.2 Disaster Response Task Force (DRTF)

TSPs shall have a Disaster Response Task Force (DRTF) at State level. DRTF teams will be responsible for immediate provisioning of emergency communication and restoration of telecom services in disaster affected areas.

DRTF shall consist of teams of experts in the following areas:

- Transmission Team
- Switching Team
- Infrastructure Team
- Mobile Service Team
- Telecom relief Team (for opening PCOs, helpline response etc.)

All teams, shall have self-contained minimum infrastructure (e.g. Satellite terminal, system related spares & stores, tools & testers, portable gen-set, vehicles, arrangements for logistics support like food, water, tents, beds blankets etc). Additional resource requirements shall be augmented by TSPs as and when required.

3.2.3 Rapid Damage Assessment Team (RDAT)

In addition to DRTF, in case the disaster affected area is large or remote area, wherever required, TSPs shall also have the Rapid Damage Assessment Team (RDAT). RDAT shall work to determine the precise nature and extent of damage so that the planning for restoration of telecommunication services can be done in the efficient and effective manner. The initial focus of the RDAT will be to identify:

- Operational telecommunications assets available for use within the affected area;
- Damaged communication facilities.
- Telecommunications assets not within the affected area that may be brought physically or employed electronically to support the affected area.

Preliminary assessment shall be carried out immediately within 24 hours for planning the response.

RDAT shall consist of members from various telecom fields such as mobile communication, switching, transmission, civil, electrical experts etc. This team shall also be well equipped with tools for quick assessment. RDAT must be a self-contained team having arrangements for logistic support like water, food [for at least 2 days], blankets, bed, satellite phone, mobile phone etc.

If the disaster is not very serious, in that case RDAT may not be required.

3.2.4 Resource Centre

Locations of the designated resource centres are to be identified at central and regional / Sate level (to be decided) for keeping the inventory taking into account the location of pre identified disaster prone areas. The inventory should include following:

- Satellite terminal (e.g. DSPT etc.)
- Satellite based Emergency Communication Terminals
- Portable/ air-transportable BTSs and satellite equipment for connecting to BSC/MSC
- OFC cable with restoration kit
- Tool kit
- Portable gen-sets/spare batteries etc.

At the resource centre locations, concerned responsible officers are required to be identified along with participation of nodal officer from the TSPs.

- In order to restore the mobile services disrupted due to damaged BTSs, TSPs should keep, as inventory, certain minimum number of portable BTSs and satellite equipment for connecting to BSC/MSC in case of disaster. Suggested number could be around 10 per TSP so that collectively sufficient numbers of portable BTSs with VSAT connectivity are available.

3.2.5 Memoranda of Understanding amongst TSPs

For restoration of Telecom services in emergencies and disaster conditions, TSPs may enter into Memoranda of Understanding (MoU) among themselves for sharing specialized resources and Intra-circle roaming for provisioning of services.

Priority user groups are to be identified, who might be involved in rescue, relief and restoration activities. These might include: different government agencies, police, Fire, Medical, civil defence, Red Cross, Army, financial institutions, NGOs, all officers and staffs engaged in restoration of telecommunication services, etc. Priority is to be given to these groups for provisioning of additional communication facilities and restoration of telecom services, if required, for quick and efficient rescue and relief operations.

3.2.6 Training and Drill

The quick & efficient response to disaster depends on availability of trained staff and inventory in immediate deployable condition. Periodic training ensures a continuous awareness of the additional demands which each individual might be confronted with in case of disaster. Also, it is necessary that the equipment meant for restoration should be kept in working condition. TSPs are required to conduct periodic mock-drills on a quarterly basis within their network and in coordination with other support agencies for testing the effectiveness of all the preparedness machinery including manpower and equipment. Further, TSPs shall also participate in the mock exercises conducted by NDMA/ State Govt./ District Administration as and when called for.

3.2.7 Directory

A directory of Nodal officers responsible for telecommunication services during disaster at various levels shall be prepared for National and State levels giving their names, addresses, telephone numbers, mobile numbers, email address, Fax numbers. Such directory will be widely circulated and updated annually. The current directory is at **Annexure –I&II.**

3.2.8 The preparedness measures taken by TSPs will be reviewed six-monthly.

CHAPTER 4

CONTROL ROOM (CR)

4.1 Emergency Control rooms will be set up at National and State levels with requisite facilities. The Control Rooms already in existence at these levels will be suitably upgraded.

4.2 Objectives of the Control Room

The Control Rooms at National and State levels will be the nerve centres for coordination and management of disasters. The objectives of the control rooms shall be to provide centralized direction and control of any or all of the following functions:

- Receive and process disaster alerts and warnings from nodal agencies and other sources and communicate the same to all designated authorities.
- Monitor emergency operations
- Facilitate Coordination among Ministries/Departments/Agencies.
- Requisitioning additional resources during the disaster phase
- Issuing disaster/incident specific information and instructions to all concerned;
- Consolidation, analysis, and dissemination of damage, loss and assessment data;
- Forwarding of consolidated reports to all designated authorities.

4.3 Location of Control Room

The control room will be set up at a suitable location and the building should be disaster resistant so as to withstand the impact of disasters and remain functional during the emergency phase.

CHAPTER 5

TRIGGER MECHANISM

5.1 This Trigger Mechanism prescribes the manner in which the disaster response system shall be activated after receiving early warning signals of a disaster happening or likely to happen or on receipt of information of an incident. Activities envisaged in this SOP under the response phase shall be initiated simultaneously without loss of time to minimize the loss and damage and mitigate the impact of disaster.

5.2 The objective of having a trigger mechanism for natural disasters is to have a suo-motto activation mechanism for spontaneous response to set in motion command, control and management of the situation.

There shall be two types of situation with different trigger mechanisms for natural disasters:

(i) Situation I – Where Early Warning signals are available
(ii) Situation II- Where Disaster occurs without early warning

5.2.1 Where Early Warning signals are available

i. At the National Level Nodal Agencies have been designated by MHA for generating/forecasting of events of natural disasters. Onset of disaster shall be indicated through forecasting by the Nodal Agencies.

ii. NTDCC, STDCC shall be fully activated and activities envisaged in this SOP under the response phase shall be initiated simultaneously without loss of time to minimize the loss and damage due to impact of likely disaster.

iii. National and State Control Centers shall be fully activated.

iv. TSPs shall inform their customers via SMS / Cell Broadcast or recorded voice messages as per instructions of Government in the affected areas.

v. TSPs shall keep all the required inventory and personnel in readiness.

vi. TSPs shall take all pre-emptive measures (based on the nature of warning) as mentioned in this SOP or in any specific instructions/ orders issued by Government and report the same to SDTCC. SDTCC shall compile the details of such measures and report to NDTCC.

vii. Advance briefing and deployment of teams may be ensured in case of early warning.

viii. TSPs should take necessary permissions from local authorities, in advance, for transportation of spares.

5.2.2 Where Disaster occurs without early warning

In disaster situations where no early warning signals are available, the primary objective of the trigger mechanism shall be to initiate immediate rescue and relief operations and set the process in as quickly as possible as per this SOP.

The following procedure shall be followed in such situations:

1. State Control Room shall be fully activated for managing the incident.
2. National Control Room shall be informed. First Information Report shall be submitted to National Control Room.
3. DRTF and RDAT teams shall be deployed by TSPs.

CHAPTER 6

RESPONSE and RECOVERY

6.1 NATIONAL LEVEL

Following shall be the sequence of actions at the national level:

i. Convening of NTDCC meeting to discuss and take stock of the situation arising due to disaster.
ii. NTDCC shall also discuss the response of various agencies.
iii. It may also depute a team of Officials to visit the affected areas for on the spot assessment of the situation and coordinate with STDCC, if required.
iv. It shall monitor and review the situation on a regular basis.
v. Central level Nodal officers of TSPs shall report to Convener, NTDCC, DoT, HQ, Delhi.

6.2 STATE LEVEL

Following shall be the sequence of actions at the State level:

i. Where Disaster strikes with/ without early warning signals, TSPs shall immediately assess damage to their network and deploy Rapid Damage Assessment Team & Disaster Response Task Force Teams (DRTF) with required inventory to provide emergency communication to priority callers like State/District Emergency Operation Centers (EOCs), Police, NDRF, Fire, Medical, Civil Defense, Red Cross, Army, Financial Institutions, NGOs, all officers and staffs engaged in restoration of telecommunication services, etc.

ii. If required portable / vehicle mounted / air-transportable BTSs / BSCs with backhaul on satellite media may be installed by TSPs.

iii. Nodal officer of TSPs of affected telecom circle level shall report to concerned DDG (TERM), DoT (Chairman of STDCC) in that circle, for sharing information and coordination related matters.

iv. TERM units of DOT shall be the single nodal point in the disaster region where representatives of TSPs shall also be present to coordinate and oversee communication restoration efforts.

v. A control room will be setup at the state HQ / nearest to affected area, as the case may be, and made operational under control of TERM cell of affected area.

vi. The DDG of concerned TERM cell shall act as interface between all TSPs and other support agencies including State Government for any coordination related issues.

vii. Meeting of STDCC shall be convened to review situations. STDCC shall report to NTDCC.

viii. TSPs shall make helpline numbers operational where the last location of the survivors or missing persons can be intimated to the relatives.

ix. Providing information through helpline numbers about the last location details or sharing of CDRs of the subscribers, in the disaster affected areas, with the state/ central agencies requires special regulatory permission. However, in case of occurrence of disaster, these permissions would be deemed automatically permitted for duration of 2 weeks.

x. TSPs shall share specialized resources and allow Intra-circle roaming as per MoU for provisioning of services to priority user groups and general public during disaster period.

xi. In order to have wider coverage from a single BTS, TSPs may radiate more radio power from the BTSs located in the disaster affected areas beyond the EMR limits prescribed for 15 days.

xii. TSPs shall broadcast messages at regular intervals, in consultation with STDCC/NTDCC, to all the subscribers in the affected areas through SMSs / Cell broadcast giving details about:

(i) Details of TSPs helpline numbers.
(ii) Details about rescue and relief activities of state government such as tentative schedule of food / water distribution / nearest shelter/ shelter camp etc. as per need of State agencies.

xiii. TSPs shall open sufficient number of PCOs, preferably free of cost, for use of general public in affected area.

xiv. TSPs shall also ensure that in the disaster affected areas, no subscriber shall be denied access to voice / SMS communication due to any commercial consideration, whatsoever, including non-payment / insufficient balance/recharge etc. This facility shall continue for at least 15 days.

xv. TSPs shall submit First Information Report as per **Annexure-III** and thereafter Daily Status Report about their network, help line and PCOs etc. as per **Annexure-IV** to concerned TERM cell on daily basis or as sought by DDG(TERM), Head of STDCC.

xvi. TERM Cell shall submit the status report in respect of overall telecom facilities in affected area to DoT, HQ on daily basis or as sought by DoT, HQ.

xvii. The Control room under respective TERM cell shall remain operational 24X7 till the telecom services are restored to near normalcy or as per instructions from DoT, HQ.

xviii. For the circle wise list of TSPs, website of DoT may be referred.

****

ABBREVIATIONS

AS	- Access Services
BSNL	- Bharat Sanchar Nigam Limited
BW	- Building Works
CS	- Carrier Services
DOT	- Department of Telecommunications
DRTF	- Disaster Response Task Force
DS	- Data Services
ESF	- Emergency Support Function
GOI	- Government of India
GSM	- Global System for Mobile communications
ICT	- Information and Communication Technology
MHA	- Ministry of Home Affairs
MTNL	- Mahanagar Telephone Nigam Limited
NDMA	- National Disaster Management Authority
NEOC	- National Emergency Operation Centre
NPDM	- National Policy on Disaster Management
NTDCC	- National Telecom Disaster Co-ordination Committee
NTP	- National Telecom Policy
PMRTS	- Public Mobile Radio trunk Service
PSU	- Public Sector Undertaking
RDAT	- Rapid Damage Assessment Team
SDH	- Synchronous Digital Hierarchy
SDMA	- State Disaster Management Authority

SMS	- Short Message Service
SOP	- Standard Operating Procedure
STDCC	- State Telecom Disaster Co-ordination Committee
TERM	- Telecom Enforcement, Resource and Monitoring
TSP	- Telecom Service Provider
UPS	- Uninterrupted Power Supply
WLL	- Wireless in Local Loop

Annexure-I

Directory of Nodal officers responsible for telecommunication services during disaster

A. Contact Details of National Emergency Telecommunication Committee

Sl. No.	Particulars	Organization / Company	Contact details
1.	Secretary (T)	DoT	23719898
2.	Member (T)	DoT	23372307
3.	Member (S)	DoT	23714644
4.	Member (F)	DoT	23716161
5.	Add. Secretary(T)	DoT	23717300
6.	Advisor (O)	DoT	23710850
7.	Advisor (T)	DoT	23718460
8.	Advisor (WPC)	DoT	23755420

B. Contact Details of Nodal officers of DoT and TERM cells

Sl. No.	Name & Designation	Organization / Company	Contact details
1.	D. Manna, DDG (DM)	DoT	23714232 (O), 23036836 (O), 23737887 (F), 9868393715 (M)
2.	Sr. DDG (TERM)	DoT	23372702
3.	A.K. Tewari, DDG (TERM)	TERM Cell, Ahmedabad, Gujarat	079-25501550 (O), 25501050 (F), 9426001199 (M)
4.	N. Hari Babu, DDG (TERM)	TERM Cell, Andaman & Nicobar (West Bengal)	03192-242727 (O), 243636 (F), 9444978118 (M)
5.	B. Suresh Krishna, DDG (TERM)	TERM Cell, Andhra Pradesh	0866-2487699 (O), 2466099 (F), 8985009999 (M)
6.	V.K. Mathur, DDG(TERM)	TERM Cell, Assam	0361-2734222 (O), 2730788 (F), 9422301900
7.	S.R.N. Prasad, DDG(TERM)	TERM Cell, Bangalore	080-26643020 (O), 22373636 (F), 9448010234

			(M)
8.	S.S. Singh, DDG (TERM)	TERM Cell, Bihar	0612-2506220 (O), 2506362 (F), 9430588736 (M)
9.	P. Jagdeesan, DDG (TERM)	TERM Cell, Chennai	044-26451000 (O), 26472255 (F), 9444219933 (M)
10.	R.K. Garhwal, DDG (TERM)	TERM Cell, Chhattisgarh	0771-2226300 (O), 2226500 (F), 9438050505 (M)
11.	N.K. Chhoker, DDG (TERM)	TERM Cell, Delhi	011-26469191 (O), 26469393 (F), 9868844455 (M)
12.	V.K. Ahuja, DDG (TERM)	TERM Cell, Haryana	0171-2602222 (O), 2602111 (F), 9478040404 (M)
13.	Sarvendra Singh, DDG (TERM)	TERM Cell, Himachal Pradesh	0177-2629700 (O), 2622599 (F), 9418419777 (M)
14.		TERM Cell, Gujarat	079-25510011 (O), 25510022 (F)
15.	G. Gaurishankar, DDG (TERM)	TERM Cell, Hyderabad	040-27543377 (O), 27543322 (F), 9490956666 (M)
16.	B.K. Gupta, DDG (TERM)	TERM Cell, Jaipur (Rajasthan)	0141-2712631 (O), 2712624 (F), 9412000556 (M)
17.	Labh Singh, DDG (TERM)	TERM Cell, Jammu & Kashmir	0191-2470111 (O), 2471998 (F), 9469797979 (M)
18.	Dilip K. Kar, DDG (TERM)	TERM Cell, Jharkhand	0651-2273444 (O), 2273900 (F), 9433000082 (M)
19.	Mohd. Aslam Khan, DDG (TERM)	TERM Cell, Karnataka	080-26631717 (O), 26638888 (F), 9448010266 (M)
20.	C Sunitha, DDG (TERM)	TERM Cell, Kerala	0484-2205900 (O), 2205903 (F), 9446016500 (M)
21.	Hasan Zaheer, DDG (TERM)	TERM Cell, Kolkata	033-23572001 (O), 23572003 (F), 8902000571 (M)
22.	Mahidhar Pant, DDG (TERM)	TERM Cell, Lucknow	0522-2727244 (O), 2727222 (F), 8004151617 (M)

23.	Vinod Gupta, DDG (TERM)	TERM Cell, Madhya Pradesh	0755-2550196 (O), 2550196 (F), 9425201947 (M)
24.	K. K. Chanda, DDG (TERM)	TERM Cell, Maharashtra	0712-2293999 (O), 2287999 (F), 9438050505 (M)
25.	M.M. Gupta, DDG (TERM)	TERM Cell, Mumbai	022-28574300 (O), 28573535 (F), 9423993636 (M)
26.	Prasanna K. Sikdar, DDG (TERM)	TERM Cell, North East-I	0364-2220418 (O), 2220415 (F). 9436784720 (M)
27.	Sidhharth Pokhrana, DDG (TERM)	TERM Cell, North East-II	03862-237171 (O), 237191 (F), 9435201212 (M)
28.	M.B. Ravichandran, DDG (TERM)	TERM Cell, Odisha	0674-2547500 (O), 2547889 (F), 9495801100 (M)
29.	S.N. Ramagopal, DDG (TERM)	TERM Cell, Pune	020-26121616 (O), 26134646 (F), 9423500400 (M)
30.	R.P. Singhal, DDG (TERM)	TERM Cell, Punjab	0172-2218540 (O), 2217700 (F), 9478844044 (M)
31.	R.D. Arya, DDG (TERM)	TERM Cell, Rajasthan	0141-2712621, 2712624 (F), 9414001910 (M)
32.	N. Asokraj, DDG (TERM)	TERM Cell, Tamilnadu	0422-2436400 (O), 2436800 (F), 9443200990 (M)
33.	V.P. Singh, DDG (TERM)	TERM Cell, Uttar Pradesh (East)	0522-2727400 (O), 2727600 (F), 9415001415 (M)
34.	A.K. Srivastava, DDG (TERM)	TERM Cell, Uttar Pradesh (West)	0121-2402200 (O), 2402256 (F), 9416005111 (M)
35.	Manoj Pant, DDG (TERM)	TERM Cell, Uttarakhand	0135-2734577 (O), 2734677 (F), 9417700600 (M)
36.	Supriya Dutta, DDG (TERM)	TERM Cell, West Bengal	033-24606575(O), 24606576 (F), 9433000045 (M)

C. Contact Details of Control Room:

Sl. No.	Particulars	Organization / Company	Contact details
1.	MHA Control Room	MHA	23093054, 23093571, 23093564, 23438252, 23092398 (F), 23093750 (F), 23093897 (F)
2.	DoT Control Room	DoT	23036836, 23737887 (F)

Annexure-II

Directory of Nodal officers of TSPs responsible for telecommunication services during disaster

Sl. No.	Name & Designation	Organization / Company	Contact details
1.	Mr. Anil Verma, Sr. G.M. (Regulations)	Bharat Sanchar Nigam Limited	Bharat Sanchar Bhawan, Harish Chandra Mathur Lane, Janpath, New Delhi-01 Ph: 011-23734097 averma@bsnl.co.in
2.	Vandana Gupta GM(RA), CO	Mahanagar Telephone Nigam Limited	MTNL Building, C.G.O Complex, New Delhi Mobile: 9868137555 gmraco@bol.net.in
3.	Sh. Ashwani Rana, Vice President, Head Regulatory Affairs (Mobile Services)	Bharti Airtel Limited	Airtel Center, Plot No.16, Udyog Vihar, Phase-IV, Gurgaon – 122015 Ph: 0124-4222222 ashwani.rana@in.airtel.com
4.	Dr. Vinod Kumar Budhiraja, Chief Regulatory Officer	Etlisalat DB Telecom Pvt. Ltd.	5th Floor, Infinity Towers, Mind Space, Link Road Malad (West),Mumbai – 400064 Ph: 022-67858888 Fax: 02267858759
5.	Sh. Rahul Vatts, Vice President – Regulatory & Corporate Affairs	Idea Cellular Limited	7th Floor, Konnectus, Bhavbhuti Marg, New Delhi – 110 001. Mob. 9891005951 e-mail: rahul.vatts@idea.adityabirla.com

6.	Sh. Harish Kapoor, Chief Regulatory Officer	Loop Mobile (India) Limited	708-709, Prakashdeep Building, Tolstoy Marg, New Delhi- 110 001 Ph: 011-43613988 E-mail: hkapoor@loopmobile.in
7.	Sh. D.K.Gupta, Vice President (Legal & Regulatory)	S Tel Pvt. Ltd.	1st Floor, Tower-B, Unitech Cyberpark, Sector-39 Gurgaon (Haryana)-122001 Ph. 0124-4740000/Fax: 0124-4740077
8.	Sh. Sanjeev Arora, DGM (Regulatory)	Vodafone Essar Ltd	7th Floor, DLF Centre, Sansad Marg, New Delhi- 110 001 Mobile: 9811918508 Fax: 23753659
9.	Sh. Ashok Sharma, National Head-Regulatory	Aircel Limited	2nd & 5th Floor, DLF Cyber City Building No. 10-A, Gurgaon-122002 Mob. 09716000008 ashok.sharma@aircel.co.in
10.	Sh. Sunil Sharma, Assistant General Manager	Videocon Telecommunications Ltd.	Plot No. 248, Udyog Vihar, Phase - IV, Gurgaon - 122015 Mob. 09999079444 E:mail: sunil.sharma@videocon.com
11.	Sh. Arun Kumar Singh, Addl. Vice President-Corporate Regulatory Affairs	Tata Teleservices Ltd	Indicom Building 2-A, Old Ishwar Nagar, Main Mathura Road,New Delhi-110065. Phone No. 011-66558923 arunkumar.singh@tatatel.co.in

12.	Sh. Mahipal Singh, AVP	Reliance Jio Infocom Limited	3rd Floor, Makers Chamber-IV, 222, Nariman Point, Mumbai-400021. M: 09717899166 E-mail: mahipal.singh@ril.com

Annexure-III

Format for First Information Report on occurrence of Disaster

(To be sent to DoT, Government of India within maximum of 24 hours of occurrence of disaster)

From: State ---------- Date of Report ---------- Name of TSP ----------

To

The DDG(TERM), ……..Concerned State (fax : ____ ;email :_____)
Director(DM), DOT (fax : _____ ; email :_____)
Convener, NTDCC, DOT (fax: _____; email:_____)

a. Nature of Disaster:_____
b. Date and time of occurrence:_____
c. Affected area (number and names of affected districts):_____
d. Initial Risk Assessment:

Sl. No.	Item	Present consequences	Future consequences	Worst case Scenario
1	Number of Priority users affected			
2	Number of subscribers affected			
3	Number of Exchanges affected			
4	Number of Towers(BTSs) affected			
5	Number of OFC/MW/Satellite links affected			
6	Any major switches affected			
7	Services affected			

e. Actions taken for restoration, alternate communication backups and opening of help lines in brief

f. Immediate resource assistance required from State / Central agencies:

g. Any other critical information:_____

h. Next information update time/date:_____

Name: Contact:

Signature Time Report Logged:

TSPs Name:

Annexure-IV

Daily Status Report

State/Circle : ---------- Date : ---------- Name of TSP :----------
To

---------- (fax ---------- Email _____)

- Mobile Services:

Sl. No.	Total No. of BTSs	Total No. of BTSs affected	No. of BTSs Restored	No. of BTSs yet to be restored	Remarks

- Landline Services :

Sl. No.	Total no. of Exchanges	Total No. of Exchanges affected	No. of Exchanges restored	No. of Exchanges yet to be restored	Remarks

- Transmission Links :

Sl. No.	Media	Total no. of links affected	No. of links restored	No. of links yet to be restored	Remarks
1	OFC				
2	Microwave				
3	Satellite				

- Help Lines & Location Services :

No. of queries on help line till date	No. of queries answered	No. of queries related to location	No of queries answered related to location	No. of free public booths opened if any

- Any Assistance required from local administration :

Signature
Name: ……
Contact: ……
Time report logged: ……
Name of TSP: ……

Government of India
Ministry of Communications
Department of Telecommunications
Access Services Wing
20, Ashoka Road, New Delhi-110001

No. 16-1/2017-AS-III/Short Codes/362/ **Dated: 28/07/2020**

To,

All Access Service Providers.

Subject: Allotment of short code '1070' as Standard Disaster Helpline to all State/UTs--reg.

This is in continuation to DoT letter No. 16-03/2011-AS-III//Vol.II/106 dated 27.03.2012 allocating short code '1070' as Disaster Management Helpline to Relief Commissioner of Natural calamities (copy enclosed). In this regard, the undersigned is directed by competent authority to state that short Code '1070' is allocated as Standard Disaster Helpline to all State/UTs with following features:

i. Category-I: Service (i.e. mandatory service to be provided by all the Access providers).
ii. Accessibility: Un-restricted i.e. available with STD code also.
iii. Metered Service: Non-metered

2. All are requested to take necessary action.
3. This is issued with the approval of competent authority.

SKgarg 28.07.2020

(Sanchit Kr. Garg)
Assistant Director General (AS-III)

Copy forwarded for information and necessary action, if any, to:

1. Secretary, TRAI.
2. CMD MTNL/ CMD BSNL.
3. Shri G.V.V. Sarma, IAS, Member Secretary, NDMA, MHA w.r.t your DO No. 29/01/2018/Comn & IT- Part-2 (FTS 111530) dated 17.07.2020 to appoint the nodal officer to implement the helpline. The Nodal officer may contact Sr. DDG (LSA), LSA Unit, concerned States/UTs.
4. Director (P&N), DoT (HQ), (respective LSA field unit may be asked to co-ordinate with respective Nodal Officers and TSPs for implementation of Helpline, as and when approached).
5. A copy for upload on DoT Website.

251314/2017/Office of DDG(AS-I) \\

261C

Government of India
Ministry of Communications & I.T.
Department of Telecommunications
Licensing Cell (AS-III Section)
Sanchar Bhawan, 20, Ashoka Road, New Delhi-1

No.16-3/2011-AS.III/Vol. II/106 Dated:27-03-2012

To

All Access Service Providers

Sub: Allotment of Short Codes'100', '101', '102', '1070', '1071', 1072', '1073', & '1077'.

As per National Numbering Plan, following help lines numbers are working for the emergency services mentioned against each:

100	Police
101	Fire
102	Ambulance
1070	Relief Commissioner of Natural Calamities
1071	Air Accident Information
1072	Train Accident
1073	Road Accident
1077	Control Room of District collector

It has been observed that Operators are implementing these services non-uniformly. Being emergency services, it is hereby clarified that above mentioned emergency help lines will be Non-Metered within the service area. Where the boundaries of the service area are not co-terminus with the boundaries of State/UT concerned, there the service will be non-metered within the entire State/UT. It implies that the service is to be provided as Non-Metered even if these State Departments/Organizations desire to collect information at a Central location outside the SDCA of origination of call.

27/3/2012

(Sudhir Kumar Sinha)
Section Officer(AS-III)

Copy to :

1.Secretary, TRAI

2.Sr.DDG, TEC, Khurshid Lal Bhawan, Janpath, New Delhi

3. All LSA TERM Cells for monitoring the compliance.

4.Dir(IT), DOT for uploading on DOT website

STANDARD OPERATING PROCEDURE

(SOP- 2020)

for

TELECOMMUNICATION SERVICES

for Responding to

DISASTERS

GOVERNMENT OF INDIA

MINISTRY OF COMMUNICATIONS

DEPARTMENT OF TELECOMMUNICATIONS

TABLE OF CONTENTS

Sl. No.	Content	Page No.
1	INTRODUCTION	1
2	INSTITUTIONAL STRUCTURE	4
3	MITIGATION and PREPAREDNESS	6
4	CONTROL ROOM	11
5	TRIGGER MECHANISM	12
6	RESPONSE and RECOVERY	14
7	ABBREVIATIONS	17
8	Annexure – I : Directory of Nodal Officers	19
9	Annexure –II : Directory of Nodal Officers of various TSPs	22
10	Annexure-III : Format for First Information Report	23
11	Annexure – IV : Daily Situation Report	24

Document History

Sr. no.	Document	Issued Date
1.	SOP-2015 for responding to Disasters	05.06.2015
2.	SOP-2017 for responding to disasters (Revision of SOP-2015)	17.03.2017
3.	SOP-2020 for responding to disasters (Revision of SOP-2017)	14.12.2020

CHAPTER 1

INTRODUCTION

The disasters may cause enormous loss of life and property. Even though the global average of number of disasters has been rising rapidly, improved warnings & mitigation programme and quick response have reduced significantly, the loss of human life. In the aftermath of a disaster, the telecommunication services are required on top priority and play a pivotal role in undertaking rescue, relief and rehabilitation measures. The role of Department of Telecommunications (DoT) is to act as a Lead/Primary agency for Emergency Support Function related to provision of telecommunication services by Telecom Service Providers and coordination with other Government agencies.

1.2. Disaster Management is broadly categorized into following 4 phases:

- Mitigation - includes any activity that prevents a disaster, reduces the chance of a disaster happening, or reduces the damaging effects of unavoidable disasters.
- Preparedness - includes plans or preparations made to save life or property, and to help the response and rescue service operations.
- Response - includes actions taken to save lives and prevent property damage, and to preserve the environment during emergencies or disasters. The response phase is the implementation of action plans.
- Recovery – includes actions that assist a community to return to a sense of normalcy after a disaster.

1.3. National Digital communication policy- 2018 has a clear objective to develop comprehensive plan for network preparedness, disaster response relief, restoration and reconstruction. National Digital Communications policy 2018 has following objectives with respect to disaster management mentioned in clause 3.4: *Developing a comprehensive plan for network preparedness, disaster response relief, restoration and reconstruction*

a) *Strengthening network resilience by:*
   i. *Framing and enforcing standard operating procedures to be followed during disasters and natural calamities, including sectoral guidelines for disaster response and recovery applicable to various service providers*
   ii. *Establishing institutional framework to promote monitoring of activities, rapid dissemination of early warning disaster notifications and better coordination and collaboration between relevant Ministries / Departments, including the National Disaster Management Authority of India*

b) *Developing a Unified Emergency Response Mechanism, by:*

i. *Creating an institutional framework with clearly defined roles and responsibilities, Standard Operating Procedures and technical guidelines*
ii. *Incorporating obligations under the license terms and conditions for implementation of Next Generation 112 services in all areas, based on geo-location technologies, and provide online access to caller location and details to authorized central and state agencies*
iii. *Enforcing obligations of service providers to share infrastructure, and ensure interoperability in emergency situations in a network-agnostic, operator-agnostic and technology-agnostic manner.*

c) *Enhancing the Public Protection and Disaster Relief (PPDR) plan for India by:*

i. *Facilitating the establishment of a Pan-India network for Public Protection and Disaster Relief (PPDR)*
ii. *Making necessary spectrum available for PPDR including by establishing INSAT satellite-based mobile communication systems*
iii. *Implementing global and regional harmonized spectrum Plans for PPDR.*

1.4. These provisions made in preamble and strategies of NDCP-2018 have far reaching implications for the nation, being prone to natural disasters like floods, earth-quakes, tsunami, coastal cyclones, urban flooding, landslides etc. Therefore, putting in place processes and structures by developing a holistic, proactive, multi disaster oriented and technology driven strategy through a culture of prevention, mitigation and preparedness in the country will definitely help in better response and recovery.

Therefore, this document deals with Standard Operating Procedures (SOP) to be followed by various stake holders providing Telecommunication services in the country, for effective mitigation and recovery during disaster and emergency.

1.5. Mission

"To minimize the impacts of disasters on life, property, environment, and the economy through prompt provisioning and restoration of Telecommunication Services in the disaster affected areas."

1.6. Objectives

- Co-ordination of national actions to ensure the provision of telecommunication support to the centre, state and district administration.
- Co-ordination of the requirement of temporary telecommunication services in the affected areas.

- Co-ordination for restoration of telecommunication services.

1.7. "Disaster Management Guidelines for Telecommunications" were issued by the Telecom Engineering Centre in Feb' 2008. These Guidelines, as amended from time to time, may also be referred for the technical details by various stake holders.

1.8. On occurrence of disasters, all stake holders make all out efforts to their own judgment for the restoration of damages caused to telecom network. However, in such times a coherent and coordinated approach among all stakeholders is the key to reap the full benefit of the efforts made for restoration of telecom networks in such emergent situations. It is, therefore, necessary to standardize the processes and actions that need to be taken.

1.9. Accordingly, in order to optimize the efforts of all stakeholders, "Standard Operating Procedure" (SOP) is required to be adopted covering all aspects of disaster management from mitigation to recovery to minimize loss due to disaster and to ensure efficient response.

1.10. For effective provisioning of emergency telecommunication services and restoration of normal telecommunication services, DoT will need support of following Ministries/ Departments/ Organizations:

(i) Telecom Service Providers / Telecom Infrastructure Providers
(ii) Department of Electronics and Information Technology
(iii) Directorate of Communication and Police Wireless
(iv) Ministry of Railways
(v) National Disaster Response Force
(vi) Ministry of Civil Aviation
(vii) Ministry of Defence
(viii) Ministry of Power
(ix) Ministry of Road Transport & Highways
(x) Ministry of Petroleum & Natural Gas
(xi) Power Grid Corporation of India Ltd.
(xii) Department of Space
(xiii) Ministry of Information & Broadcasting
(xiv) State and District Administration / Agencies
(xv) Indian Meteorological Department
(xvi) Ministry of Water Resources
(xvii) Ministry of Earth Sciences (INCOIS)

CHAPTER 2

INSTITUTIONAL STRUCTURE

2.1 In the aftermath of a disaster, the primary responsibility for organizing response and relief as well as recovery in disaster affected areas lies with the state government. However, sectoral institutional structures are needed to strengthen and supplement the efforts of rescue, relief and restoration.

2.2 The Disaster Management Act, 2005, lays down a three tier institutional structure for disaster management at the national, state and district levels in the form of NDMA, SDMA and DDMA. National Policy on Disaster Management (NPDM) has further specified the roles and responsibilities of various organizations for disaster response. Accordingly, to fulfil roles and responsibility of Department of Telecommunications, institutional structure has been set up in the following paragraphs.

2.3 Organizational Set-up for Telecommunications Sector

The Department of Telecommunications will render Emergency Support Functions (ESF) for telecommunication services, wherever Central intervention and support are needed by the State Governments. An effective organizational set-up for telecom services during disasters and emergencies at all levels viz. Central, State and District is very important for implementing and monitoring the disaster management plans.

2.3.1 NATIONAL LEVEL

National Telecom Disaster Co-ordination Committee (NTDCC):

At National or central level, all Disaster related activities will be coordinated and monitored by National Telecom Disaster Co-ordination Committee (NTDCC). The committee will be chaired by the Member (T), DOT and will have the following constitution:

- Member (Technology) - Chairman
- DG(T) / DDG (IT&S), DGT HQ
- DDG(AS), DOT HQ
- DDG(CS), DOT HQ
- Joint Wireless Advisor, WPC, DOT
- DDG (DM), DOT HQ – Member Convener
- Members from Telecom Service Providers (Public & Private)
- Representative from MHA/NDMA

It will issue guidelines from time to time as required for effective preparedness and response to disasters.

2.3.2 STATE LEVEL

State Telecom Disaster Co-ordination Committee (STDCC):

At State or telecom circle level, all Disaster related activities will be coordinated and monitored by State Telecom Disaster Co-ordination Committee (STDCC). The Committee will be chaired by the respective DDG (Technology) and will have members from:

(i) All the Telecom Service Providers.
(ii) Representative from concerned disaster management unit of State Government.
(iii) Representative of concerned NDRF Battalion.

The Telecom Service Providers shall comply with the instructions of STDCC. The head of STDCC will also co-ordinate with State Government and State District Management Authority (SDMA), its concerned organizations and other agencies for support functions including transportation and allocation of petrol/diesel on priority to the TSPs.

Further, the STDCC, in consultation with District Collector, will earmark petrol pumps in the districts which will supply petrol/diesel on priority to TSPs.

2.3.3 DISTRICT LEVEL

In the telecom sector, the organizational set up of most of the Telecom Service Providers is at State / Telecom circle level. It is therefore expected that in the eventuality of a disaster, STDCC shall oversee the sectoral requirement for the rescue, relief and restoration by State / District. STDCC shall also promptly establish a camp office at the respective District HQ., if needed, for effective co-ordination and support with District administration and District Disaster Management Authority (DDMA) in disaster management activities. Such Camp office of STDCC shall be headed by the concerned DDG (Technology) or his nominated officer, as the case may be.

All the above Co-ordination committees shall meet at least once in 6 months to review the plans and activities related to disaster management.

CHAPTER 3

MITIGATION AND PREPAREDNESS

3.1 Telecommunication Systems/Networks

Robustness and preventive measures of the telecommunications systems are keys for their effective utilization during disaster. Telecom Service providers (TSPs) are required to implement the following measures in order to increase the robustness and prevent failures of their networks during disasters and emergencies.

3.1.1 Physical Infrastructure Safety

TSPs should follow norms and standards in building their physical infrastructure. In addition, the following practices help in the safety of the primary network elements and infrastructure:

- Telecommunication equipment should be installed at suitable locations in disaster prone areas to be able to withstand impacts of any disaster. e. g. in flood prone areas location of exchanges/ critical equipment to be preferably at higher altitude area to avoid inundation of water. The plinth should be kept high in coastal and flood prone areas.

- Wherever feasible, critical equipment should not be concentrated in one building.

- All buildings, towers and equipment sites should be equipped with adequate fire protection measures like detection and extinguishing systems etc.

- All buildings, towers and equipment site structure should comply with building bylaws prescribed for earthquake resistant building depending upon seismic zones.

- As far as possible, communication cables should be buried underground in ducts to reduce their vulnerability (it is also advisable to have all disaster management centers connected through underground cables).

3.1.2 Redundancy

It is well established that redundancy in traffic management is essential to the integrity and robustness of the telecom networks. Sufficient redundancy prevents total network failure due to a single point of failure.

TSPs should ensure that transmission links between main Network Elements and switching equipment are redundant through two distinct geographical paths. Some of the key aspects of a robust and resilient telecom network are:

- Alternative telecommunications links (such as SDH ring on optical fibre) between primary switches
- Connection of main switches and Network Elements through mesh and ring transmission networks
- Redundant microwave, aerial or underground links and other network elements such as switches etc. should be secured in alternative locations.
- In hilly and remote area satellite connectivity should be preferred.

3.1.3 Vulnerable and critical network components

- According to hazard profile of the area, TSPs will identify vulnerability of their respective telecom infrastructure and accordingly prepare plan for emergency situations. All the vulnerable critical network components should have sufficient redundancy including transmission links and power backups in terms of battery storage capacity and diesel / fuel availability.

- Low power consumption equipment should be preferred at all vulnerable / critical locations.

3.1.4 Backup of network elements

Provision of sufficient backups of network elements, gen-sets/batteries and fuel can prevent total failures from minor equipment damage. TSPs should ensure sufficient fuel, power and essential equipment backups. Some of the key actions recommended are:

- Provide an uninterrupted power supply (UPS) along with sufficient External Battery support to ensure that the power supply is not interrupted to the key equipment in the event of a main power supply failure.
- Ensure supply of fuel for back-up generators.
- Ensure availability of spares on site during emergency.
- Ensure enough spares in air conditioning equipment to serve the peak hour's load.
- Store backup spares and fuel in an accessible and secured area.
- Use alternate means of power like solar panel etc., wherever possible.

3.1.5 Overload Prevention Measures

Emergency situation often triggers overload of the network due to high traffic, anxiety calls and repeated call attempts. TSPs should ensure provision of an effective solution to prevent the crash of the network in such cases and develop effective congestion management processes which should be reviewed and tested periodically.

Public is also required to be made aware to use alternate mode of communications such as SMS or internet media whenever congestion in the voice calling in mobile network is experienced.

3.2 Preparedness for Handling Disasters

All TSPs shall prepare their Disaster Management Plan and submit the same to DoT HQ Nodal officer i.e. DDG (DM).

3.2.1 Nodal Officers:

TSPs shall identify Main and alternate Nodal officers at central level and at every telecom circle level and publicize their full contact details for coordination related to disaster management prominently on their website. Same shall also be informed to DoT HQ Nodal officer and respective LSA head.

3.2.2 Disaster Response Task Force (DRTF)

TSPs shall have a Disaster Response Task Force (DRTF) at State level. DRTF teams will be responsible for immediate provisioning of emergency communication and restoration of telecom services in disaster affected areas.

DRTF shall consist of teams of experts in the following areas:

- Transmission Team
- Switching Team
- Infrastructure Team
- Mobile Service Team
- Telecom relief Team (for opening PCOs, helpline response etc.)

All teams, shall have self-contained minimum infrastructure (e.g. Satellite terminal, system related spares & stores, tools & testers, portable gen-set, vehicles, arrangements for logistics support like food, water, tents, beds blankets etc). Additional resource requirements shall be augmented by TSPs as and when required.

3.2.3 Rapid Damage Assessment Team (RDAT)

In addition to DRTF, in case the disaster affected area is large or remote area, wherever required, TSPs shall also have the Rapid Damage Assessment Team (RDAT). RDAT shall work to determine the precise nature and extent of damage so that the planning for restoration of telecommunication services can be done in the efficient and effective manner. The initial focus of the RDAT will be to identify:

- Operational telecommunications assets available for use within the affected area;
- Damaged communication facilities.
- Telecommunications assets not within the affected area that may be brought physically or employed electronically to support the affected area.

Preliminary assessment shall be carried out immediately within 24 hours for planning the response.

RDAT shall consist of members from various telecom fields such as mobile communication, switching, transmission, civil, electrical experts etc. This team

shall also be well equipped with tools for quick assessment. RDAT must be a self-contained team having arrangements for logistic support like water, food [for at least 2 days], blankets, bed, satellite phone, mobile phone etc.

3.2.4 Resource Centre

Locations of the designated resource centres are to be identified at central and regional / Sate level (to be decided) for keeping the inventory taking into account the location of pre identified disaster prone areas. The inventory should include following:

- Satellite terminal (e.g. DSPT etc.)
- Satellite based Emergency Communication Terminals
- Portable/ air-transportable BTSs and satellite equipment for connecting to BSC/MSC
- OFC cable with restoration kit
- Tool kit
- Portable gen-sets/spare batteries etc.

At the resource centre locations, concerned responsible officers are required to be identified along with participation of nodal officer from the TSPs.

- In order to restore the mobile services disrupted due to damaged BTSs, TSPs should keep, as inventory, certain minimum number of portable BTSs and satellite equipment for connecting to BSC/MSC in case of disaster. Suggested number could be around 10 per TSP so that collectively sufficient numbers of portable BTSs with VSAT connectivity are available.

3.2.5 Memoranda of Understanding amongst TSPs

For restoration of Telecom services in emergencies and disaster conditions, TSPs may enter into Memoranda of Understanding (MoU) among themselves for sharing specialized resources and Intra-circle roaming for provisioning of services.

Priority user groups are to be identified, who might be involved in rescue, relief and restoration activities. These might include: different government agencies, police, Fire, Medical, civil defence, Red Cross, Army, financial institutions, NGOs, all officers and staffs engaged in restoration of telecommunication services, etc. Priority is to be given to these groups for provisioning of additional communication facilities and restoration of telecom services, if required, for quick and efficient rescue and relief operations.

3.2.6 Training and Drill

The quick & efficient response to disaster depends on availability of trained staff and inventory in immediate deployable condition. Periodic training ensures a continuous awareness of the additional demands which each individual might be confronted with in case of disaster. Also, it is necessary that the equipment meant for restoration should be kept in working condition. TSPs are required to conduct periodic mock-drills on a quarterly basis within their network and in coordination with other support agencies for testing the effectiveness of all the preparedness machinery including manpower and equipment. Further, TSPs shall also participate in the mock exercises conducted by NDMA/ State Govt./ District Administration as and when called for.

3.2.7 Directory

A directory of Nodal officers responsible for telecommunication services during disaster at various levels shall be prepared for National and State levels giving their names, addresses, telephone numbers, mobile numbers, email address, Fax numbers. Such directory will be widely circulated and updated annually. The current directory is at **Annexure –I & II.**

3.2.8 The preparedness measures taken by TSPs will be reviewed six-monthly.

CHAPTER 4

CONTROL ROOM (CR)

4.1 Emergency Control rooms will be set up at National and State levels with requisite facilities. The Control Rooms already in existence at these levels will be suitably upgraded.

4.1.1 DoT has already issued instructions for the implementation of 'Single Number Based Integrated Emergency Communication and Response System' for improving communication network during emergency/disasters. The single emergency number will be '112'. The existing emergency numbers 100, 101, 102 and 108 will be treated as secondary numbers and will be connected to the PSAP (Public Safety Answering Point) under the control of respective State/UT and are to be re-routed to '112'. A four digit help line number '1938' is to be made operational at Control Room of MHA for coordination.

4.2 Objectives of the Control Room

The Control Rooms at National and State levels will be the nerve centres for coordination and management of disasters. The objectives of the control rooms shall be to provide centralized direction and control of any or all of the following functions:

- Receive and process disaster alerts and warnings from nodal agencies and other sources and communicate the same to all designated authorities.
- Monitor emergency operations
- Facilitate Coordination among Ministries/Departments/Agencies.
- Requisitioning additional resources during the disaster phase
- Issuing disaster/incident specific information and instructions to all concerned;
- Consolidation, analysis, and dissemination of damage, loss and assessment data;
- Forwarding of consolidated reports to all designated authorities.

4.3 Location of Control Room

The control room will be set up at a suitable location and the building should be disaster resistant so as to withstand the impact of disasters and remain functional during the emergency phase.

CHAPTER 5

TRIGGER MECHANISM

5.1 This Trigger Mechanism prescribes the manner in which the disaster response system shall be activated after receiving early warning signals of a disaster happening or likely to happen or on receipt of information of an incident. Activities envisaged in this SOP under the response phase shall be initiated simultaneously without loss of time to minimize the loss and damage and mitigate the impact of disaster.

5.2 The objective of having a trigger mechanism for natural disasters is to have a suo-motto activation mechanism for spontaneous response to set in motion command, control and management of the situation.

There shall be two types of situation with different trigger mechanisms for natural disasters:

(i) Situation I – Where Early Warning signals are available
(ii) Situation II- Where Disaster occurs without early warning

5.2.1 Where Early Warning signals are available:

i. At the National Level Nodal Agencies have been designated by MHA for generating/forecasting of events of natural disasters. Onset of disaster shall be indicated through forecasting by the Nodal Agencies.

ii. NTDCC, STDCC shall be fully activated and activities envisaged in this SOP under the response phase shall be initiated simultaneously without loss of time to minimize the loss and damage due to impact of likely disaster.

iii. National and State Control Centers shall be fully activated.

iv. TSPs shall inform their customers via SMS / Cell Broadcast or recorded voice messages as per instructions of Government in the affected areas.

v. TSPs shall keep all the required inventory and personnel in readiness.

vi. TSPs shall take all pre-emptive measures (based on the nature of warning) as mentioned in this SOP or in any specific instructions/ orders issued by Government and report the same to SDTCC. SDTCC shall compile the details of such measures and report to NDTCC.

vii. Advance briefing and deployment of teams may be ensured in case of early warning.

viii. TSPs should take necessary permissions from local authorities, in advance, for transportation of spares.

5.2.2 Where Disaster occurs without early warning:

In disaster situations where no early warning signals are available, the primary objective of the trigger mechanism shall be to initiate immediate rescue and relief operations and set the process in as quickly as possible as per this SOP.

The following procedure shall be followed in such situations:

i. State Control Room shall be fully activated for managing the incident.
ii. National Control Room shall be informed. First Information Report shall be submitted to National Control Room.
iii. DRTF and RDAT teams shall be deployed by TSPs.

CHAPTER 6

RESPONSE and RECOVERY

6.1 NATIONAL LEVEL

Following shall be the sequence of actions at the national level:

i. Convening of NTDCC meeting to discuss and take stock of the situation arising due to disaster.
ii. NTDCC shall also discuss the response of various agencies.
iii. It may also depute a team of Officials to visit the affected areas for on the spot assessment of the situation and coordinate with STDCC, if required.
iv. It shall monitor and review the situation on a regular basis.
v. Central level Nodal officers of TSPs shall report to Convener, NTDCC, DoT, HQ, Delhi.

6.2 STATE LEVEL

Following shall be the sequence of actions at the State level:

i. Where Disaster strikes with/ without early warning signals, TSPs shall immediately assess damage to their network and deploy Rapid Damage Assessment Team & Disaster Response Task Force Teams (DRTF) with required inventory to provide emergency communication to priority callers like State/District Emergency Operation Centers (EOCs), Police, NDRF, Fire, Medical, Civil Defence, Red Cross, Army, Financial Institutions, NGOs, all officers and staffs engaged in restoration of telecommunication services, etc.

ii. If required portable / vehicle mounted / air-transportable BTSs / BSCs with backhaul on satellite media may be installed by TSPs.

iii. Nodal officer of TSPs of affected telecom circle level shall report to concerned DDG (Technology), DoT (Chairman of STDCC) in that circle, for sharing information and coordination related matters.

iv. LSA of DOT shall be the single nodal point in the disaster region where representatives of TSPs shall also be present to coordinate and oversee communication restoration efforts.

v. A control room will be setup at the state HQ / nearest to affected area, as the case may be, and made operational under control of LSA of affected area.

vi. The DDG (Technology) of concerned LSA shall act as interface between all TSPs and other support agencies including State Government for any coordination related issues.

vii. Meeting of STDCC shall be convened to review situations. STDCC shall report to NTDCC.

viii. TSPs shall make helpline numbers operational where the last location of the survivors or missing persons can be intimated to the relatives.

ix. Providing information through helpline numbers about the last location details or sharing of CDRs of the subscribers, in the disaster affected areas, with the state/ central agencies requires special regulatory permission. However, in case of occurrence of disaster, these permissions would be deemed automatically permitted for duration of 2 weeks.

x. TSPs shall share specialized resources and allow Intra-circle roaming as per MoU for provisioning of services to priority user groups and general public during disaster period.

xi. In order to have wider coverage from a single BTS, TSPs may radiate more radio power from the BTSs located in the disaster affected areas beyond the EMR limits prescribed for 15 days.

xii. TSPs shall broadcast messages at regular intervals, in consultation with STDCC/NTDCC, to all the subscribers in the affected areas through SMSs / Cell broadcast free of cost during disaster period based upon instructions of Nodal authorities as per DM act 2005 i.e. National Executive Committee(NEC)/National Crisis Management Committee(NCMC)/State Executive Committee(SEC). This shall provide details about:

a) Details of TSPs helpline numbers.
b) Details about rescue and relief activities of state government such as tentative schedule of food / water distribution / nearest shelter/ shelter camp etc. as per need of State agencies.

xiii. TSPs shall open sufficient number of PCOs, preferably free of cost, for use of general public in affected area.

xiv. TSPs shall also ensure that in the disaster affected areas, no subscriber shall be denied access to voice / SMS communication due to any commercial consideration, whatsoever, including non-payment / insufficient balance/recharge etc. This facility shall continue for at least 15 days.

xv. TSPs shall submit First Information Report as per **Annexure-III** and thereafter Daily Status Report about their network, help line and PCOs etc. as per **Annexure-IV** to concerned LSA unit on daily basis or as sought by DDG (Technology), Head of STDCC.

xvi. LSA shall submit the status report in respect of overall telecom facilities in affected area to DoT, HQ on daily basis or as sought by DoT, HQ.

xvii. The Control room under respective LSA shall remain operational 24X7 till the telecom services are restored to near normalcy or as per instructions from DoT, HQ.

xviii. For the circle wise list of TSPs, website of DoT may be referred.

xix. Every LSA shall have a Satellite Phone for proper coordination with State Govt. during disasters.

xx. HAM operators are allowed to operate anywhere in India except those locations restricted by GOI from time to time. Officer in Charge (disaster Management) at State or the District Magistrate of disaster affected district, as the case may be, should prescribe minimum level of officer for signing a message, to be passed on HAM network during disaster.

****

ABBREVIATIONS

AS	- Access Services
BSNL	- Bharat Sanchar Nigam Limited
BW	- Building Works
CS	- Carrier Services
DOT	- Department of Telecommunications
DRTF	- Disaster Response Task Force
DM	- Disaster Management
ESF	- Emergency Support Function
GOI	- Government of India
GSM	- Global System for Mobile communications
ICT	- Information and Communication Technology
MHA	- Ministry of Home Affairs
MTNL	- Mahanagar Telephone Nigam Limited
NCMC	- National Crisis Management Committee
NDMA	- National Disaster Management Authority
NEOC	- National Emergency Operation Centre
NPDM	- National Policy on Disaster Management
NTDCC	- National Telecom Disaster Co-ordination Committee
NTP	- National Telecom Policy
PMRTS	- Public Mobile Radio Trunk Service
PSU	- Public Sector Undertaking
RDAT	- Rapid Damage Assessment Team
SDH	- Synchronous Digital Hierarchy
SDMA	- State Disaster Management Authority
SMS	- Short Message Service

SOP	- Standard Operating Procedure
STDCC	- State Telecom Disaster Co-ordination Committee
LSA	- License Service Area
TSP	- Telecom Service Provider
UPS	- Uninterrupted Power Supply
WLL	- Wireless in Local Loop

Annexure-I

Directory of Nodal officers responsible for telecommunication services during disaster

A. Contact Details of National Emergency Telecommunication Committee

Sl. No.	Particulars	Organization / Company	Contact details
1.	Secretary (T)	DoT	23719898
2.	Member (T)	DoT	23372307
3.	Member (S)	DoT	23714644
4.	Member (F)	DoT	23716161
5.	Add. Secretary(T)	DoT	23717300
6.	Advisor (O)	DoT	23714644
7.	Advisor (T)	DoT	23372307
8.	Advisor (WPC)	DoT	23755444

B. Contact Details of Nodal officers of DoT HQ and LSAs

Sl. No.	Name & Designation	Organization / Company	Contact details
1.	Shri Chandra Shekhar, DDG (DM)	DoT, HQ	011-23372702 (O), 8700878787 (M)
2.	Shri K. Raja Sekhar DDG (Technology)	Andhra Pradesh LSA	040-27722713 (O) 040- 27833322 (F) 9490000248 (M)
3.	Shri Sanjeev Peeka DDG (Technology)	Assam LSA	0361-2736999(O), 0361-2730788 (F), 9446455500 (M)
4.	Shri A.K. Sinha, DDG (Technology)	Bihar LSA	0612-2506139 (O), 0612-2506362 (F), 9470000500 (M)
5.	Y. K. Singh, DDG(Technology)	Delhi LSA	011-26469393 (O), 9868133266(M)
6.	Shri Roshan Lal Meena, DDG(Technology)	Gujrat LSA	9414001115 (M)

7.	Shri O.P.Manhas, DDG (Technology)	Haryana LSA	9868131123 (M)
8.	Shri Ranveer Singh, DDG (Technology)	Himachal Pradesh LSA	0177-2628500 (O), 0177-2622599 (F), 9414000124(M)
9.	Shri Radheshyam Parmar, DDG (Technology)	Jammu & Kashmir LSA	0191-2471117 (O), 9425001088 (M)
10.	Shri R K Dubey, DDG (Technology)	Karnataka LSA	080-26642626 (O), 080-26633010 (F), 9448010625 (M)
11.	Smt. V. Shobhna, DDG (Technology)	Kerala LSA	0484-2207255 (O), 0484-2203955 (F), 9910395165 (M)
12.	Shri R.K.Singh DDG(Technology)	Madhya Pradesh LSA	0755- 2555854 (O), 9868218150 (M)
13.	Shri V.K. Mathur DDG (Technology)	Maharashtra, LSA	020-26129988 (O), 020-26129966 (F), 9422301900 (M)
14.	Shri A K Jain, DDG (Technology)	Mumbai, LSA	022-28573530 (O), 022-28573535 (F), 9869033066 (M)
15.	Shri R R Mittar, DDG (Technology)	NE, Shillong LSA	9868137776 (M)
16.	Shri Basudev Mallick, DDG (Technology)	Odisha, LSA	0674-2544050(O) 9437066000 (M)
17.	Smt. Rekha Singh, DDG (Technology)	Punjab LSA	0172-2218094 (O), 9417210020 (M)
18.	Shri R G Yadav, DDG (Technology)	Rajasthan, LSA	0141-2712620(O), 9414001310 (M)
19.	Smt R Radha, DDG (Technology)	Tamilnadu, LSA	044-28251118 (O), 044-28251120 (F), 9443200606(M)
20.	Shri S. K. Mittal, DDG (Technology)	UP(E) LSA	0522-2725501 (O), 9868136621 (M)

21.	Shri Virendra Kumar, DDG(Technology)	UP(W) LSA	0121-2402266 (O), 0121-2402202(F), 9412200056 (M)
22.	Shri R K Sharma, DDG (Technology)	West Bengal LSA	033- 24402027 (O), 033-24402029 (F), 9667116790 (M)
23.	Shri R K Sharma, DDG (Technology)	Kolkata LSA	033- 24402027 (O), 033-24402029 (F), 9667116790 (M)

C. Contact Details of Control Room:

Sl. No.	Particulars	Organization / Company	Contact details
1.	MHA Control Room	MHA	011-23438252, 011-23438253, 011-23438254 (F)
2.	DoT Control Room	DOT	011-23372098 (Operational during disasters only)

Annexure-II

Directory of Nodal officers of TSPs responsible for telecommunication services during disaster

Sl. No.	Name & Designation	Organization / Company	Contact details
1	Ms. Madhu Arora, PGM	BSNL	9868133388
2	Shri Tarun Chitkara	Airtel	9871106558
3	Shri Sandeep Kathuria	Vodafone-Idea	9811918500
4	Shri Mahipal Singh	Reliance Jio	9717899166
5	Shri O P Misra	MTNL	9868132819

Annexure-III

Format for First Information Report on occurrence of Disaster

(To be sent to DoT, Government of India within maximum of 24 hours of occurrence of disaster)

From: State ---------- Date of Report ---------- Name of TSP ----------

To,

The DDG(Technology), ……. Concerned State (fax: ____ ; email :_____)
Director(DM), DOT (fax : _____ ; email :_____)
Convener, NTDCC, DOT (fax: _____; email: _____)

a. Nature of Disaster: _____
b. Date and time of occurrence: _____
c. Affected area (number and names of affected districts):_____
d. Initial Risk Assessment:

Sl. No.	Item	Present consequences	Future consequences	Worst case Scenario
1	Number of Priority users affected			
2	Number of subscribers affected			
3	Number of Exchanges affected			
4	Number of Towers(BTSs) affected			
5	Number of OFC/MW/Satellite links affected			
6	Any major switches affected			
7	Services affected			

e. Actions taken for restoration, alternate communication backups and opening of help lines in brief
f. Immediate resource assistance required from State / Central agencies:
g. Any other critical information: _____

h. Next information update time/date:_____

Name: Contact:
Signature Time Report Logged:

TSPs Name:

Annexure-IV

Daily Status Report

State/ Circle: ---------- Date : ---------- Name of TSP:----------

To,

---------- (fax ---------- Email _____)

- Mobile Services:

Sl. No.	Total no. of Towers	Total towers down	Total No. of BTSs	Total No. of BTSs non-functional	Percentage of BTS non-functional	No. of OFC link affected	CoW deployed

- Landline Services:

Sl. No.	Total no. of Exchanges	Total No. of Exchanges Down	Percentage of Exchanges Down

- Transmission Links:

Sl. No.	Media	Total no. of links affected	No. of links restored	No. of links yet to be restored	Remarks
1	OFC				
2	Microwave				
3	Satellite				

- Help Lines & Location Services:

No. of queries on help line till date	No. of queries answered	No. of queries related to location	No of queries answered related to location	No. of free public booths opened if any

- Any Assistance required from local administration:

Signature
Name: ……
Contact: ……
Time report logged: ……

Name

F.No. 47-1/2020-DGTHQ/4
Government of India
Ministry of Communications
Department of Telecommunications
(DGT-HQ)

Dated: 12.05.2022

OFFICE MEMORANDUM

Subject: Procurement of Satellite Phone in LSAs. – reg.

Ref: SOP-2020 for Telecommunications Services for Responding to Disasters issued on 14.12.2020

With reference to the SOP-2020 for Telecommunications Services for Responding to Disasters issued on 14.12.2020, various requests have been received from LSAs regarding purchase of Satellite phones centrally at Head quarter level.

2. In this regard,the matter has been examined at DGT Hq and accordingly administrative approval is hereby accorded to all LSAs to purchase 28 Satellite phones from BSNL in which 22 Satellite Phones may be procured for each LSA and additional 6 satellite phones may be procured for State Coordination offices at Manipur, Mizoram, Tripura, Nagaland, Arunachal Pradesh and Andaman & Nicobar Islands.

3. The LSAs may purchase the Satellite Phones as per the financial powers delegated to them.

4. This issues with the approval of DG (Telecom).

12.05.2022

(Khushboo Sharma)
ADG (New Technology)

Enclosure: The list of Satellite Phones to be purchased in each LSA

To:
All Head of LSAs, DoT

CamScanner

List of Satellite Phones to be purchased in each LSA

S.No.	Name of the LSA	Number of Satellite Phones proposed	Remarks
1	Andhra Pradesh	1	
2	Bihar	1	
3	Haryana	1	
4	Gujarat	1	
5	Kerala	1	
6	Himachal Pradesh	1	
7	Delhi	1	
8	Karnataka	1	
9	Assam	1	
10	Jammu & Kashmir	1	
11	Kolkata	1	
12	Madhya Pradesh	1	
13	Maharashtra	1	
14	Mumbai	1	
15	North-East	6	Meghalaya-1, Arunachal Pradesh-1, Mizoram-1, Manipur-1, Tripura-1, Nagaland-1
16	Odisha	1	
17	Punjab	1	
18	Rajasthan	1	
19	Tamil Nadu	1	
20	UP(East)	1	
21	UP(West)	1	
22	West Bengal	2	West Bengal-1, Andaman & Nicobar Islands-1
Total		**28**	

12.05.2022

CamScanner

F.No.15-2/2021-DM
Government of India
Ministry of Communications
(Disaster Management Unit)
606, Mahanagar Sanchar Bhawan, JLN Marg, New Delhi-110002

New Delhi, Dated: 26 December, 2022

Office Memorandum

Subject: Tariff for SMS and Cell Broadcast Alerts/Messages Disseminated through CAP Platform regd.

Vide letter no. 15-2/2017/DS-I/DM(Vol-II) dated 25th March, 2021, DoT requested TRAI to provide tariff for SMS and Cell Broadcast Alerts/Messages Disseminated through CAP SACHET Platform. In this regard, TRAI has notified Telecom Tariff (Sixty ninth Amendment) order in the Gazette on dated 6th December, 2022. The prescribed tariff for SMS and Cell Broadcast Alerts/ messages to be sent through CAP SACHET platform are as below:

Sr. no.	Item	Tariff	
		Per SMS (In Rs.)	Per Cell Broadcast (In Rs.)
a.	Alerts or messages sent during non-disaster situations, other than those sent as per directions issued under the Disaster Management Act, 2005 (53 of 2005)	0.02	Nil
b.	Alerts or messages sent prior to notification of disaster or after the expiry of disaster, as per directions issued under the Disaster Management Act, 2005 (53 of 2005)	Nil	Nil
c.	Alerts or messages sent during a disaster situation, as per directions issued under the Disaster Management Act, 2005 (53 of 2005)	Nil	Nil
d.	Alerts or messages sent during a disaster situation, other than those sent as per directions issued under the Disaster Management Act, 2005 (53 of 2005)	0.02	Nil

2. While forwarding the Telecom Tariff order (Sixty ninth Amendment), TRAI has desired to devise a mechanism for ensuring following provisions through SOP please:

i. The difference in tariff may give options to relevant agencies to opt for sending messages through SMS or CB or combination thereof depending upon the context.

ii. The usage of CAP platform may be administered keeping in view the primary objective of developing a comprehensive plan for network preparedness, disaster response relief, restoration and reconstruction.

iii. SOP may be suitably designed to facilitate primacy of disaster-related alerts/ messages (b & c above) to be disseminated from the CAP system. The SOP should ensure that alerts/ messages (as per b & c above) may be broadcasted using the CB system, and only in rare cases, provide for other users that too, backed by proper justification and subject to rigorous prior scrutiny may use CB system.

iv. Dissemination of messages may be strictly incentivized according to previously determined priority duly announced in the SOP.

v. No political/ commercial messages/ alerts to be disseminated through CAP platform.

vi. Feedback mechanism to be developed vide which LSAs (through dashboard) may detect and provisions for actions in cases of misuse or abuse of CAP Platform or Cell Broadcast.

vii. Custodians of CAP i.e. Alert Authorizing Agencies should ensure that traffic which is supposed to be routed through the ecosystem built under Telecom Commercial Communication Customer Preference Regulation (TCCCPR)-2018 may not find its way for delivery through CAP Platform.

3. I am hereby directed to attach Gazette Notification on Telecom Tariff order (Sixty ninth Amendment) dated 6th December, 2022 with a request to incorporate provisions mentioned in para 2 above in the SOP and for taking up the matter with States/ UTs please.

Encl: As above.

26/12/2022

(Shriphal Meena)
Asstt. Director General (DM-A)

To,

Member Secretary, NDMA,

New Delhi-110029

Copy to:

1) Secretary and Chairman, Digital Communications Commission, Department of Telecommunications, New Delhi – 110001.
2) Member (T), Digital Communication Commission, Department of Telecommunications, New Delhi – 110001.
3) Director General (Telecom), DGT Hq, New Delhi-110001.
4) Regulatory heads of all Telecom Service Providers.

सी-डॉट
C-DOT

सेंटर फॉर डेवलपमेंट ऑफ टेलीमैटिक्स
भारत सरकार का दूरसंचार प्रौद्योगिकी केन्द्र

Centre for Development of Telematics
Telecom Technology Centre of Govt. Of India

Letter No: C-DOT/DSA/CAP-SACHET/DOT-LSA/2023/1/1 Date 27th January, 2023

To,

All LSA Heads,
Department of Telecommunications (DOT).

Subject: Regarding Tariff for SMS and Cell Broadcast message disseminated through CAP-based Integrated Alert System

Dear Sir,

This is brought to your kind notice that CAP based Integrated Alert system is operational PAN-India with geo targeted SMS dissemination in 36 States / UTs.

With reference to earlier communication C-DOT/DSA/CAP-SACHET/DoT-HQ/2022/07/1 dated 18th July 2022 on "Training and Access of the DOT LSA units on CAP-based Integrated Alert System", the e-trainings were conducted by C-DOT for DOT LSA officials on 4th Aug 2022 and 26th Aug 2022 and live access has been provided to the LSA Officials. The DOT LSA units play a vital role in the monitoring the SMS based dissemination done through CAP based Integrated Alert System where LSA units can view and monitor the SMS dissemination and dissemination statistics submitted by the TSPs in DOT Dashboard.

The Office Memorandum (OM) dated 26th Dec 2022 issued by DOT HQ with respect to the charges illustrates the modalities and operationalisation of SMS based dissemination amongst various stakeholders. The above referred instruction in OM may be brought to the notice of all the TSPs under your jurisdiction for ensuring compliance (copy enclosed).

The URL for accessing the CAP Platform is https://platform.sachetdashboard.ndma.gov.in and kindly contact us for any queries on cap.support@cdot.in regarding access to the platform.

With Regards,

Saurabh 27.01.2023

Saurabh Basu
Group Leader (A), DSA
C-DOT, New Delhi.

CC: DDG (DM), DOT HQ
: DDG (IT), DOT HQ
: Advisor (OPS), NDMA

Enclosures: As above

सी-डॉट परिसर, महरौली, नई दिल्ली -110 030 भारत फोन : +91-11-2680 2856 फैक्स: +91-11-2680 3338
C-DOT Campus, Mehrauli, New Delhi-110 030 India Phone: +91-11-26802856 Fax: +91-11-2680 3338
वेबसाईट /Website : www.cdot.in

F. No. 4-1/2023-QoS
Government of India
Ministry of Communications
Department of Telecommunications
(QoS Division)
6th Floor, MTNL Building, JLN Marg, New Delhi-110002

Dated 10th July, 2024

To,

1. DGT HQ, DoT
2. DDG (NBM), O/o DGT HQ
3. DDG(AS), DoT HQ
4. DDG (Security), DoT HQ
5. DDG (DM), DoT HQ

Subject: Standard Operating Procedure (SOP) for maintaining Quality of Telecom Services during Significant Social/ Cultural/ Religious/ Commercial/ International Events etc.

Please find attached the Standard Operating Procedure (SoP) for maintaining Quality of Telecom Services during Significant Social/ Cultural/ Religious/ Commercial/ International Events etc.

2. This SOP outlines a framework for collaboration between various stakeholders like License Service Areas (LSAs), Telecom Service Providers (TSPs), State Authorities, etc. during the events of both National and State-level Importance.

3. This SOP serves as a guideline/reference outlining steps to be taken by the concerned LSA during execution of these events. It is important to note that the concerned LSA isn't bound to adhere rigidly to every detail outlined in this SOP. As deemed necessary, LSA can use its discretion to selectively utilize SOP's components based on the nature of various events. This SOP provides a broad framework to guide the implementation/execution process of such events.

This has the approval of the Competent Authority.

Encl: As above.

Vivek Khare
10/7/2024.

(Vivek Khare)
DDG (QoS), DoT HQ
Tel. No. 011-23210182

Copy To:

1. PSO to Member (T), DoT for kind information please.
2. TSPs - for compliance and cooperation with LSAs.

Standard Operating Procedure (SOP) for maintaining Quality of Telecom Services during Significant Social/Cultural/ Religious/Commercial/International Events etc.

Introduction –

In order to ensure seamless and reliable telecom services during events with significant social, cultural, religious, commercial and/or international importance etc., this Standard Operating Procedure (SOP) outlines a framework for collaboration between various stakeholders like License Service Areas (LSAs), Telecom Service Providers (TSPs), State Authorities, etc.

The SOP encompasses proactive measures and planning for optimal network connectivity and quality of telecom services. It is important to emphasize that this SOP remains a dynamic document. Events like the Kumbh Mela, Magh Mela, Trade Fairs, National and International Sport events etc. or events hosting delegations from other countries and experiences gained from managing such events will serve as resources to shape this SOP continously. It is important to flag that this SOP is advisory in nature in terms of "what is to be done for a specific event". As deemed necessary, LSA can use its discretion to selectively utilize SOP's components based on the nature of various events.

Each step outlined in this SOP, from listing upcoming events to coordinating with State Authorities, obtaining necessary permissions, and conducting network assessments, is tailored to address the unique challenges posed by events with substantial public gatherings and/or international significance. Thus this SOP aims to enhance the efficiency of telecom services and mitigate potential challenges, ensuring a seamless experience for users during such events.

(A) Preparatory Phase

1. Event Calendar Compilation:

- All the LSAs should compile a complete calendar, delineating the probable dates/months of upcoming events significant in terms of social, cultural, religious, commercial and/or International importance for the entire calendar year. This calendar must include events of importance where ensuring the quality of telecom services is crucial, such as gatherings with substantial public attendance like the inauguration of the Shree Ram Mandir Temple in Ayodhya, Kumbh Mela, Trade Fairs, National and International Sport events etc. or events hosting delegations from other countries, like G-20 Summit, meeting with Important Foreign Delegations, etc.
- The State Broadband Committee would serve as a valuable resource for gathering additional such information.

- In cases of last-minute events of importance, the LSA head, in consultation with DGT HQ, can make judicious decisions.
- Events conveyed by ministries or other sources may periodically be shared with the respective LSAs by DoT HQ/DGT HQ. Typically, ministries directly engage with TSPs for telecom service requirements, especially considering the regular occurrence of events. DoT intervention should be reserved for special events.
- There are many events that happen every year where VVIPs may visit or not, so the calendar here is targeted for all such events which occur on regular basis. The LSA head in coordination with DGT HQ may take judicious decision for last minute confirmations.
- Event calendars prepared by LSAs should be shared with the DGT, Central team and nodal officers of TSPs well in advance.

(Responsibility Centre - DGT HQ, LSAs and QoS Division, DoT HQ)

2. Site Identification and Network Evaluation with TSPs:

- Ahead of the commencement of any event, the LSA is required to pinpoint the specific location(s) or venue site(s) where the event is scheduled. LSA shall assess the approximate gathering of the event in consultation with State authorities, TSPs, media reports etc.
- To ensure effective network connectivity and coverage during the event, the LSA should conduct meetings with all TSPs well in advance. These meetings aim to assess the present network status at the venue site(s) and nearby areas, including current network load, facilitating optimal performance planning. Current network evaluations can be done by LSAs through any mechanism, be it report or in performa or through meetings as deemed necessary by the concerned LSA. The primary objective is to gain a comprehensive understanding of the present status of TSP's networks so that the proper future planning can be done for optimal performance.
- LSA shall send the Network evaluation report to DGT HQ and DoT HQ.
- Preparatory activities for events should be initiated well in advance of events.
- The QoS Division, DoT HQ in its discretion may also arrange meetings with the Central Team of TSPs to gather essential information on network conditions in the field, enhancing overall preparedness.
- LSA heads have the authority to deploy officers or establish task forces based on their discretion and sound judgment.
- For last-minute events, the LSA head, in consultation with DGT HQ, can make judicious decisions.

(Responsibility Centre – DGT HQ, concerned LSA and QoS Division, DoT HQ)

(B) Planning Phase

1. Planning and Coordination with State Authorities:

- To proactively plan for periods of increased mobile/telecom traffic during event, LSA may coordinate with the concerned State Authority, if required, to assess anticipated network traffic demand, gather information on establishing temporary accommodations for the public, if any, like Tent Cities etc., establishment of centers for media, important routes and location for drive tests/spot/walk test etc..
- Furthermore, the State Government should be asked to designate a Single Point of Contact (SPOC) for the event. Discussions regarding the appointment of this SPOC can be conducted with relevant authorities during meetings pertaining to the respective events.
- The State Broadband Committee can also be used as a platform for appointment of this SPOC by the concerned authorities.
- Additionally, both LSAs and TSPs should appoint their own Single Point of Contact (SPOC) for the event. If deemed fit, the LSA Head may constitute a committee under the chairmanship of the concerned DDG for better coordination.

(Responsibility Centre – DGT HQ and concerned LSA)

(C) Deployment and Monitoring Phase

1. Instructions for Telecommunication Infrastructure Deployment:

- Based on the anticipated surge in mobile/telecom traffic demand during events at the venue site(s) and/or nearby, LSA may instruct all the TSPs to deploy supplementary infrastructures, such as Cell on Wheels (CoWs), additional BTSs (Base Transceiver Stations), In-Building Solutions (IBS), Wi-Fi, etc. at the venue site(s) and/or nearby areas. The goal is to enhance network capacity and performance to meet the increased mobile/telecom traffic needs during such events.
- To ensure timely and efficient deployment of telecommunication infrastructure, the LSA must oversee the timely submission of requisite applications by all the TSPs through designated online portals (ex: Gati Shakti Portal, etc.) to the relevant authorities and secure Right of Way (RoW) permissions and lay Optical Fiber Cables (OFCs)/installation of BTSs.
- In instances where necessary, DoT HQ may convene meetings with the Central Team of TSPs to monitor the status of telecommunication infrastructure deployment and to address bottlenecks or challenges faced by TSPs and LSAs, thus ensuring an efficient deployment process.
- In cases where there is lack of response or initiation from TSPs, the same may be brought up and escalated to DGT HQ, so that the matter may be taken up with the central team of TSPs to resolve any issues.

- For indoor events, typically, the property owner or relevant department/ministry is responsible for planning the provision of IBS within event venues/buildings like Pragati Maidan, Bharat Mandapam, or similar locations in other states. Any supplementary needs such as IBS/WiFi are to be addressed by the property owner or relevant department/ministry through commercial agreements with TSPs. The role of the DoT and LSAs is to facilitate these arrangements.
- As is feasible, the NBM Division may prioritize applications on the RoW Portal intended for temporary arrangements during Significant Social/ Cultural / Religious / Commercial / International Events, so that they appear at the top of the list.

(Responsibility Centre – DGT HQ, concerned LSA, NBM Division and QoS Division, DoT HQ)

2. Control Room/Camp Office Setup:

- Control Room/Camp Office, if required, may be setup by LSA at the venue site for the effective coordination and swift resolution of issues. An illustrative example is the Camp Office set up by Uttar Pradesh (East) LSA during the inauguration of the Shree Ram Mandir Temple in Ayodhya. The space required for the control room/camp office may be arranged by the LSA from relevant authorities, state or district management, local BSNL office space, or other suitable sources.
- LSA must ensure that all the TSPs deploy their project team, along with the project nodal officer, to this Control Room/Camp Office. This centre serves as a focal point for addressing concerns with both State Authorities and TSPs, as well as for conducting drive tests/walk tests by LSA and TSPs at crucial important locations.
- LSA should also confirm that the contact details of the Senior Level Technical Officer (Single Point of Contact or SPOC) as Project nodal officer, are communicated by all the TSPs. The LSA, in turn, may share this information with the O/o DGT HQ for comprehensive coordination. LSA and TSP may also appoint their own Single Point of Contact (SPOC) for the specific event, ensuring that any concerns can be addressed to the primary designated authority
- Additionally, LSA must check that all the TSPs have provided the names and contact details of their teams, including drivers, to the concerned Authority, ensuring the unhindered movement of their personnel.
- DGT HQ retains the authority to make decisions regarding the allocation of manpower, budgetary allocations and other necessary resources to address event requirements.

(Responsibility Centre – DGT HQ and concerned LSA)

3. Emergency Preparedness:

- Necessary instructions for enabling Intra-Circle Roaming (ICR) among Telecom

Service Providers may be issued by Disaster Management Division of DoT HQ for handling emergency situations. In such events, the decision will be taken as the situation warrants. When necessary, the LSA may send a request to the DM Division, DoT HQ with a copy to the QoS Division/DGT HQ, for issuing the necessary instructions. However, the LSA(s) should ensure that ICR testing is conducted by all the TSPs beforehand.

- Priority Call Routing (PCR), if necessary, may be implemented as per extant instructions issued by DoT HQ for emergency situation, thereby enhancing the efficiency of call handling during critical situations. List of numbers to be enabled for PCR may be sought from concerned State Government/Central government Agencies/autonomous bodies etc. by necessary coordination by the respective LSA(s).
- It is to note that this SOP is not intended for managing disasters, for which there is a separate SOP in place.

(Responsibility Centre – DGT HQ, concerned LSA and DM Division, DoT HQ)

4. Coordination with State Broadband Committee:

Within the framework of the National Broadband Mission, the State Broadband Committee has been setup which is presided over by the Chief Secretary of the respective State Government, serving as the Chairman with Advisor/Sr.DDG/Addl. DG(T) in LSAs as the Member Convener. All the requirements for ensuring uninterrupted Telecom Services during events of importance, if required, may be made part of the agenda of these State Broadband Committee Meetings.

(Responsibility Centre – concerned LSA)

5. Comprehensive Drive Tests and Analysis:

- LSAs must ensure drive tests are carried out to assess network performance along crucial routes, with the information of these routes obtainable from the relevant State Authority.
- The benchmarks for these drive tests are already established by TRAI but LSAs can incorporate additional parameters like Download/Upload Speeds, depending on requirements of the events. Based on this analysis, LSAs may issue directives to TSPs for necessary enhancements or installations aimed at improving mobile/telecom network performance, thereby optimizing based on Drive Test assessments.
- The findings from these tests are also to be communicated by LSA to the O/o DGT HQ.

(Responsibility Centre – DGT HQ and concerned LSA)

6. Walk Test/Spot Test and Analysis:

- LSA and TSPs in coordination will conduct Walk Test/Spot Test to evaluate network quality in pedestrian-heavy areas such as event venue(s) and/or nearby or other vital locations. These walk test/spot test reports are to be communicated by LSA to the O/o DGT HQ.
- The frequency of walk test/spot test may be determined by the LSA based on the event's characteristics. The main goal is to evaluate actual network performance on-site, whether indoors, outdoors, or in nearby areas, whenever feasible. Any deficiencies discovered during these tests will be conveyed to the TSP by the LSA for possible enhancements.
- Annexure-I contains a sample walk test report, while Annexure-II offers analysis, providing insights into the methodology and results of these assessments. These reports provide valuable information on network performance in real-world scenarios and are instrumental in network planning and improvement. LSAs have the flexibility to customize them as needed, with the attached sample format serving solely as a reference.

(Responsibility Centre – DGT HQ, concerned LSA and QoS Division, DoT HQ)

7. Timely Execution through Checklists:

- LSA is required to develop a thorough checklist, complete with specified deadlines, covering all crucial activities to ensure the timely completion of activities. These activities encompass timelines for deploying CoWs/BTSs/IBS/Wi-Fi, securing RoW permissions, laying OFCs, conducting drive tests, establishing camp offices, etc. It is imperative for LSA to ensure the timely completion of these tasks by TSPs, avoiding last-minute rushes and facilitating effective network planning.
- Annexure-III includes a sample checklist provided for LSA's reference, subject to modification as required. LSA must communicate both the checklists and their associated deadlines to the O/o DGT HQ for comprehensive coordination and oversight. The attached sample checklist serves as a reference, yet it may vary based on the event's requirements. LSA is authorized to make necessary adjustments as deemed fit.

(Responsibility Centre – DGT HQ and concerned LSA)

8. Deployment of Additional Officials:

When deemed necessary, the LSA may seek the deployment of additional officers from DGT HQ. Upon evaluation, DGT HQ may deploy additional officers from other LSAs to reinforce current teams.

(Responsibility Centre – DGT HQ and concerned LSA)

9. Vigilance and Incident Response:

During significant events, it's crucial for all TSPs/ISPs, in collaboration with the LSA, to remain vigilant and promptly address any adverse incidents concerning mobile network security and/or cyber security. Responses should align with directives from the Security Wing of the DoT. It is mandatory for all TSPs/ISPs to strictly adhere to all periodically issued instructions by the Department of Telecommunications in respect of network security and/or cyber security.

(Responsibility Centre – concerned LSA and Security Wing, DoT HQ)

10. Progress and Traffic Monitoring:

- To ensure effective monitoring and analysis of telecom traffic conditions, LSAs should ensure that all TSPs to share Progress Reports/Traffic Monitoring Reports. The sharing of these reports enables LSAs to make informed decisions for optimizing network performance and promptly addressing challenges.
- During events, LSAs plays a pivotal role in monitoring various aspects, including traffic and signalling channels w.r.t. to various technologies, due to potential surges in network subscribers. LSAs are authorized to request specific reports from TSPs to analyse traffic and signalling patterns specific to the event.
- Subsequently, the organizing LSA may share its experiences with other LSAs through O/o DGT HQ. This knowledge-sharing facilitates collective learning and improvement across LSAs.
- Sample TSP-Wise Traffic Monitoring Reports and their analyses, provided as Annexures-IV and Annexure-V respectively, serve as examples, with LSA can modify as per the nature of the event.

(Responsibility Centre – DGT HQ, concerned LSA and QoS Division, DoT HQ)

*****

Sample Format for Walk Test

Location	Lat	Long	TSP	Date	Time	2G - Rx level	3G- RSCP	3G-DL	3G - UL	Latency	Packet loss	4G - RSRP	4G - DL	4G - UL	Latency	Packet loss	5G - RSRP	5G -DL	5G - UL	Latency	Packet loss	video call status	youtube streaming status	whatsapp messaging	file transfer status	sms testing status	voice call status	Remark
Anawa Ram mandir	26.79466	82.19751	TSP1	30-Jan-24	01:13pm	-51	NA	NA	NA	NA	NA	-60	30.46	27.28	58.3	0	-69	258.97	86.71	51.7	0	OK	OK	OK	OK	OK	OK	
Anawa Ram mandir	26.79532	82.19528	TSP2	30-Jan-24	12:58 PM	NA	NA	4.8	4.08	75	0	NA	NA	NA	NA	NA	NA	NA	NA	NA	NA	OK	OK	OK	OK	OK	OK	
Anawa Ram mandir	26.79467	82.19736	TSP3	30-Jan-24	1:15 PM	NA	NA	NA	NA	NA	NA	-76	23.26	5.92	44.8	0%	-76	123.41	74.84	39.7	0%	OK	OK	OK	OK	OK	OK	
Anawa Ram mandir	26.79461	82.1975	TSP4	30-Jan-24	13:13	-73	NA	NA	NA	NA	NA	-67	46.87	17.27	33.5	0	NA	NA	NA	NA	NA	OK	OK	OK	OK	OK	OK	
Bhagya shree guest house Rampath	26.78967	82.19459	TSP1	30-Jan-24	11:06am	-55	NA	NA	NA	NA	NA	-77	25.57	18.26	44	0	-81	278.84	20.55	74.4	0	OK	OK	OK	OK	OK	OK	
Bhagya shree guest house Rampath	26.78966	82.19464	TSP2	30-Jan-24	11:06 AM	NA	NA	5.58	4.97	66	0	NA	NA	NA	NA	NA	NA	NA	NA	NA	NA	OK	OK	OK	OK	OK	OK	
Bhagya shree guest house Rampath	26.78966	82.19475	TSP3	30-Jan-24	11:06 AM	NA	NA	NA	NA	NA	NA	-76	188.16	87.91	39.5	0%	-76	31.07	15.31	51.7	0%	OK	OK	OK	OK	OK	OK	
Bhagya shree guest house Rampath	26.78956	82.1945	TSP4	30-Jan-24	11:06	-63	NA	NA	NA	NA	NA	-70	50.63	17.14	37.6	0	NA	NA	NA	NA	NA	OK	OK	OK	OK	OK	OK	
Birla mandir rampath	26.79286	82.20055	TSP1	30-Jan-24	11:35am	-51	NA	NA	NA	NA	NA	-76	30.46	6.44	64.2	0	-74	236.54	70.78	0	0	OK	OK	OK	OK	OK	OK	
Birla mandir rampath	26.79301	82.20055	TSP2	30-Jan-24	11:34 AM	NA	NA	2.93	5.34	76	0	NA	NA	NA	NA	NA	NA	NA	NA	NA	NA	OK	OK	OK	OK	OK	OK	
Birla mandir rampath	26.79296	82.20095	TSP3	30-Jan-24	11:38 AM	NA	NA	NA	NA	NA	NA	-76	44	57.61	42.6	0%	-76	64.36	87.54	33.2	0%	OK	OK	OK	OK	OK	OK	
Birla mandir rampath	26.79285	82.20042	TSP4	30-Jan-24	11:39	-61	NA	NA	NA	NA	NA	-62	41.87	26.44	37.6	0	NA	NA	NA	NA	NA	OK	OK	OK	OK	OK	OK	
Dasrath mahal	26.79654	82.19958	TSP1	30-Jan-24	02:42pm	-53	NA	NA	NA	NA	NA	-83	76.84	22.23	36.2	0	-77	202.08	88.92	28.9	0	OK	OK	OK	OK	OK	OK	
Dasrath mahal	26.79654	82.19958	TSP2	30-Jan-24	02:42pm	NA	NA	2.87	4.98	92	0	NA	NA	NA	NA	NA	NA	NA	NA	NA	NA	OK	OK	OK	OK	OK	OK	
Dasrath mahal	26.79637	82.19982	TSP3	30-Jan-24	NA	NA	NA	NA	NA	NA	NA	NA	NA	NA	NA	NA	NA	NA	NA	NA	NA	NA	NA	NA	NA	NA	NA	Fiber Cut
Dasrath mahal	26.79637	82.19982	TSP4	30-Jan-24	14:11	-73	NA	NA	NA	NA	NA	-84	45.45	7.5	62.5	0	NA	NA	NA	NA	NA	OK	OK	OK	OK	OK	OK	
Hanuman garhi main gate	26.79443	82.20204	TSP1	30-Jan-24	01:40pm	-55	NA	NA	NA	NA	NA	-68	29.42	3.27	72	0	-79	121.99	30.41	91.4	0	OK	OK	OK	OK	OK	OK	
Hanuman garhi main gate	26.79446	82.20203	TSP2	30-Jan-24	1:41 PM	NA	NA	2.32	11.06	63	0	NA	NA	NA	NA	NA	NA	NA	NA	NA	NA	OK	OK	OK	OK	OK	OK	
Hanuman garhi main gate	26.79526	82.20432	TSP3	30-Jan-24	2:18 PM	NA	NA	NA	NA	NA	NA	-76	1.04	6.3	NA	33%	NA	NA	NA	NA	NA	NA	NA	NA	NA	NA	NA	Fiber Cut
Hanuman garhi main gate	26.79442	82.202	TSP4	30-Jan-24	13:40	-57	NA	NA	NA	NA	NA	-85	18.01	16.33	35.9	0	NA	NA	NA	NA	NA	OK	OK	OK	OK	OK	OK	
Hanuman garhi temple	26.79577	82.20194	TSP1	30-Jan-24	02:34pm	-65	NA	NA	NA	NA	NA	-83	36.48	5.28	34.7	0	-82	149.84	28.8	42	0	OK	OK	OK	OK	OK	OK	
Hanuman garhi temple	26.79584	82.20193	TSP2	30-Jan-24	2:34 PM	NA	NA	5.6	5.1	127	0	NA	NA	NA	NA	NA	NA	NA	NA	NA	NA	OK	OK	OK	OK	OK	OK	
Hanuman garhi temple	26.79575	82.20189	TSP3	30-Jan-24	2:34 PM	NA	NA	NA	NA	NA	NA	-58	0.82	7.26	N.A	0%	-73	0.66	81.54	59.9	9	OK	OK	OK	OK	OK	OK	
Hanuman garhi temple	26.79583	82.20188	TSP4	30-Jan-24	14:34	-69	NA	NA	NA	NA	NA	-81	47.35	16	33.2	0	NA	NA	NA	NA	NA	OK	OK	OK	OK	OK	OK	
Hanumanth market ram path	26.79724	82.20339	TSP1	30-Jan-24	02:30pm	-69	NA	NA	NA	NA	NA	-58	39.88	26.95	40.6	0	-85	266.15	60.69	31.1	0	OK	OK	OK	OK	OK	OK	
Hanumanth market ram path	26.79628	82.20296	TSP2	30-Jan-24	2:25 PM	NA	NA	4.6	9.42	74	0	NA	NA	NA	NA	NA	NA	NA	NA	NA	NA	OK	OK	OK	OK	OK	OK	
Hanumanth market ram path	26.7962	82.20284	TSP3	30-Jan-24	2:22 PM	NA	NA	NA	NA	NA	NA	-76	2.62	1.93	67.3	16%	-76	1.04	6.3		33%	OK	OK	OK	OK	OK	OK	
Hanumanth market ram path	26.79621	82.20294	TSP4	30-Jan-24	14:27	-55	NA	NA	NA	NA	NA	-93	32.05	14.08	35.9	0	NA	NA	NA	NA	NA	OK	OK	OK	OK	OK	OK	
Kajayana	26.79007	82.19547	TSP1	30-Jan-24	03:10pm	-51	NA	NA	NA	NA	NA	-69	44.42	4.18	77.2	0	-74	250.33	68.46	70.1	0	OK	OK	OK	OK	OK	OK	
Kajayana	26.79005	82.19548	TSP2	30-Jan-24	3:10 PM	NA	NA	2.87	5.58	99	0	NA	NA	NA	NA	NA	NA	NA	NA	NA	NA	OK	OK	OK	OK	OK	OK	
Kajayana	26.78999	82.19554	TSP3	30-Jan-24	NA	NA	NA	NA	NA	NA	NA	NA	NA	NA	NA	NA	NA	NA	NA	NA	NA	NA	NA	NA	NA	NA	NA	Fiber Cut
Kajayana	26.78999	82.19554	TSP4	30-Jan-24	15:44	-75	NA	NA	NA	NA	NA	-89	23.47	12.6	36.7	0	NA	NA	NA	NA	NA	OK	OK	OK	OK	OK	OK	
Ram mandir Gate no 3 rampath	26.79137	82.19826	TSP1	30-Jan-24	11:21am	-51	NA	NA	NA	NA	NA	-67	67.51	23.04	26.6	0	-65	304.34	59.08	34.3	0	OK	OK	OK	OK	OK	OK	
Ram mandir Gate no 3 rampath	26.79135	82.19828	TSP2	30-Jan-24	11:21 AM	NA	NA	4.52	5.32	56	0	NA	NA	NA	NA	NA	NA	NA	NA	NA	NA	OK	OK	OK	OK	OK	OK	
Ram mandir Gate no 3 rampath	26.79059	82.19709	TSP3	30-Jan-24	11:14 AM	NA	NA	NA	NA	NA	NA	-76	4.4	5.52	43.2	0%	-76	162.57	40.45	33.6	0%	OK	OK	OK	OK	OK	OK	
Ram mandir Gate no 3 rampath	26.79142	82.19831	TSP4	30-Jan-24	11:21	-65	NA	NA	NA	NA	NA	-82	45.75	19.64	41.6	0	NA	NA	NA	NA	NA	OK	OK	OK	OK	OK	OK	
Ram mandir locker gate	26.79426	82.1982	TSP1	30-Jan-24	01:26pm	-51	NA	NA	NA	NA	NA	-82	26.93	7.33	88.1	0	-77	254.38	68.1	0	0	OK	OK	OK	OK	OK	OK	
Ram mandir locker gate	26.79428	82.19817	TSP2	30-Jan-24	1:27 PM	NA	NA	7.2	5.76	75	0	NA	NA	NA	NA	NA	NA	NA	NA	NA	NA	OK	OK	OK	OK	OK	OK	
Ram mandir locker gate	26.79113	82.19812	TSP3	30-Jan-24	11:25 AM	NA	NA	NA	NA	NA	NA	-76	18.99	20.27	34.4	0%	-76	141.51	41.12	37.6	0%	OK	OK	OK	OK	OK	OK	
Ram mandir locker gate	26.79422	82.19817	TSP4	30-Jan-24	13:27	-69	NA	NA	NA	NA	NA	-83	44.01	9.12	42.3	0	NA	NA	NA	NA	NA	OK	OK	OK	OK	OK	OK	
Rampath mandir Gate no 2	26.79063	82.1965	TSP1	30-Jan-24	11:14am	-57	NA	NA	NA	NA	NA	-73	24.48	10.65	53.4	0	-72	275.82	94.43	0	0	OK	OK	OK	OK	OK	OK	
Rampath mandir Gate no 2	26.79061	82.1965	TSP2	30-Jan-24	11:14 AM	NA	NA	7.12	5.55	128	0	NA	NA	NA	NA	NA	NA	NA	NA	NA	NA	OK	OK	OK	OK	OK	OK	
Rampath mandir Gate no 2	26.79442	82.19826	TSP3	30-Jan-24	1:26 PM	NA	NA	NA	NA	NA	NA	-76	14.92	2.84	46.2	0%	-76	87.88	57.17	62.3	0%	OK	OK	OK	OK	OK	OK	
Rampath mandir Gate no 2	26.79071	82.19644	TSP4	30-Jan-24	11:14	-69	NA	NA	NA	NA	NA	-84	39.25	17.67	37.1	0	NA	NA	NA	NA	NA	OK	OK	OK	OK	OK	OK	
Tedi bajar chauraha ram path	26.7879	82.19241	TSP1	30-Jan-24	10:58am	-51	NA	NA	NA	NA	NA	-76	34.89	9.2	57.5	0	-80	202.16	42.2	53.9	0	OK	OK	OK	OK	OK	OK	
Tedi bajar chauraha ram path	26.78474	82.1876	TSP2	30-Jan-24	10:50 AM	NA	NA	6.48	10.7	74	0	NA	NA	NA	NA	NA	NA	NA	NA	NA	NA	OK	OK	OK	OK	OK	OK	
Tedi bajar chauraha ram path	26.78716	82.19251	TSP3	30-Jan-24	10:57 AM	NA	NA	NA	NA	NA	NA	-76	14.12	5.18	45.2	0%	-76	184.23	109.39	32.7	0%	OK	OK	OK	OK	OK	OK	
Tedi bajar chauraha ram path	26.78786	82.19239	TSP4	30-Jan-24	10:57	-51	NA	NA	NA	NA	NA	-72	40.39	9.42	70.2	0	NA	NA	NA	NA	NA	OK	OK	OK	OK	OK	OK	
Ayodhya Dham Rly Station near old building	26.78874	82.2008	TSP1	30-Jan-24	11:07am	-55	NA	NA	NA	NA	NA	-70	33.21	7.25	41.1	0	-70	174.07	24.68	67.1	0	OK	OK	OK	OK	OK	OK	
Ayodhya Dham Rly Station near old building	26.78838	82.20071	TSP2	30-Jan-24	11:08 AM	-59	-75	1.37	1.83	85.5	0	NA	NA	NA	NA	NA	NA	NA	NA	NA	NA	OK	OK	OK	OK	OK	OK	
Ayodhya Dham Rly Station near old building	26.78819	82.20098	TSP3	30-Jan-24	11.07AM	NA	NA	NA	NA	NA	NA	-54	29.21	7.85	51.5	0%	-61	159.19	91.36	30.9	0	OK	OK	OK	OK	OK	OK	
Ayodhya Dham Rly Station near old building	26.78863	82.20063	TSP4	30-Jan-24	11:06	-57	NA	NA	NA	NA	NA	-70	53.21	21.89	36	0	NA	NA	NA	NA	NA	OK	OK	OK	OK	OK	OK	
Ayodhya Dham rly. Station Entry Gate	26.78764	82.19912	TSP1	30-Jan-24	10:46am	-61	NA	NA	NA	NA	NA	-80	23.13	15.35	18.8	0	-78	209.22	22.74	64.4	0	OK	OK	OK	OK	OK	OK	
Ayodhya Dham rly. Station Entry Gate	26.78795	82.19981	TSP2	30-Jan-24	10:47 AM	-67	-67	6.35	2.26	78.8	0	NA	NA	NA	NA	NA	NA	NA	NA	NA	NA	OK	OK	OK	OK	OK	OK	
Ayodhya Dham rly. Station Entry Gate	26.78792	82.19982	TSP3	30-Jan-24	10.47AM	NA	NA	NA	NA	NA	NA	-78	29.87	27.98	38.7	0%	-77	98.56	58.45	38.3	0	OK	OK	OK	OK	OK	OK	
Ayodhya Dham rly. Station Entry Gate	26.78796	82.19974	TSP4	30-Jan-24	10:47	-67	NA	NA	NA	NA	NA	-88	43.63	6.18	36.5	0	NA	NA	NA	NA	NA	OK	OK	OK	OK	OK	OK	
Ayodhya Dham rly. Station Entry Gate5	26.78795	82.19997	TSP1	30-Jan-24	10:56am	-57	NA	NA	NA	NA	NA	-72	21.29	6.64	16.2	0	-83	179.85	60.71	36.2	0	OK	OK	OK	OK	OK	OK	
Ayodhya Dham rly. Station Entry Gate5	26.78796	82.19999	TSP2	30-Jan-24	11:00 AM	-71	-77	3.55	2.47	72.4	0	NA	NA	NA	NA	NA	NA	NA	NA	NA	NA	OK	OK	OK	OK	OK	OK	
Ayodhya Dham rly. Station Entry Gate5	26.78796	82.2	TSP3	30-Jan-24	10.58AM	NA	NA	NA	NA	NA	NA	-63	28.15	26.49	45.9	0%	-73	149.45	79.09	41.8	0	OK	OK	OK	OK	OK	OK	
Ayodhya Dham rly. Station Entry Gate5	26.78796	82.19997	TSP4	30-Jan-24	10:59	-71	NA	NA	NA	NA	NA	-67	44.47	15.63	39.7	0	NA	NA	NA	NA	NA	OK	OK	OK	OK	OK	OK	
District Hospital Ayodhya Emergency Gate	26.77889	82.14032	TSP1	30-Jan-24	04:37pm	-65	NA	NA	NA	NA	NA	-79	44.17	10.59	53.6	0	NA	NA	NA	NA	0	OK	OK	OK	OK	OK	OK	FZD51 5G deployement in WIP
District Hospital Ayodhya Emergency Gate	26.77887	82.13991	TSP2	30-Jan-24	4:41 PM	-69	-81	3.46	2.16	113.7	0	NA	NA	NA	NA	NA	NA	NA	NA	NA	NA	OK	OK	OK	OK	OK	OK	
District Hospital Ayodhya Emergency Gate	26.77878	82.14089	TSP3	30-Jan-24	04.40PM	NA	NA	NA	NA	NA	NA	-72	27.12	28.45	36.3	NA	-79	167.49	38.51	24.8	0	OK	OK	OK	OK	OK	OK	
District Hospital Ayodhya Emergency Gate	26.77888	82.14056	TSP4	30-Jan-24	14:41	-77	NA	NA	NA	NA	NA	-92	51.82	20.52	31.4	0	NA	NA	NA	NA	NA	OK	OK	OK	OK	OK	OK	
District Hospital Ayodhya Entry Gate	26.77892	82.13978	TSP1	30-Jan-24	04:35pm	-67	NA	NA	NA	NA	NA	-82	26.18	15.8	59.1	0	NA	NA	NA	NA	0	OK	OK	OK	OK	OK	OK	FZD51 5G deployement in WIP
District Hospital Ayodhya Entry Gate	26.77892	82.13999	TSP2	30-Jan-24	4:38 PM	-63	-79	1.64	1.16	125.1	0	NA	NA	NA	NA	NA	NA	NA	NA	NA	NA	OK	OK	OK	OK	OK	OK	
District Hospital Ayodhya Entry Gate	26.7789	82.14005	TSP3	30-Jan-24	04.36PM	NA	NA	NA	NA	NA	NA	-67	20.54	26.45	36.2	0%	-75	150.3	64.71	41.8	0	OK	OK	OK	OK	OK	OK	
District Hospital Ayodhya Entry Gate	26.77899	82.14049	TSP4	30-Jan-24	14:37	-67	NA	NA	NA	NA	NA	-88	39.21	20.48	31.8	0	NA	NA	NA	NA	NA	OK	OK	OK	OK	OK	OK	
Gujrati Dharmshala	26.78926	82.20047	TSP1	30-Jan-24	11:23am	-63	NA	NA	NA	NA	NA	-73	21.3	2.42	44.1	0	-85	136.91	14.2	24.8	0	OK	OK	OK	OK	OK	OK	
Gujrati Dharmshala	26.78909	82.20033	TSP2	30-Jan-24	11:23 AM	-73	-73	3.26	2.33	93	0	NA	NA	NA	NA	NA	NA	NA	NA	NA	NA	OK	OK	OK	OK	OK	OK	
Gujrati Dharmshala	26.7891	82.20033	TSP3	30-Jan-24	11.22AM	NA	NA	NA	NA	NA	NA	-63	28.06	29.95	36.7	0%	-74	178.11	74.11	36.1	0	OK	OK	OK	OK	OK	OK	
Gujrati Dharmshala	26.78917	82.20039	TSP4	30-Jan-24	11:24	-67	NA	NA	NA	NA	NA	-86	33.2	16.97	33.8	0	NA	NA	NA	NA	NA	OK	OK	OK	OK	OK	OK	
Head Post Office	26.77885	82.13388	TSP1	30-Jan-24	02:55pm	-59	NA	NA	NA	NA	NA	-83	48.29	9.42	81.6	0	-89	202.37	21.95	66.5	0	OK	OK	OK	OK	OK	OK	
Head Post Office	26.77864	82.13389	TSP2	30-Jan-24	2:57 PM	-71	-77	6.96	2.48	73.5	0	NA	NA	NA	NA	NA	NA	NA	NA	NA	NA	OK	OK	OK	OK	OK	OK	
Head Post Office	26.77864	82.13389	TSP3	30-Jan-24	02.55PM	NA	NA	NA	NA	NA	NA	-98	10.11	7.54	53	0%	-98	162.08	3.31	33.8	0	OK	OK	OK	OK	OK	OK	
Head Post Office	26.77863	82.13394	TSP4	30-Jan-24	14:56	-73	NA	NA	NA	NA	NA	-99	46.09	15.52	30.1	0	NA	NA	NA	NA	NA	OK	OK	OK	OK	OK	OK	

Annexure-II

Analysis of Walk Test Report

Low Download and Upload Speeds (in mbps)

Locations	Average of 4G - DL
Media Centre	19.21
Ayodhya Dham rly. Station Entry Gate5	21.29
Gujrati Dharmshala	21.3
Shabri Rasoi Naya Ghat	21.33
Tulsi Udhyan	21.73
Ayodhya Dham rly. Station Entry Gate	23.13
stn. Marry school, army cant	23.33
Rampath mandir Gate no 2	24.48
Ramkatha Park	24.78
Bhagya shree guest house Rampath	25.57

Locations	Average of 4G - UL
Gujrati Dharmshala	2.42
Hanuman garhi main gate	3.27
Ayodhya Bus Stand	3.9
Kajayana	4.18
stn. Marry school, army cant	4.3
Hotel Park inn.	4.88
Hanuman garhi temple	5.28
Ramkatha Park	6.17
Birla mandir rampath	6.44
Ayodhya Dham rly. Station Entry Gate5	6.64

Locations	Average of 5G -DL
Ram Ghat Halt	56.02
Rajkiya Udhaan Ayodhya ,Guptar Ghat	91.27
Lata Mangeshkar chowk	114.59
Tent no. 124 Praveg Tent City	117.32
Praveg Tent City Reception Area , Bramhak	120.83
Praveg Tent City Outdoor Bramhakund	120.87
Hanuman garhi main gate	121.99
Ramkatha Park	122.51
Ram ki Paidi Turn Guptar Ghat	129.05
Guptar Ghat Tent City	129.3

Locations	Average of 5G - UL
Hotel shane Avadh	5.46
Nagar Nigam Ayodhya Gate	5.58
Rikabganj near laxmi sweets	7.78
Shabri Rasoi Naya Ghat	10.9
Near Rikabganj hanuman garhi	11.91
Indiniyat Tent City , Behind Rammandir	12.04
stn. Marry school, army cant	13.45
Tulsi Udhyan	13.61
Gujrati Dharmshala	14.2
Bhajan Sandhya Sthal Naya Ghat	16.46

Check List with Target Dates

S. No.	Activity	Action By	Target Date
1	**Action Plan of TSPs**		
a	Existing connectivity (Capacity and Loading technology-wise)	All TSPs	15.12.2023
b	Planned sites (Technology-wise)		07.01.2024
c	RoW application and approval status (Site-wise)		18.12.2023
d	Backhaul media (Redundancy and upgradation)		07.01.2024
e	Availability of COW BTSs at Ayodhya		07.01.2024
2	**Setting up of Camp Office / Control Room at Ayodhya**	UP East LSA	26.12.2023
a	Space for Camp Office/ Control Room	BSNL, Ayodhya	22.12.2023
b	Ground Assessment of requirements – 1. Mobile Tower/ BTS 2. OFC 3. Drive tests etc.	ALL TSPs in coordination with UP East LSA	19th and 20th Dec 2023
c	Nodal Officers of LSA – UP(E) at Ayodhya	UP East LSA	26.12.2023
d	Project in-charge of TSPs at Ayodhya including OFC/ Media maintenance teams.	All TSPs	18.12.2023
3	**Vehicular Traffic Plan of Ayodhya Region**	UP East LSA	26.12.2023
4	**Map of Ayodhya Region with marking of important places such as Airport, Railway Station, Bus Stand, Shree Ram Mandir site, Other pilgrimage sites with radial distance marking.**	UP East LSA, TSPs	20.12.2023

Sample Daily Traffic Monitoring Report

TSP Name	Site ID (sector-wise)	Site location	Area being served	Date	Start Time	End Time	Total Traffic (Erl)	Maximum Provisioned Capacity (i.e. Maximum no. of customers which can be served)	Data Capacity (in GB)	Payload (in GB)	Call Success Rate	Call drop Rate	Total Calls attempted	Total Number of Users Latched	Backhaul type	Backhaul capacity	Remarks (eg. Sectors being added or upgraded or additional sites added)
TSP Name	FZD01	Civil Lines	Bus Stop-3	30-Jan	9:00	10:00	27.1	6300	1230	17.55	99.90	0.35	865	400	NA	NA	
TSP Name	FZD01	Civil Lines	Bus Stop-3	30-Jan	10:00	11:00	51.4	6300	1230	29.47	99.86	0.00	1856	636	NA	NA	
TSP Name	FZD01	Civil Lines	Bus Stop-3	30-Jan	11:00	12:00	64.2	6300	1230	35.25	99.79	0.08	2503	748	NA	NA	
TSP Name	FZD01	Civil Lines	Bus Stop-3	30-Jan	12:00	13:00	70.0	6300	1230	36.78	99.74	0.07	2963	870	NA	NA	
TSP Name	FZD01	Civil Lines	Bus Stop-3	30-Jan	13:00	14:00	69.1	6300	1230	41.99	99.90	0.17	2914	876	NA	NA	
TSP Name	FZD01	Civil Lines	Bus Stop-3	30-Jan	14:00	15:00	65.9	6300	1230	40.38	99.89	0.07	2807	850	NA	NA	
TSP Name	FZD01	Civil Lines	Bus Stop-3	30-Jan	15:00	16:00	66.3	6300	1230	43.85	99.75	0.00	2881	849	NA	NA	
TSP Name	FZD01	Civil Lines	Bus Stop-3	30-Jan	16:00	17:00	63.3	6300	1230	40.62	99.88	0.07	2776	796	NA	NA	
TSP Name	FZD01	Civil Lines	Bus Stop-3	30-Jan	17:00	18:00	66.0	6300	1230	40.11	99.87	0.04	2658	784	NA	NA	
TSP Name	FZD01	Civil Lines	Bus Stop-3	30-Jan	18:00	19:00	59.2	6300	1230	34.47	99.96	0.17	2401	693	NA	NA	
TSP Name	FZD01	Civil Lines	Bus Stop-3	30-Jan	19:00	20:00	55.5	6300	1230	30.81	99.86	0.00	1857	605	NA	NA	
TSP Name	FZD01	Civil Lines	Bus Stop-3	30-Jan	20:00	21:00	47.4	6300	1230	29.84	99.74	0.00	1376	520	NA	NA	
TSP Name	FZD01	Civil Lines	Bus Stop-3	30-Jan	21:00	22:00	42.6	6300	1230	32.36	99.59	0.00	1068	491	NA	NA	
TSP Name	FZD01	Civil Lines	Bus Stop-3	30-Jan	22:00	23:00	25.4	6300	1230	33.08	100.00	0.14	730	480	NA	NA	
TSP Name	FZD03	Ayodhya	Mandir Area	30-Jan	9:00	10:00	34.3	6300	2186	17.66	99.45	0.25	789	368	NA	NA	
TSP Name	FZD03	Ayodhya	Mandir Area	30-Jan	10:00	11:00	46.1	6300	2186	18.08	99.08	0.39	1026	378	NA	NA	
TSP Name	FZD03	Ayodhya	Mandir Area	30-Jan	11:00	12:00	51.6	6300	2186	19.04	99.62	0.21	1401	401	NA	NA	
TSP Name	FZD03	Ayodhya	Mandir Area	30-Jan	12:00	13:00	53.1	6300	2186	22.67	99.70	0.07	1513	420	NA	NA	
TSP Name	FZD03	Ayodhya	Mandir Area	30-Jan	13:00	14:00	58.2	6300	2186	27.12	99.76	0.15	1353	419	NA	NA	
TSP Name	FZD03	Ayodhya	Mandir Area	30-Jan	14:00	15:00	55.1	6300	2186	25.48	99.92	0.08	1271	428	NA	NA	
TSP Name	FZD03	Ayodhya	Mandir Area	30-Jan	15:00	16:00	47.2	6300	2186	26.00	99.83	0.16	1283	418	NA	NA	
TSP Name	FZD03	Ayodhya	Mandir Area	30-Jan	16:00	17:00	51.1	6300	2186	25.19	99.92	0.08	1207	433	NA	NA	
TSP Name	FZD03	Ayodhya	Mandir Area	30-Jan	17:00	18:00	53.1	6300	2186	25.06	99.71	0.00	1152	431	NA	NA	
TSP Name	FZD03	Ayodhya	Mandir Area	30-Jan	18:00	19:00	58.2	6300	2186	24.90	99.28	0.08	1269	429	NA	NA	
TSP Name	FZD03	Ayodhya	Mandir Area	30-Jan	19:00	20:00	70.3	6300	2186	27.24	99.68	0.00	1365	448	NA	NA	
TSP Name	FZD03	Ayodhya	Mandir Area	30-Jan	20:00	21:00	58.7	6300	2186	32.05	99.71	0.09	1088	459	NA	NA	
TSP Name	FZD03	Ayodhya	Mandir Area	30-Jan	21:00	22:00	57.7	6300	2186	39.32	99.88	0.00	833	453	NA	NA	
TSP Name	FZD03	Ayodhya	Mandir Area	30-Jan	22:00	23:00	40.9	6300	2186	41.50	99.61	0.18	564	435	NA	NA	
TSP Name	FZD103	RAMGHAT	Mandir Area	30-Jan	9:00	10:00	33.9	6300	1797	21.15	99.25	0.00	911	404	NA	NA	
TSP Name	FZD103	RAMGHAT	Mandir Area	30-Jan	10:00	11:00	38.8	6300	1797	20.22	99.38	0.36	1115	396	NA	NA	
TSP Name	FZD103	RAMGHAT	Mandir Area	30-Jan	11:00	12:00	41.7	6300	1797	21.69	99.82	0.09	1126	394	NA	NA	
TSP Name	FZD103	RAMGHAT	Mandir Area	30-Jan	12:00	13:00	42.9	6300	1797	24.74	99.52	0.16	1246	479	NA	NA	
TSP Name	FZD103	RAMGHAT	Mandir Area	30-Jan	13:00	14:00	44.0	6300	1797	25.56	99.92	0.16	1251	429	NA	NA	
TSP Name	FZD103	RAMGHAT	Mandir Area	30-Jan	14:00	15:00	42.1	6300	1797	24.62	99.60	0.00	1213	420	NA	NA	
TSP Name	FZD103	RAMGHAT	Mandir Area	30-Jan	15:00	16:00	40.6	6300	1797	26.48	99.82	0.36	1102	428	NA	NA	
TSP Name	FZD103	RAMGHAT	Mandir Area	30-Jan	16:00	17:00	43.4	6300	1797	24.76	99.44	0.24	1276	445	NA	NA	
TSP Name	FZD103	RAMGHAT	Mandir Area	30-Jan	17:00	18:00	42.7	6300	1797	24.60	99.58	0.00	1186	458	NA	NA	
TSP Name	FZD103	RAMGHAT	Mandir Area	30-Jan	18:00	19:00	54.1	6300	1797	26.15	99.92	0.23	1289	428	NA	NA	

Annexure-V

Analysis of Traffic Monitoring Report

Maximum No. of Users Latched in an Hour		
TSPs	**Time Duration**	**Max. No. of Users Latched**
TSP1	3:00 PM to 4:00 PM	79883
TSP2	6:00 PM to 7:00 PM	22445
TSP3	9:00 PM to 10:00 PM	83704
TSP4	6:00 PM to 7:00 PM	22526
	Total	**208558**

TSP Name	Site ID (sector-wise)	Site location	Area being served	Start Time	End Time	Data Capacity (in GB)	Payload (in GB)	% Loading
TSP Name	TCOW04	Cantt Board Stadium Sadar Bazar	Rest Ayodhya	22:00	23:00	1776	92.85	52.28%
TSP Name	TCOW04	Cantt Board Stadium Sadar Bazar	Rest Ayodhya	21:00	22:00	1776	90.81	51.13%
TSP Name	FZD63	Kaushalpuri Colony	Rest Ayodhya	22:00	23:00	1594	78.01	48.94%
TSP Name	FZD63	Kaushalpuri Colony	Rest Ayodhya	21:00	22:00	1594	74.83	46.94%

Highest % Loading at Important Locations								
TSP Name	**Site ID (sector-wise)**	**Site location**	**Area being served**	**Start Time**	**End Time**	**Data Capacity (in GB)**	**Payload (in GB)**	**% Loading**
TSP Name	FZD57	CHANDPURHARBANSHPUR	Shane Avadh	22:00	23:00	1607	64.22	39.96%
TSP Name	UFZD53	Purani Sabji Mandi	Main Ayodhya	22:00	23:00	1051	40.32	38.36%
TSP Name	FZD122	Dabhasemar	Hotel Krinoscco	21:00	22:00	1594	57.51	36.08%
TSP Name	FZD01	Civil Lines	Bus Stop-3	15:00	16:00	1230	43.85	35.65%
TSP Name	UFZD53	Purani Sabji Mandi	Main Ayodhya	21:00	22:00	1051	37.04	35.24%
TSP Name	FZD57	CHANDPURHARBANSHPUR	Shane Avadh	21:00	22:00	1607	56.42	35.11%
TSP Name	FZD42	Railway Chauraha	Outside Ayodhya Cannt.	16:00	17:00	1230	42.62	34.65%

TSP Name	BTS NAME	Site location	Area being served	Start Time	End Time	Call drop rate
TSP Name	Z2FZFZBP7FAI003B	RSU AYOYDHYA_UPE-0557	Ram Mandir	09:00	10:00	2.04%
TSP Name	Z2FZFZBP7FAI026C	Brahmbaba_UPE-2966	Airport	11:00	12:00	2.04%
TSP Name	Z2FZFZBP7FAI010C	Nirala Nagar_UPE-2959	Ayodhya City	10:00	11:00	2.04%
TSP Name	Z2FZFZBP8FZBBNGB	Boniganj_UPE-10034	Rampath	11:00	12:00	2.04%

F. No. 4-1/2023-QoS
Government of India
Ministry of Communications
Department of Telecommunications
(QoS Division)

6th Floor, MTNL Building,
Old Minto Road, New Delhi-110002
Dated: 22nd January, 2025

Subject: Clarification regarding Para C(3) "Emergency Preparedness" of Standard Operating Procedure (SOP) dated 10.07.2024 for maintaining Quality of Telecom Services during significant Social/Cultural/Religious/Commercial/ International Events etc.

This is with reference to the Standard Operating Procedure (SOP) for maintaining Quality of Telecom Services during significant Social/Cultural/Religious/Commercial/International Events etc. issued by QoS Division, DoT HQ on 10.07.2024. Regarding Para C(3) "*Emergency Preparedness*" of the said SOP, the undersigned is directed to provide the following clarifications:

"As per Clause No. 3 of (C) of SOP dated 10.07.2024 for maintaining Quality of Telecom Services during Significant Social/Cultural/ Religious/Commercial/ International Events etc., the position of implementation of ICR is as given below -

- **Action by LSA Head** - LSA(s) should ensure that ICR testing is conducted by all the TSPs beforehand.
- **Action by DM Division DoT HQ** - Necessary instructions for enabling Intra-Circle Roaming (ICR) among Telecom Service Providers may be issued by Disaster Management Division of DoT HQ for handling emergency situations. In such events, the decision will be taken as the situation warrants. When necessary, the LSA may send a request to the DM Division, DoT HQ with a copy to the QoS Division/DGT HQ, for issuing the necessary instructions.

Based on above SOP, the following steps may be taken by different units –

- **Testing of ICR by LSA** - Based on assessment of estimated footfall in Social/Cultural/ Religious/Commercial/International Events etc. and network assessment of TSPs along with inputs from State Government, LSA may ensure that ICR testing is conducted by all the TSPs in advance, only if it is anticipated by LSA that emergency situation may arise due to disruption of the telecom network during the event.

- **Issue of instructions by DM Division, DoT HQ** - During the event itself, if it is necessary to implement ICR (in case of disruption of the telecom network), LSA may send a request to the DM Division, DoT HQ, for issuing the necessary instructions. Necessary instructions for enabling Intra-Circle Roaming (ICR) among Telecom Service Providers may be issued by Disaster Management Division of DoT HQ for handling emergency situations arising out of disruption to the telecom network so that in case of any major disruption to the telecom network due to a disaster, the ICR may be implemented immediately.

Key Guidelines for ICR Implementation:

- **Primary Purpose**: ICR is primarily intended for managing telecom services during network outages. This ensures that the mechanism is used to maintain network stability during critical situations such as network failures or emergencies.
- **Congestion Issues**: ICR is not intended for addressing network congestion or anticipated congestion. It is a tool for emergency situations related to outages, not for routine network load management."

This is issued with the approval of Member (T), DoT.

22/1/25

ADET (QoS)
O/o DDG (QoS), DoT HQ

To,

1) DGT HQ, DoT / LSA Heads
2) DDG (Security), DoT HQ
3) DDG (NBM), O/o DGT HQ
4) DDG (AS), DoT HQ
5) DDG (DM), DoT HQ
6) All TSPs (M/s Bharti Airtel Limited, M/s BSNL/MTNL, M/s Reliance Jio Infocomm Limited, M/s Vodafone Idea Limited)

No. 22016/05/2019-CIS-IV-Pt.
Government of India
Ministry of Home Affairs
(CIS Division/Monitoring Unit)

Hall No. 350222
Kartavya Bhawan-03, New Delhi
Dated: 8th December, 2025

OFFICE MEMORANDUM

Subject: NOC for use of BSNL-GSPS Satellite Phone connection in Barred areas by Department of Telecommunications (DoT) –reg:

The undersigned is directed to refer to DoT's letter No. 60-1/satellite phones/2022-SAT dated 19.05.2025 on the subject cited above and to convey the DoT for use of **one** GSPS Satellite phone of BSNL by each State coordination offices of DoT in Manipur, Mizoram, Tripura, Nagaland and Arunachal Pradesh subject to the following conditions:

i BSNL should ensure that the NOC is for use of the each of the satellite phone in the respective barred area of Manipur, Mizoram, Tripura, Nagaland or Arunachal Pradesh only and hence cannot be used in other barred areas. The same may also be ensured by DoT through sample check/ audits.

ii Sensitive matter must not be communicated through the satellite phone. Also, the phone must not be used at sensitive locations or places where strategic assets are deployed as the communication content as well as location of satellite phone might be intercepted by adversaries and foreign intelligence/ security agencies.

iii DoT must take all measures to safeguard the satellite phone and SIM card from unauthorized access. Theft or loss should be immediately brought to the notice of BSNL, under intimation to MHA, to deactivate the GSPS phone.

iv The use of satellite phones should be thoroughly monitored through proper maintenance of a log book. Besides it should be kept under the custody of a senior officer with periodic inspections.

v DoT should adopt suitable security measures to avoid misuse of the handsets.

vi There should be thorough Character & Antecedents verifications and supervision of the user by DoT with the exact demarcation of the area where the satellite phone will be used.

2. This issue with the approval of Competent Authority.

(Mahendra Vikram Singh)
Under Secretary to the Govt. of India
Tele: 011-24010221

Secretary
Department of Telecommunications
Sanchar Bhawan, Ashoka Road
New Delhi- 110001

Copy to:

Director (Satellite), Department of Telecommunications
Sanchar Bhawan, Ashoka Road
New Delhi- 110001

CAP- Integrated Alert System (सचेत)

for Disaster Alert Dissemination to Citizens of Targeted Area

Presented By :

C-DOT

Thursday, 26 May 2022

Agenda/Contents

- Overview of CAP Integrated Alert System सचेत
- Purpose of CAP सचेत
- Working of CAP Integrated Alert System सचेत
- Milestones for CAP सचेत Phase 1
- Usage of CAP सचेत by SDMA
- Possible use case scenarios
- Deliverables of CAP सचेत
- Role of SDMA
- Logistics Requirement
- Support and Escalation Mechanism
- Demo of CAP Integrated Alert System सचेत

Centre for Development of Telematics (C-DOT)

- C-DOT is an autonomous Telecom R&D centre of DOT, Govt. of India established in 1984
- Known for indigenous design, development and production of telecom technologies for more than 3 decades
- Product portfolios includes Optical Communication, Quantum based communication, 4G-LTE, 5G, AI, WiFi, M2M/IOT, NMS, etc.

CONTACT US

C-DOT Campus, Mehrauli, New Delhi - 110030, : +91-11-26598262/26802856, 8800092683

C-DOT Campus, Electronic City Phase 1, Bengaluru-560100, : +91-80-28520050

Gour Niketan(GF), P108, Lake Town(A Block), Kolkata-700089, : +91-33-25210674

Email : cdotweb@cdot.in, : +91-11-26803477 (International Calls Only)

Overview of CAP-Integrated Alert System सचेत

- Conceptualized & Funded by **NDMA** based on Globally accepted **Common Alerting Protocol (CAP)** standard.
- Facilitates **MHA** and all **State & UT Govts.** to disseminate **location- based Disaster Alerts** to Citizens in **vernacular languages** in **near real time** for enabling people to act timely and reduce the possibility of harm, loss or risk.
- Integrates Alert Generating Agencies like **IMD, CWC, INCOIS, DGRE (**formerly **SASE)** and **FSI** on a **National level** common platform.
- Field Proven Technology, Operational for more than **1 Year** in **Tamil Nadu** as Pilot Project.
- **MoU signed between NDMA and CDOT on 23rd Aug,2021** for Pan-India implementation of CAP Project.
- Selected by Hon'ble **Minister of Communications** as a part of **Azadi ka Amrit Mahotsav India @75**

Phase – I Plan

- **Location based SMS** via Airtel, BSNL, MTNL, Reliance Jio, Vodafone-Idea
- Geo-Targeted Alert Notification in **Internet Browsers**
- Common **Website** for Citizen for any kind of Disaster related Information
- Location specific **RSS Feed** for **news media** for automatic disaster bulletins & updates
- Integrated **Group SMS** to notify Disaster Managers in one go.
- **Mobile App** for Citizens localized alerts
- **GAGAN & NavIC** – Satellite based messaging for **Fisherman, Avalanche warning** or Alert citizen where no traditional network available

Plan for Subsequent Phases

- **Cell Broadcast** in Mobile Phones
- **Radio broadcast** in specific region (by All India Radio & private FM Channels)
- **TV Broadcast** in Specific region (by DTH & Local TV Operators)
- Integration of **Coastal Sirens**
- Alerting through **Indian Railways Passenger Information announcement System.**

Project Finance & Operational Cost

- Project's cost will be **funded by NDMA** for **10 years** (under 'Preparedness and Capacity Building' funding window of NDRF).
- **No financial implication** on the **States & UTs**
- As per **SoP 2020** circulated by **DoT**, TSPs shall send SMS and CB in affected areas **free of cost** during **disaster period** based upon instruction from concerned Nodal authority
- However, the cost of dissemination of **non-disaster messages** would be borne by the States/UTs, as fixed by TRAI/DoT (TRAI OHD 19th Jan 2022)

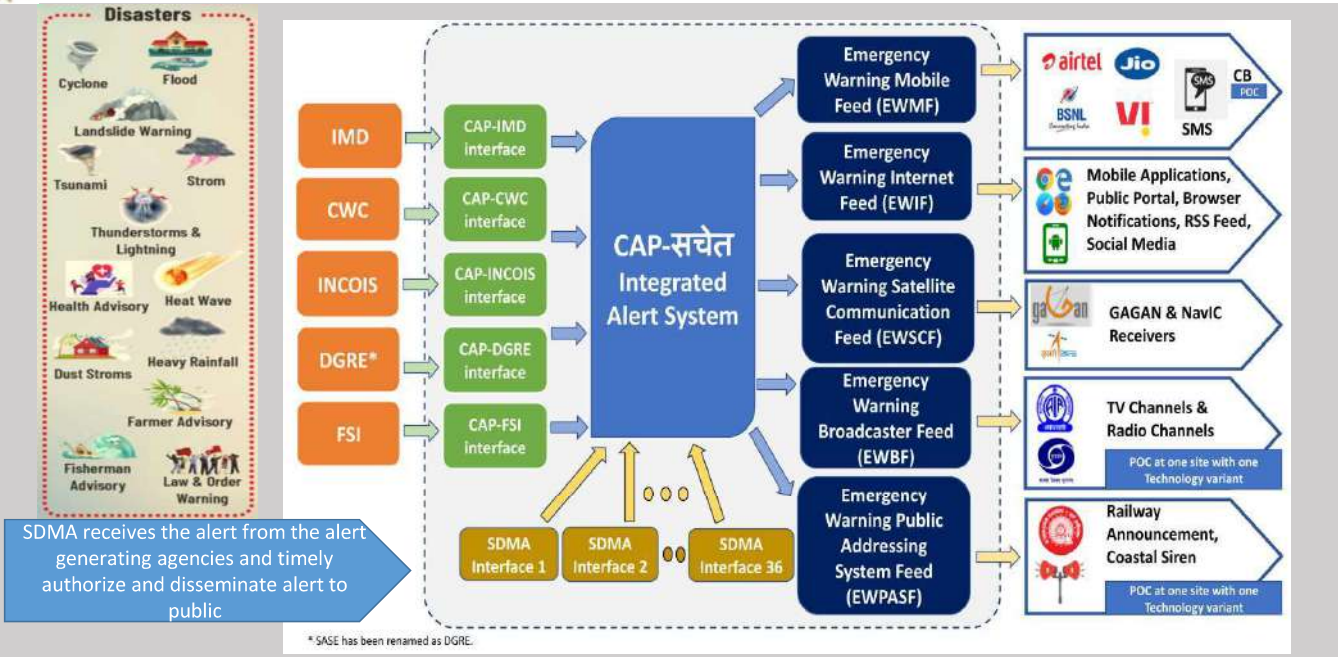
Purpose of CAP-Integrated Alert System सचेत

- Globally accepted **ITU-T x.1303 CAP standard based system** for disaster alert information exchange
- **Integration of all Agencies under one umbrella** saving the precious time used in communication.
- Ensures **Last mile Reachability to the Citizens** using all possible Dissemination methods for **Any Hazard, To anyone.**
- Also covers **seasonal or tourist population** of the area to be affected.
- **Geo-targeted warning**, increase **effectiveness** and **sanctity** as Information reaching only to the **targeted public**.
- **Vernacular languages Support** to ensure **understandability** among local population.
- **Impact based Alerting** to send **different alert message** in **different parts** of the affected area
- Using **Satellite Communication** to cover area where **terrestrial communication is not available or is disrupted.**
- **Multilevel Secure authentication** and **Trusted Device Mechanism** to prevent any unauthorized access.
- Supports different **voice notification tone** to cater to **visually impaired persons** and **for addressing odd hours**.

Increase Effectiveness

ENCRYPTION

Working of CAP-Integrated Alert System सचेत



Milestones and Timeline of CAP SACHET Pan India Phase I

T0 + 6 Months — Feb 2022

- 6 SDMAs (Total - 6/36)
- 4 IMD centres (Total- 4/29)
- CWC HQ
- 5 Telecom Operator for SMS.
- Draft SOP circulation.
- Project Management.
- Operation Dashboard for NDMA and 6 SDMAs.

T0 + 12 Months — Aug 2022

- 12 SDMA (Total – 18/36)
- 8 IMD centres (Total 12/29)
- INCOIS
- Test bed establishment.
- RSS feed, Internet browser notifications.
- PCR
- Mobile Application
- Operation Dashboard for 12 SDMAs.
- PoC for AIR and Railways

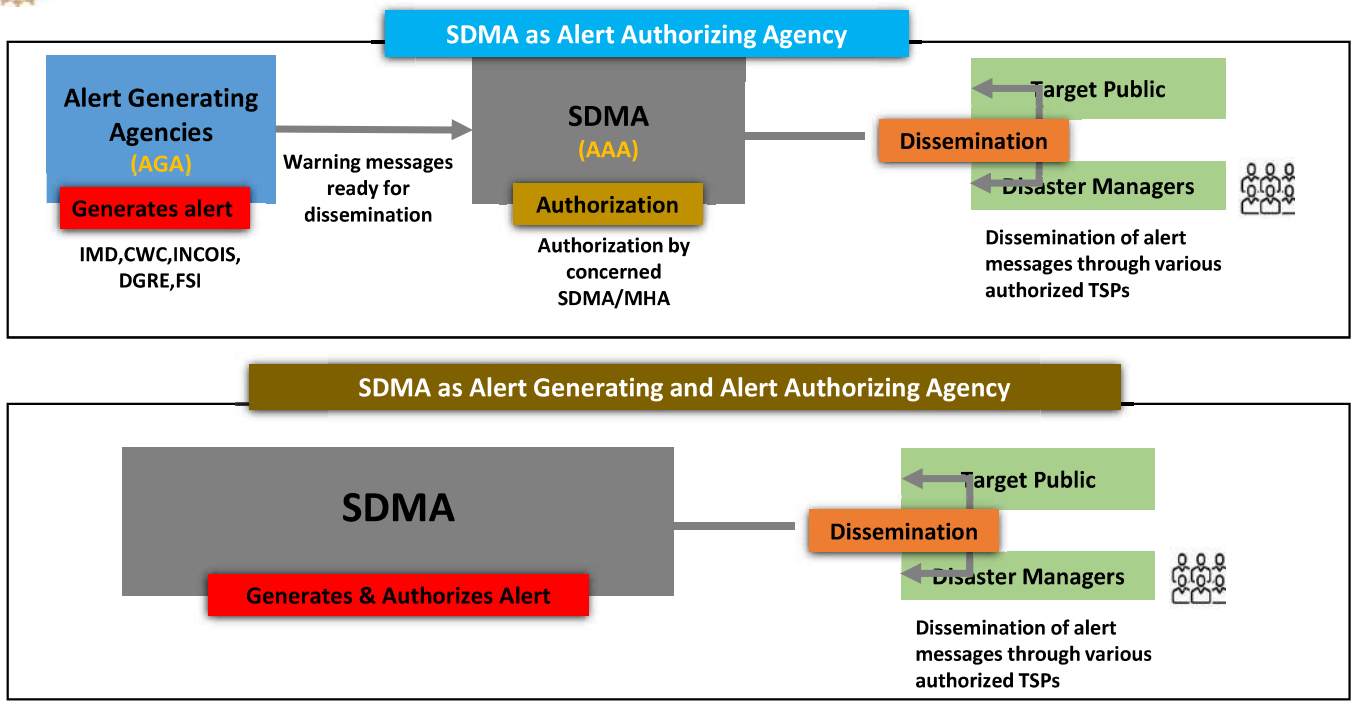
T0 + 18 Months — Feb 2023

- 18 SDMA (36/36)
- 17 IMD (29/29)*
- DGRE and FSI.
- GAGAN and NavIC messaging.
- PoC for TV service providers (DTH/Cable Operator), Coastal Siren.
- PoC for Cell broadcast.
- Google alert.

**IMD Centres have been increased to 29 from 26*

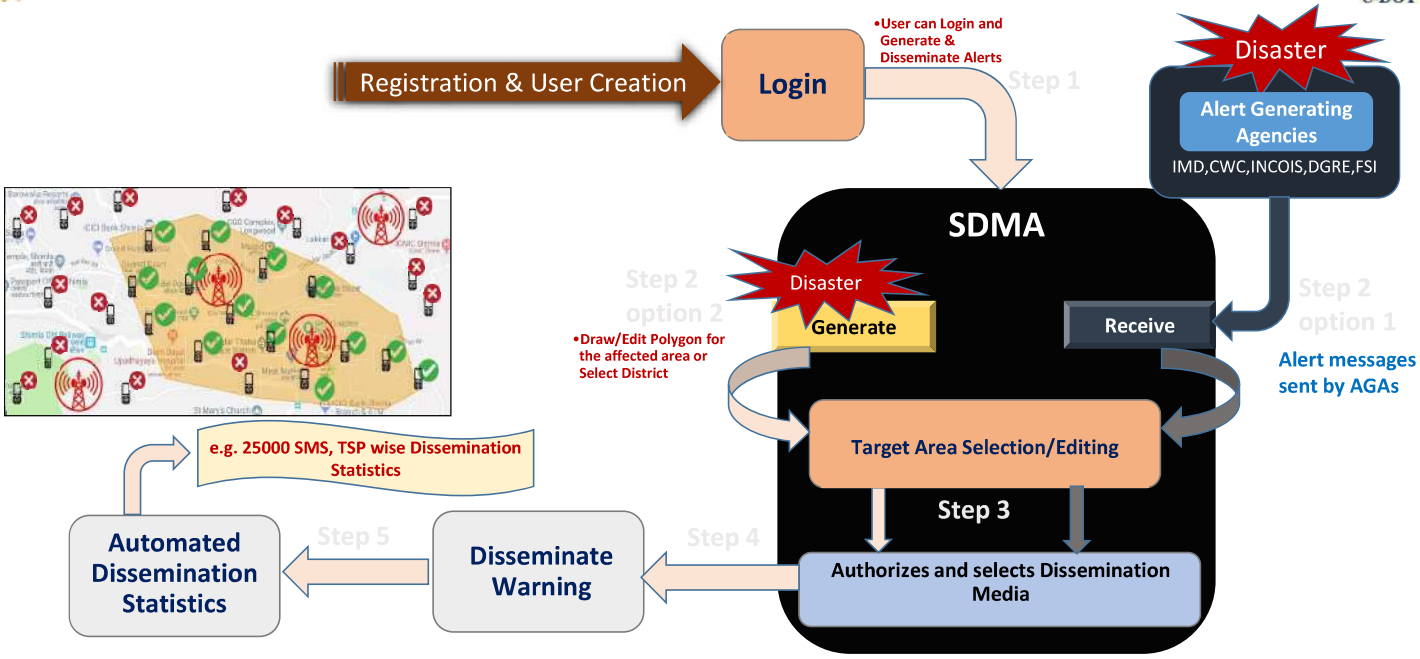
7

Usage of CAP-Integrated Alert System सचेत



8

Usage of CAP-Integrated Alert System सचेत



CAP SMS Dissemination Process & SMS Receiving Criterion सचेत

CAP SMS Dissemination Process

- Receive CAP XML from CAP platform
- Parse of CAP XML for desired area / polygon and Message body
- Fetch BTS IDs within the selected area / polygon in sequential order
- Store polygon wise BTS cell site count
- Fetch subscribers mobile number latched to each BTS cell site in sequential manner
- Store polygon wise subscriber count
- Push subscriber list and message to SMSC for dissemination
- Get statistics through API

Condition for Receiving SMS

- Mobile Handset is turned ON
- Mobile Handset having active network coverage from service provider.
- User having active subscription plan (Prepaid/Postpaid)
- Mobile Handset latched to valid BTS in selected polygon area
- Mobile Handset not in Dark Spot of Mobile Network

Possible Use case scenarios of CAP-Integrated Alert System सचेत

Pre-disaster	During the Disaster	Post-Disaster
• Flood Alerts • Cyclone alerts • Avalanche alerts • Tsunami alerts • Lightning/ Thunderstorm related advisories • Fishermen Advisories • Helpline nos. • Shelter/ Evacuation Instruction	• Alert updates • Information about safe places • Do's and Don'ts • Helpline nos.	• Relief related info • Landslides location • Earthquake rescue and relief ops • Helpline nos.

- **Location Based SMS** to disseminate Geo targeted Alerts.
- Location specific Regular weather warnings and advisories through **Mobile App**.
- Advisories to fishermen in sea where other type of communication is not possible through **NavIC and GAGAN**.
- **Evacuation plan and safe places** related information sharing over all media.

Deliverables for the States & UTs

Dashboard for SEOCs for Disaster Alert Dissemination related monitoring and control purpose.

SDMA Login for dissemination of Geo-targeted Disaster Alerts received alerts from **IMD,CWC, INCOIS, DGRE, FSI** etc.

Mobile App for citizens to receive Geo-targeted Disaster Alerts and to receive alerts from **GAGAN** and **NavIC** messaging system

Browser Plug-in for internet users to receive alerts notification in their Desktop/ Laptop

- ❑ **User Training for the Users of CAP सचेत**
- ❑ **Necessary technical support & assistance.**
- ❑ **24x7 support on Phone and Email.**
- ❑ **Mock Drills**

Training and Access Schedule

Training and Access provided to Disaster Management Department, Maharashtra:

- Orientation conducted on 6th January 2022 (1 Participant)
- Hands-on provided on 11th May 2022 (15 Participants)
- Access to Live Platform provided on 18th May 2022
- On Site Training & Mock drill conducted on 26-27 May 2022

Training and Access provided to Meteorological Centre Mumbai:

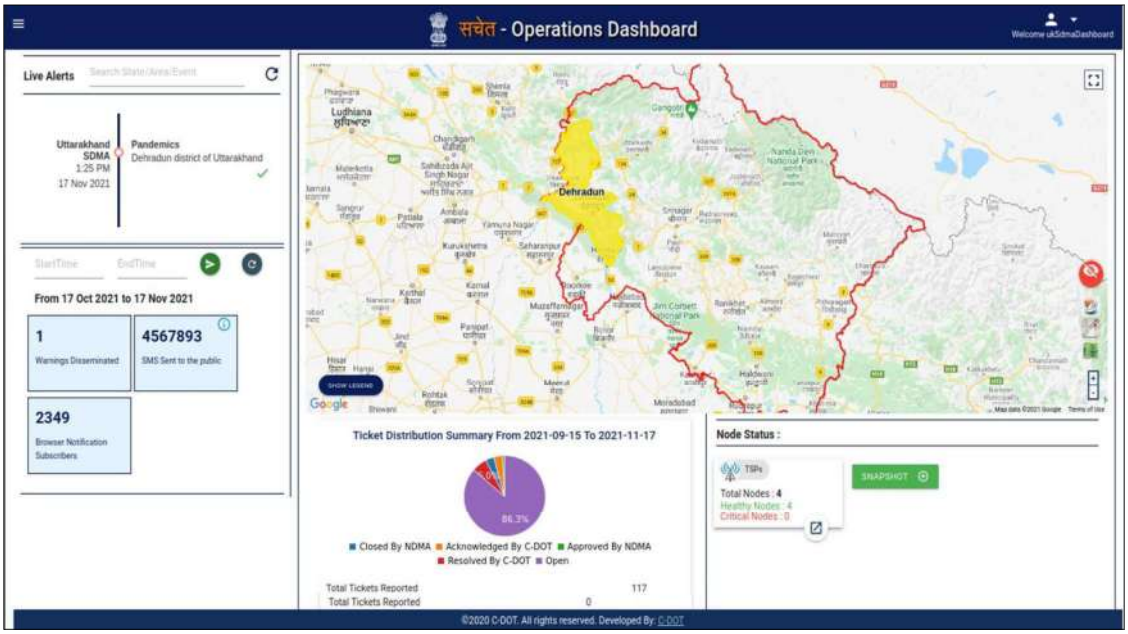
- Orientation & Hands-on conducted on 9th December 2021
- Access to Live Platform provided on 4th February 2022 for 3 Officials

Training and Access provided to Meteorological Centre Nagpur:

- Orientation & Hands-on conducted on 15th December 2021
- Access to Live Platform provided on 5th February 2022 for 5 Officials

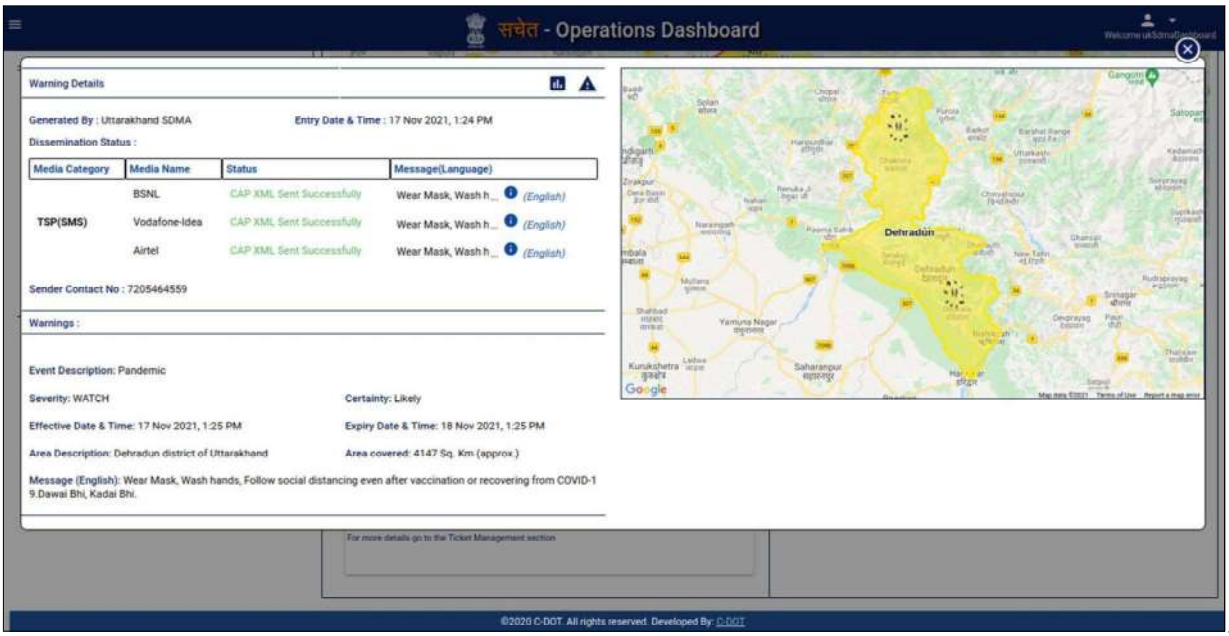
Deliverables for the States & UTs

Dashboard for SEOCs for Disaster Alert Dissemination related monitoring and control purpose.



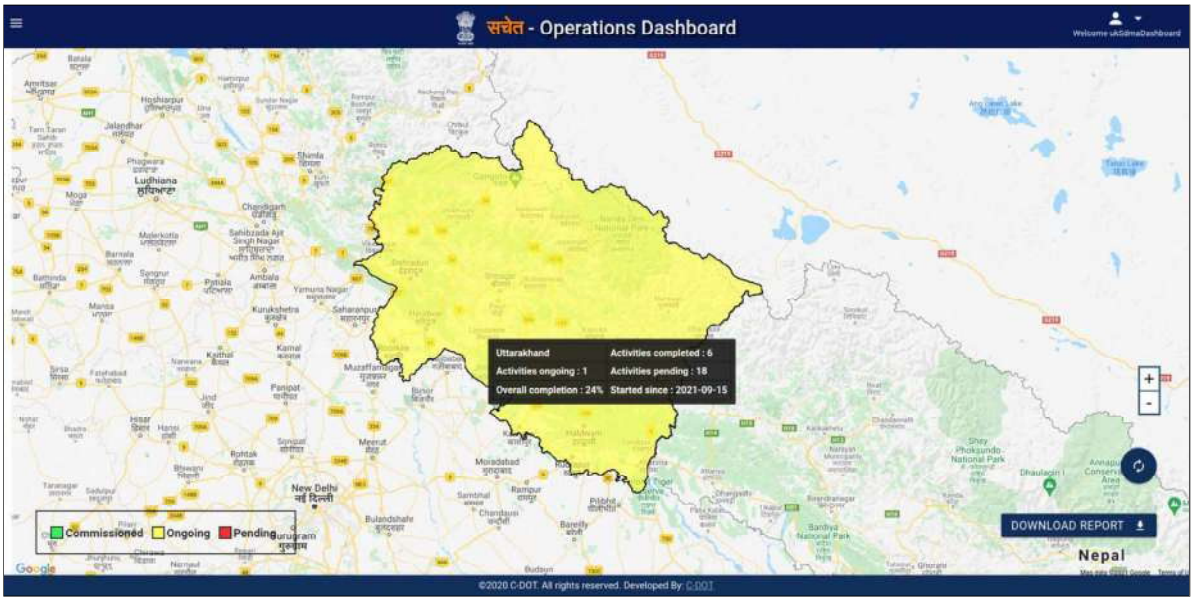
Deliverables for the States & UTs

Dashboard for SEOCs for Disaster Alert Dissemination related monitoring and control purpose.



Deliverables for the States & UTs

Project Management Dashboard for SEOCs



Role of SDMA

- Extend full support during implementation, testing and operation phase and provide feedback, utilization report of platform to C-DOT/NDMA when required.
- Providing Inputs and Feedback on Draft SOP (Standard Operating Procedures) circulated by NDMA
- Receive the alert from Alert Generating Agencies, timely authorize and disseminate to citizens/disaster management authorities using CAP System after moderation of area and message over dissemination media as per SOP.
- For Group SMS, SDMAs will provide existing SMS Gateway API to C-DOT for integration with CAP System.
- Sensitization of various modes of CAP System for receiving Location Specific Disaster Alerts among citizens for public awareness.

Logistics Requirements (For SDMA)

System Requirements:

1. Desktop PC or Laptop in working condition having latest version of Google chrome/Mozilla Firefox browser with uninterrupted power supply and minimum 10 Mbps internet connectivity to access the CAP Integrated Alert System (Sachet Platform).

2. Mobile with SMS accessibility for One Time Password (OTP) authentication.

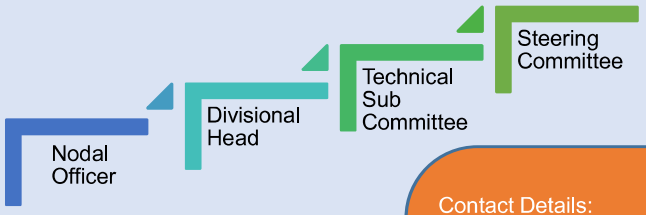
User Requirements:

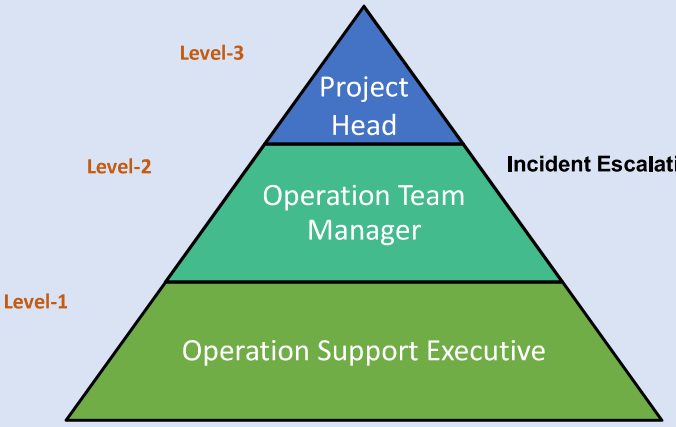
1. Officials responsible for feeding the alert information in the Sachet Platform must be having basic knowledge of working on a PC or Laptop with Internet browsing to access the CAP Integrated Alert System.

2. Should have a Mobile phone with SMS and email accessibility for receiving One Time Password (OTP) authentication.

Support and Escalation Mechanism

Escalation process for Project Implementation





Incident Escalation matrix in CDOT

Contact Details:
01126598547/ 8368654258
Email: cap.support@cdot.in

Operations Support Executive (Level 1)
Ph: 9289044996

Shri. Prashant Bansal (Level 2)
Ph: 9650627675

Shri. Sabyasachi Majumdar (Level 3)
Ph. 9818580435

In the event that target resolution times are not being met or an incident is of a particularly sensitive nature, these escalation levels will apply.

Demo of CAP-Integrated Alert System सचेत

Login Page

SDMA Dashboard

- Home
- Send Warning
- All Warnings
- Group SMS
- Statistics
- Templates
- Help

1 Home

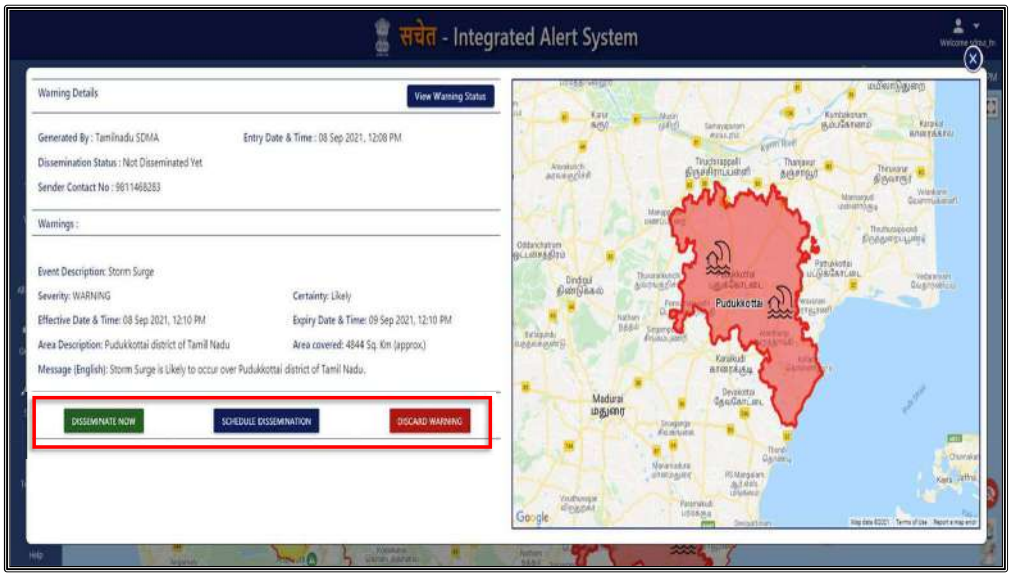
The **Home Page** shows all the Live Warnings present i.e., the warnings which are not expired received from Alert Generating Agencies like IMD, CWC, State Authorities or generated by itself.

SDMA can take action on warnings which are not disseminated

1a Dissemination by SDMA

Depending upon the criticality of the Warning, following actions can be taken by SDMA

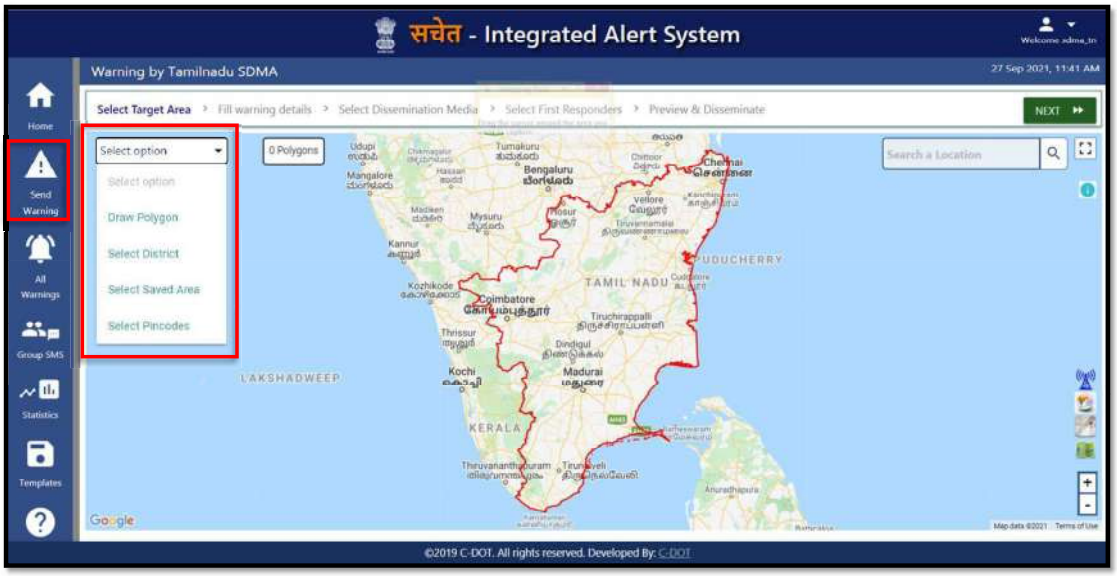
- ❑ Disseminate Now
- ❑ Schedule Dissemination
- ❑ Discard Warning



2 Send Warning

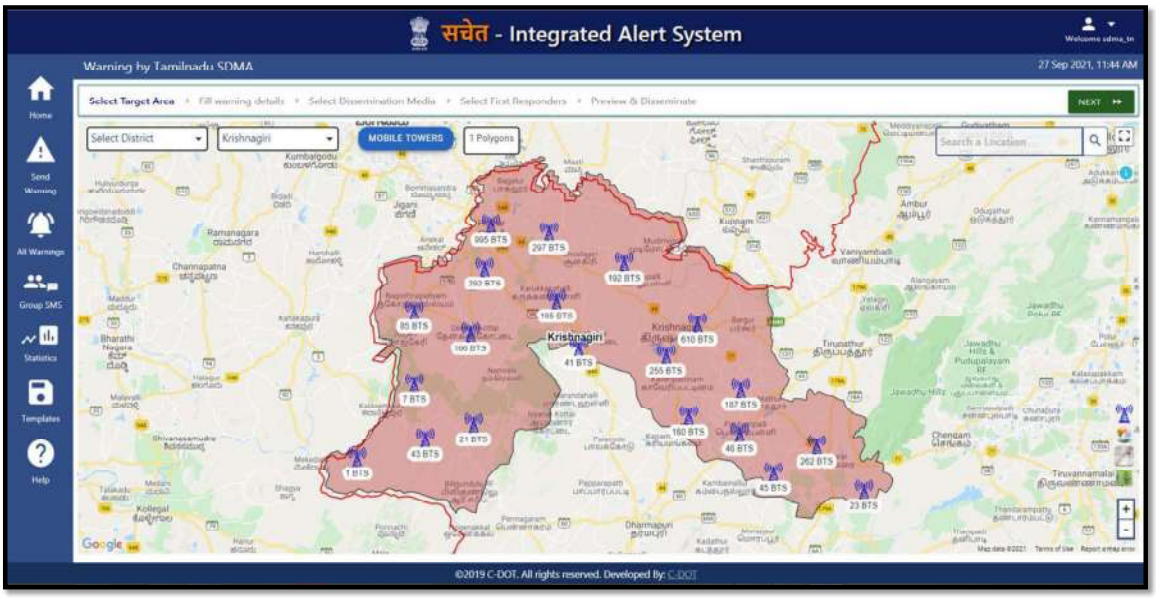
SDMA can select the target area for dissemination from the options given in the drop down window

- ❑ **Draw Polygon**
- ❑ **Select District**
- ❑ **Select Saved Area**
- ❑ **Select Pincodes**



2a Select Target Area

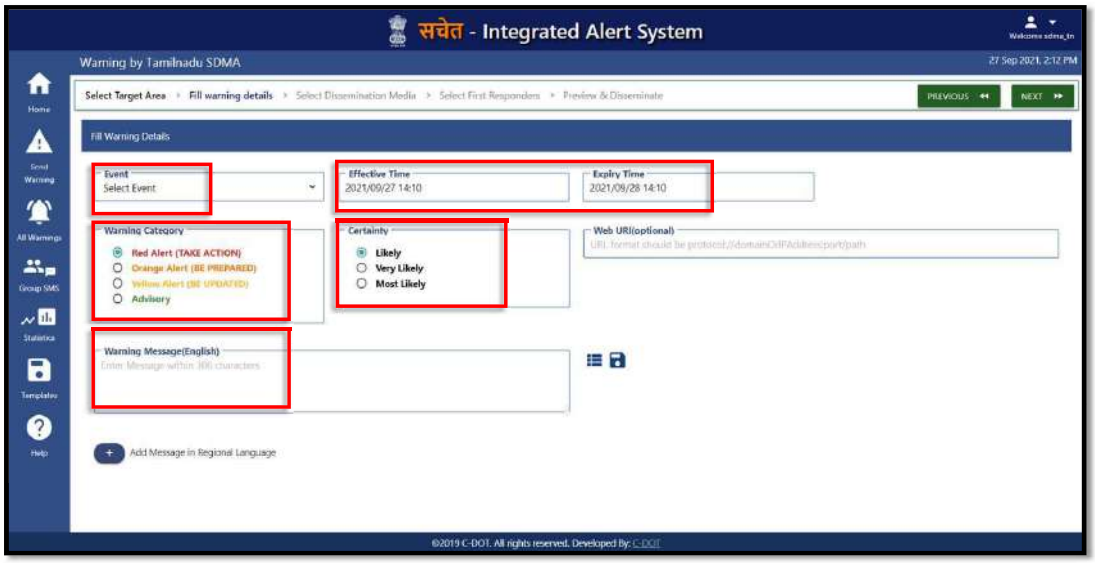
Location based **mobile towers** are displayed in the selected area



2b Fill Warning Details

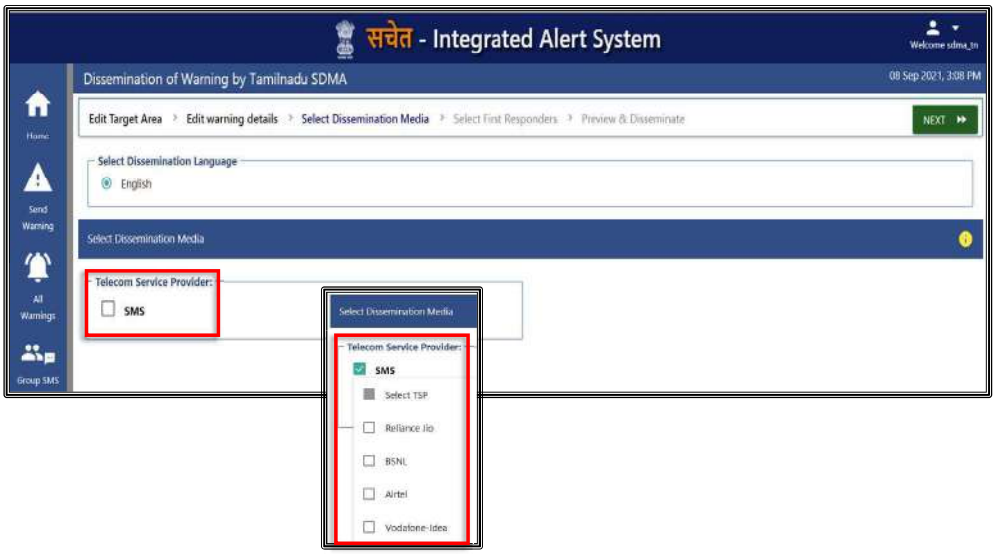
Fill Warning Details

- ❑ **Event**
- ❑ **Effective Time and Expiry Time.**
- ❑ **Warning Category**
- ❑ **Message (English)**
- ❑ **Message (Regional Language) -** ***Optional***



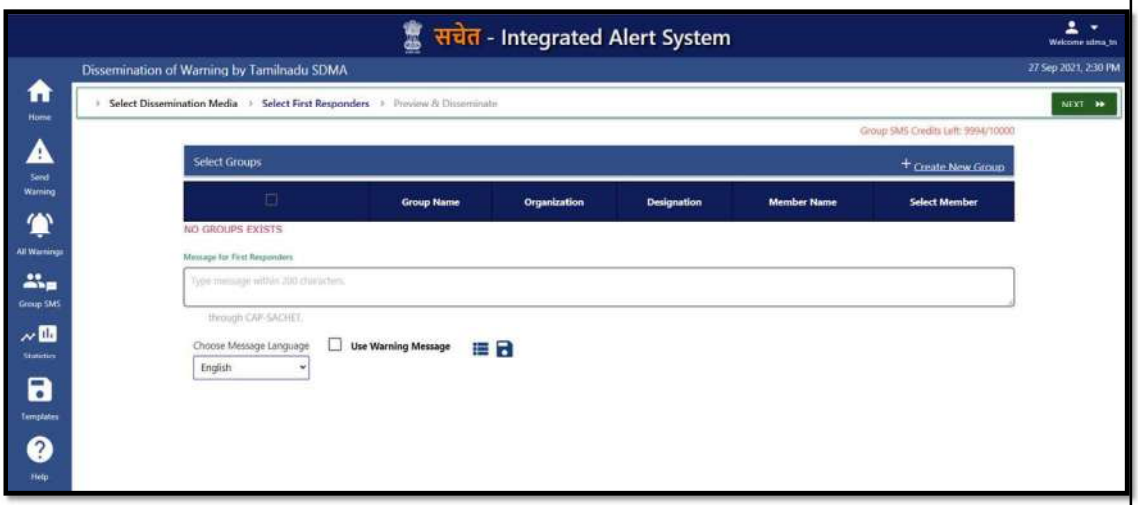
2c Select Dissemination Media

Select the **Telecom Service Providers (TSPs)** for the Dissemination of Warning Message



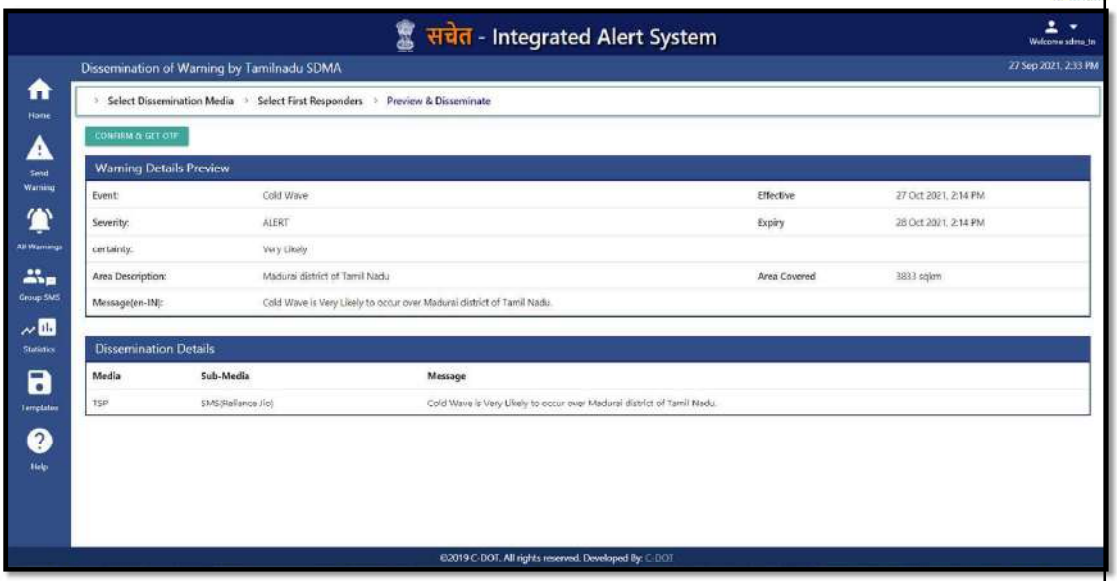
2c Select First Responders

Select **the First Responders** and Write **Message for the First Responders**



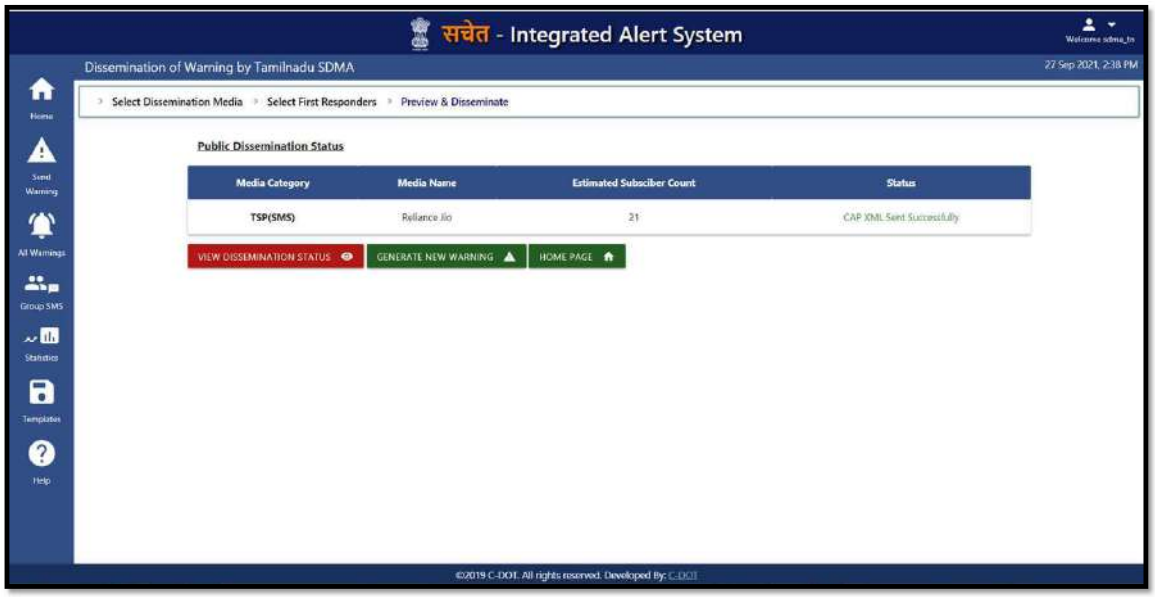
2d Preview & Disseminate

Verify the Warning and Dissemination Details Confirm and Authenticate via **OTP**



2e Public Dissemination Status

View Dissemination Status



31

3 All Warnings

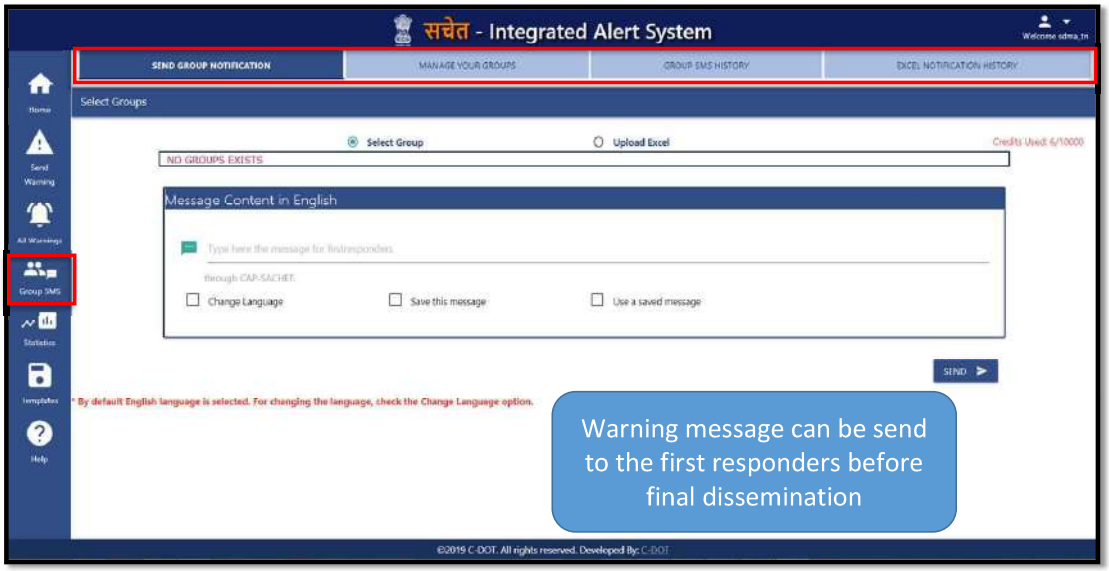
The **All-Warnings** Page shows the list of the warnings which have been generated successfully, saved, discarded and expired.

32

4 Group SMS

User can send **Group SMS** to the First Responders or higher authorities, Manage the Groups and Members in the Group.

User can also check the history of the group messages sent by SDMA to the First Responders.



5 Statistics

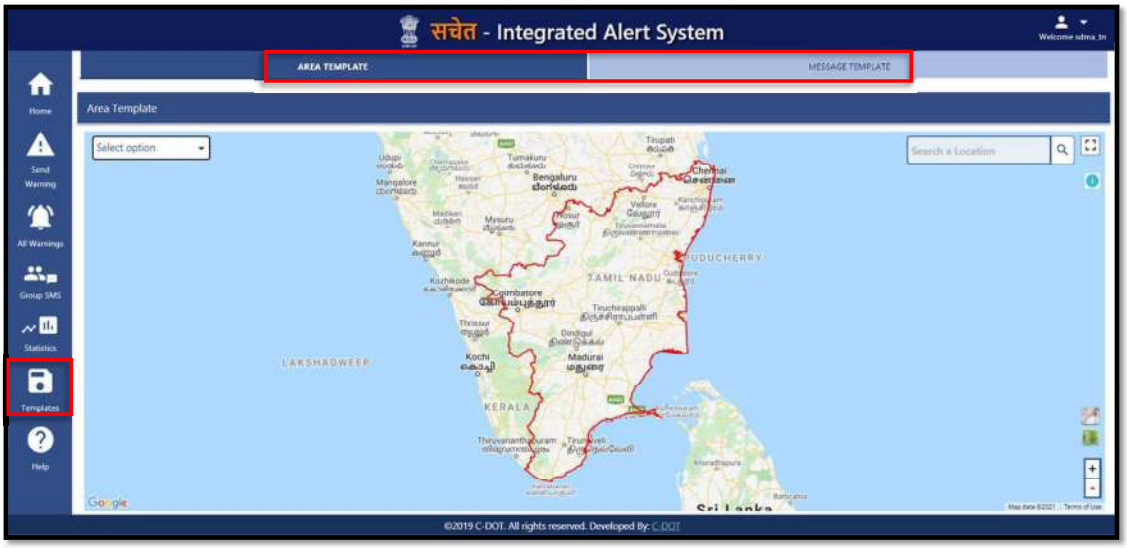
The **Statistics Page** shows the list of the Warnings Received and the Warnings Disseminated by SDMA.

User can choose the particular time period to check statistics of warnings and also use the status filter.

6 Templates

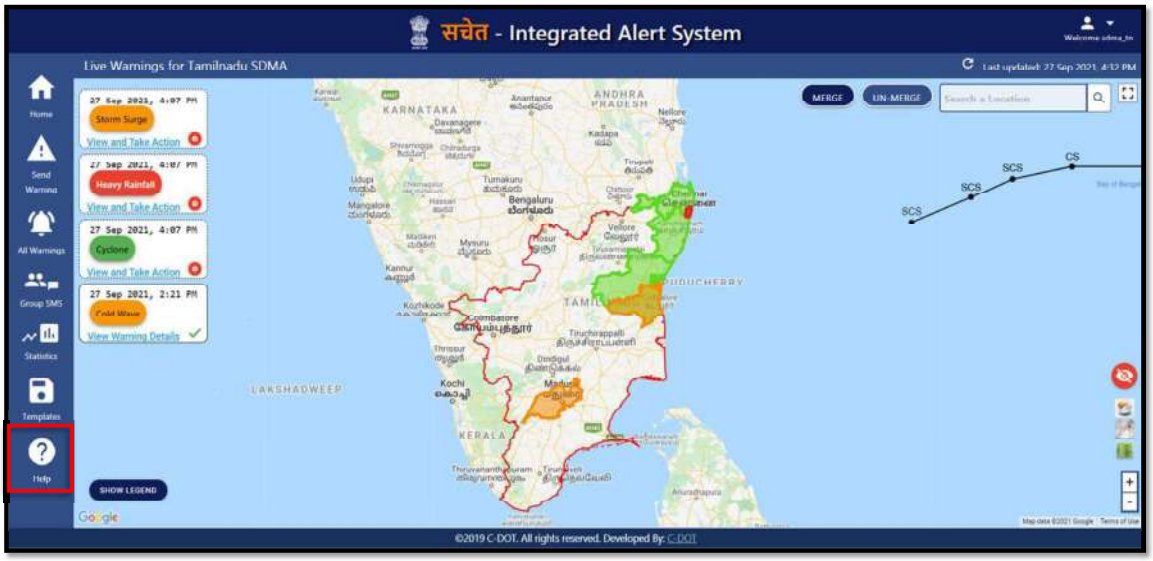
Templates are used to create and save the warning messages to be used later anytime.

- ❑ **Area Template**
- ❑ **Message Template**



7 Help

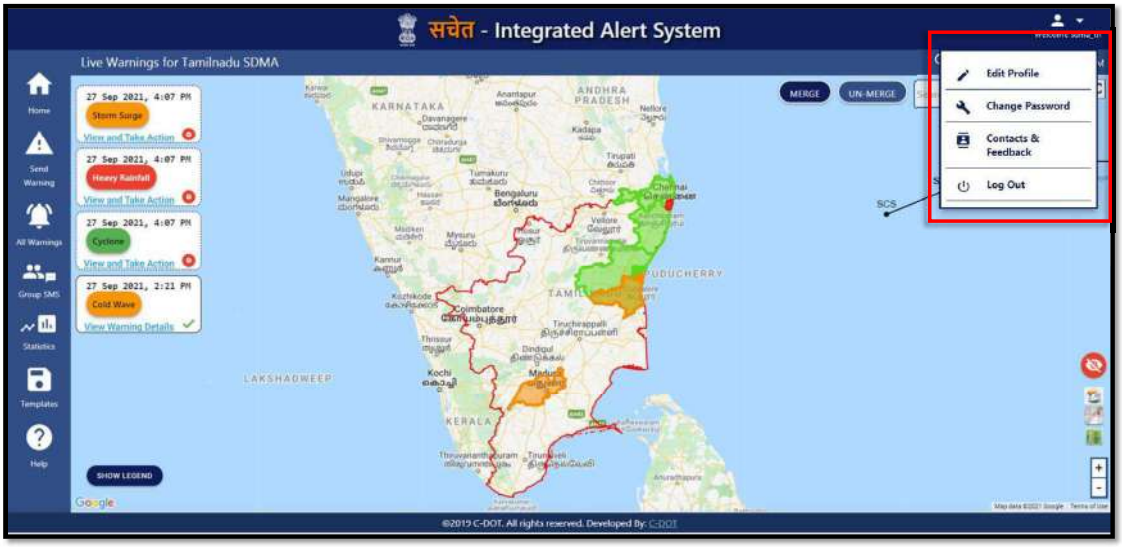
The **Help Page** contains **the Step by Step instructions** for working on the platform.



User Management

User Management allows account owners to manage their accounts.

- ❑ **Edit Profile**
- ❑ **Change Password**
- ❑ **Contacts & Feedback**
- ❑ **Log Out**



Questions

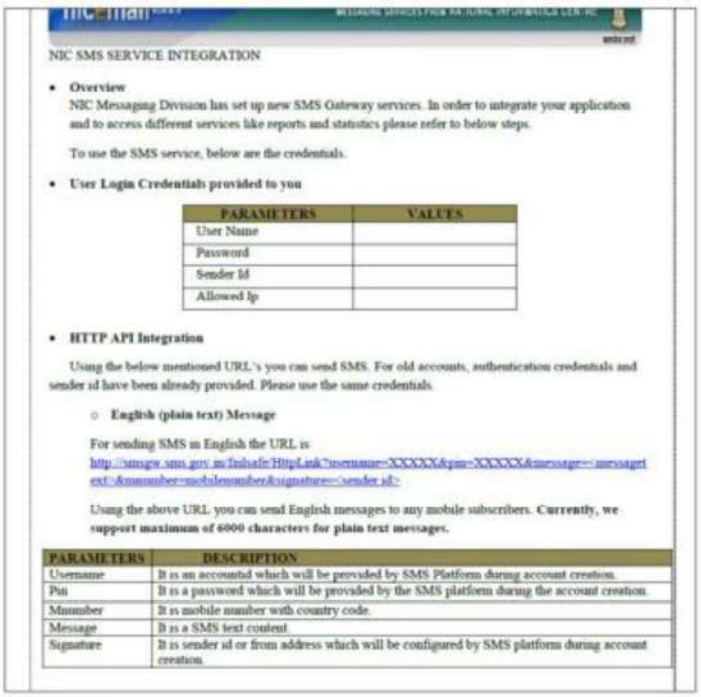
THANKS!

39

SMS Gateway API details for integration and sending bulk SMS

1. To use Group SMS feature for **better coordination** between **disaster managers, States/UTs** needs to provide **SMS Gateway API** details for integration with CAP Integrated Alert System.
   - API endpoint URL
   - API Key and/or API access credentials
   - Details of parameters to store list of recipients mobile numbers, SMS content in English and Regional languages
   - Share the technical specification document of API, if any.
2. **SMS Headers** that will be used for sending Group Messages.
3. The access of bulk SMS gateway provided by a state/UT will be **restricted to the concerned State/UT only** for sending Group SMS through Integrated Alert System.
4. State/UT nodal needs to maintain **sufficient SMS credits** in the SMS Gateway account to facilitate bulk messaging and will be notified in case of credit deficiency by Integrated Alert System.

Sample Example of NIC SMS Gateway API Details



NIC SMS SERVICE INTEGRATION

- **Overview**

NIC Messaging Division has set up new SMS Gateway services. In order to integrate your application and to access different services like reports and statistics please refer to below steps.

To use the SMS service, below are the credentials.

- **User Login Credentials provided to you**

PARAMETERS	VALUES
User Name	
Password	
Sender Id	
Allowed Ip	

- **HTTP API Integration**

Using the below mentioned URL's you can send SMS. For old accounts, authentication credentials and sender id have been already provided. Please use the same credentials.

- **English (plain text) Message**

For sending SMS in English the URL is

http://smgw.sms.gov.in/failsafe/HttpLink?username=XXXXX&pass=XXXXX&message=<messagetext>&mnumber=<mobilenumber&signature=<sender id>

Using the above URL you can send English messages to any mobile subscribers. **Currently, we support maximum of 6000 characters for plain text messages.**

PARAMETERS	DESCRIPTION
Username	It is an accountid which will be provided by SMS Platform during account creation.
Pin	It is a password which will be provided by the SMS platform during the account creation.
Mnumber	It is mobile number with country code.
Message	It is a SMS text content
Signature	It is sender id or from address which will be configured by SMS platforms during account creation.

Copyright © C-DOT

1

43

:3133804/2023

Ministry of Communications
Department of Telecommunications
Director General Telecom-HQ
Khurshid Lal Bhawan New Delhi-110001

File No: 47-1/2022-DGTHQ/4 **Dated**: . 10. 2023

To,

The Heads of Licensed Service Areas (Andhra Pradesh, Bihar, Madhya Pradesh, Maharashtra, UP(West), Kolkata and West Bengal)

Subject: Procurement of Satellite Phones in LSAs-reg.

Madam/Sir,

In continuation to letter of even number dated 12.05.2022, the undersigned is directed to convey the administrative approval from DGT HQ for procurement of additional satellite phones (Total 5) from BSNL by the LSAs as per following details:

S.No.	Name of the LSA	Additional Satellite Phone sanctioned	Remarks
1	Andhra Pradesh	1	For Telangana SC unit
2	Bihar	1	For Jharkhand SC unit
3	Madhya Pradesh	1	For Chhattisgarh SC unit
4	Maharashtra	1	For Goa SC unit
5	UP(West)	1	For Uttarakhand SC unit

2. As Kolkata and WB LSAs are located in the same city, the Satellite phone sanctioned for Kolkata LSA vide letter No. 47-1/2020-DGTHQ/4 dated 12.05.2022 is to be transferred to the Sikkim unit of West Bengal LSA and Satellite Phone sanctioned for WB LSA is to be used for both Kolkata & WB LSA.

3. The LSAs may procure the satellite phones as per the financial powers delegated to them.

This issues with the approval of the Director General (Telecom).

Signed by Sumit Singh
Date: 18-10-2023 17:38:41
Reason: Approved

Assistant Director General (Technology)

CamScanner

The Disaster Communication Challenge

- Natural and man-made disasters can severely damage telecom infrastructure.
- Immediate communication is critical for rescue, coordination, and resource allocation.
- Ham radio (amateur radio) serves as a resilient, independent communication tool during outages.

What is Ham Radio (Amateur Radio)?

Hams

- Non-commercial, licensed radio service with unique Call Sign
- Operated by trained volunteers
- Uses HF, VHF, UHF, digital, and satellite bands

Capable of:

- Voice, text, image, GPS data transmission
- Functioning without internet or cellular networks

Role of Ham Radio in Disasters

- Operates independently of commercial telecom networks.
- Licensed operators can deploy quickly with minimal setup.

Supports:

- Emergency coordination
- Health and welfare traffic
- Situation reports to/from command centers

Initial Provisioning Steps

Immediately after disaster:

- Activate emergency nets (HF/VHF/UHF frequencies)
- Set up portable/field ham stations
- Deploy battery or solar-powered gear
- Coordinate with local emergency response centers (EOCs)

Network Restoration via Ham Radio

- Provide communication bridges between isolated areas
- Transmit data via packet radio APRS.
- Relay messages to regional or national agencies
- Help identify critical network damage locations

Equipment & Technologies Used

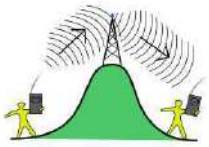
- Handheld transceivers (HTs)
- Mobile/base station radios (HF/VHF/UHF)
- Laptop/mini computers with digital modes app (APRS, FT8 etc.)
- Power sources: Solar panels, battery packs.
- Antennas: Dipole, Yagi, mobile whips

HF/VHF Communication in Emergencies

Band	Use Case	Range
HF (High Frequency)	Long-distance communication (100–1000+ km)	Inter-state / national
VHF/UHF (Very/Ultra High Frequency)	Local coverage (5–100 km)	Regional communication

- HF ideal for rural & inter-district coordination
- VHF/UHF best for urban and field team links

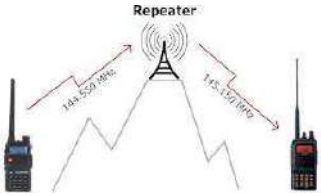
Role of Repeaters in Coverage Extension



- Repeaters receive and rebroadcast VHF/UHF signals
- Installed on high ground / towers

Essential for:

- Mountainous/remote terrain
- Urban zones with signal blockage



Proposal: Government-funded repeaters powered by **solar + battery backup** in disaster-prone areas

APRS – Real-Time Tracking & Messaging

Automatic Packet Reporting System (APRS):

- Sends GPS, short text, sensor data over radio
- Uses VHF/UHF (no internet needed)

Ideal for:

- Tracking rescue teams & vehicles
- Sending updates from the field
- Managing resource movement

SSTV – Slow Scan Television for Image Sharing

- SSTV (Slow Scan TV) transmits still images via radio

Useful for:

- Sharing flood maps, damage visuals
- Remote assessment from field teams
- Works over HF/VHF without data lines

Quick Response Teams (QRTs)

Role	Equipment	Function
Lead Operator	VHF/HF Rig	Establish comms, link to EOC
Tech Support	Tools, antenna kits	Setup, repair, troubleshoot
Field Messenger	HT Radio, APRS Tracker	Relay info, scout, track
Government Liaison	Tablet, reporting forms	Coordinate with agencies

QRTs are ready-to-deploy units capable of setting up a communication hub in

Establish Ham Radio Quick Response Teams (QRTs)

Equip QRTs with:

- Portable VHF/HF radios
- Handheld APRS trackers
- Solar panels, battery kits
- SSTV image tools
- Pre-positioned in strategic locations
- Linked to EOCs and control rooms

National/State Ham Operator Registry

Proposed initiative:

- Create a verified database of all active ham operators
- Includes:
  - Call sign, location, gear owned
  - Emergency response availability
  - Shared with: SDMAs, NDRF, DoT zones
- Train and support Quick Response Teams
- Host annual simulation drills with ham coordination

Government Support Request

- Infrastructure Support
- Fund:
  - Solar-powered repeater networks
  - Portable ham radio kits
- MOU with ham organizations for mobilization

 Policy Support

- Recognize ham radio as auxiliary emergency service
- Include in NDMA/DoT/SDMA disaster protocols

Case Example – Wayanad Landslide 2024

Event Details

- **Date:** August 2024
- **Location:** Mundakkai & Chooralmala, Wayanad district, Kerala
- **Impact:**
  - Over **150 people displaced**
  - Mobile and internet services **completely disrupted** in many areas
  - **Severe road blockages** and isolation of remote communities

Action	Details
Emergency Station Setup	District Collector's Office, Kalpetta – Operated by volunteer ham network
Repeater Deployment	Installed at **Ponmudi Kotta (Ambalavayal)** by Sultan Bathery DX Assoc.
Operators on Ground	**Sabu Mathew (VU2WMU)** and **Dr. Abraham Jacob** aided rescue ops
Coverage Achieved	Communication restored in **5 remote villages**
Time to Setup	Less than **12 hours** after the disaster struck

APRS(Automatic Packet Reporting System)

APRS is a real-time digital communication system used by amateur radio operators to share location, messages, and telemetry using packet radio.

Key Components:

- GPS Receiver: Determines current location.
- TNC (Terminal Node Controller): Converts data into AX.25 packet format.
- Radio Transmitter: Sends the packet over VHF/UHF frequencies.
- Digipeaters: Relay packets over greater distances.
- iGate Stations: Receive packets via radio and upload them to the internet.
- APRS-IS: Global internet-based system storing and distributing APRS data.

How It Flows:

- Position/Message is generated by a GPS or input by the user.
- TNC encodes the data into a packet (AX.25 format). Radio transmits the packet.
- Digipeater receives and retransmits the packet.
- iGate receives the packet and forwards it to APRS-IS.
- APRS Applications (like aprs.fi) visualize it on maps.

What Can It Send?

- Position/location
- Weather data
- Short messages
- Telemetry (voltage, temp, etc.)
- Emergency/priority alerts

Details of Chairperson of State Telecom Disaster Co-ordination Committee (STDCC)

SL. NO.	LSA NAME	DESIGNATION	NAME`	EMAIL ID	MOBILE NO
1	Andhra Pradesh	DDG(T)	Shri K. Raja Sekhar	ddgt.ap-dgt-dot@gov.in	9490000248
2	Assam	DDG(T)	Shri Vijay Agrawal	ddgt.as-dgt-dot@gov.in	9871465155
3	Bihar	DDG(T)	Shri Arun Kumar	ddgt.br-dgt-dot@gov.in	9431008989
4	Delhi	DDG(T)	Ms. Deepika Pokharna	ddgt.dl-dgt-dot@gov.in	9910809863
5	Gujarat	DDG(T)	Shri G N Kataria	ddgt.gj-dgt-dot@gov.in	9429403799
6	Himachal Pradesh	DDG(T)	Shri Sandeep Arya	ddgt.hp-dgt-dot@gov.in	8800850497
7	Haryana	DDG(T)	Shri Sanjay Kumar	ddgt.hr-dgt-dot@gov.in	9463000044
8	Jammu & Kashmir	DDG(T)	Shri Rahul Kumar Sachan	ddgt.jk-dgt-dot@gov.in	9415922999
9	Karnataka	DDG(T)	Shri Gopal Krushna Sutar	ddgt.ktk-dgt-dot@gov.in	9448010333
10	Kerala	DDG(T)	Shri Thomas Varghese	ddgt.krl-dgt-dot@nic.in	9013475879
11	Madhya Pradesh	DDG(T)	Shri Bhagwat Prasad Vishwakarma	ddgt.mp-dgt-dot@gov.in	9425001200
12	Maharashtra	DDG(T)	Shri Ujwal Gulhane	ddgt-mh-dot@gov.in	9427602244
13	Mumbai	DDG(T)	Shri Rakesh Shrivastava	ddgt.mb-dgt-dot@gov.in	9466119900
14	North East	DDG(T)	Shri Mukesh Kumar	ddgt.shillong-dot@gov.in	9413394072
15	Odisha	DDG(T)	Shri Yogesh Kumar	ddgt.dot.od@gov.in	9868254041
16	Punjab	DDG(T)	Shri Ravinder Kumar	ddgt.pb-dgt-dot@gov.in	9478800088
17	Rajasthan	DDG(T)	Shri Ram Kumar	ddgt.rj-dgt-dot@gov.in	7383001592
18	Tamilnadu	DDG(T)	Shri K S R Suresh Kumar	ddgt.tn-dgt-dot@gov.in	9444030201
19	UP(E)	DDG(T)	Shri Deep Krishna Misra	ddgt.upe-dgt-dot@gov.in	9415022325
20	UP(W)	DDG(T)	Shri Ashok Kumar	ddgt.upw-dgt-dot@gov.in	9818655056
21	West Bengal (Incl. Kolkata)	DDG(T)	Shri Tarak Nath Das	ddgt.wb-dgt-dot@gov.in	9435599499

Details of Satellite Phones available with LSAs			
S.No.	**LSA Name**	**Location**	**Satellite Phone No.**
1	AP	Hyderabad	8991114092
2	Assam	Guwahati	8991115521
3	Bihar	Bihar	8991125065
4	Delhi	New Delhi	8991115539
5	Gujarat	Ahmedabad	8991124463
6	Himachal Pradesh	Shimla	8991125182
7	Haryana	Ambala	8991115792
8	J&K LSA	Jammu	8991115777
9	Karnataka	Bengaluru	8991125184
10	Kerala	Ernakulam	8991114090
	Madhya Pradesh	Bhopal	8991124464
11		Raipur	8991124432
	Maharashtra	Pune	8991125213
12		Goa	8991124434
13	Mumbai	Mumbai	8991125070
	North East	Meghalaya	8991115684
14		Nagaland	8991115556
15	Odisha	Bhubaneswar	8991115695
16	Punjab	Mohali	8991115682
17	Rajasthan	Jaipur	8991125186
18	Tamil Nadu	Chennai	8991115751
19	UP(E)	Lucknow	8991115731
20	UP(W)	Dehradun	8991115515
21	West Bengal	Kolkata	8991124519
		Andaman & Nicobar Islands	8991125192
		Gangtok	8991115535